

GRADE 5 COMMON CORE MATH

Question Collection

Main • Backward Branches • Forward 1 (Grade 6) • Forward 2 (Grade 7)
57 question sets • 1580 questions

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Evaluate expressions that use parentheses

MAIN	Grade 5 main question	5.OA.A.1
BACKWARD 1	Do the work in parentheses first	3.OA.D.8
BACKWARD 2	Multiplication and division facts	3.OA.C.7
BACKWARD 3	Adding and subtracting multi-digit numbers	4.NBT.B.4
FORWARD 1	Evaluate expressions with variables	6.EE.A.2.c
FORWARD 2	Evaluate expressions with negative numbers	7.NS.A.3

SET 1 • 5.OA.A.1

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$6 \times (8 + 5)$$

Value: _____

Evaluate.

$$(35 - 11) \div 4$$

Value: _____

Evaluate.

$$50 - (6 \times 7)$$

Value: _____

Evaluate.

$$3 + (24 \div 6) \times 5$$

A. 35

B. 135

C. 27

D. 23

Evaluate.

$$(9 + 3) \times (10 - 4)$$

Value: _____

SET 1 • 5.OA.A.1

BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

True or false?

In $4 \times (2 + 5)$, you add $2 + 5$ first.

A. True

B. False

Evaluate.

$$(6 + 4) \times 2$$

Value: _____

Which step comes first in $18 - (3 \times 4)$?

- A.** Subtract $18 - 4$
- B.** Multiply 3×4
- C.** Subtract $18 - 3$
- D.** Add $3 + 4$

Evaluate.

$$20 - (5 + 5)$$

Value: _____

True or false?

$$(10 - 4) + 3 = 10 - 7$$

A. True

B. False

SET 1 • 5.OA.A.1

BACKWARD 2

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$7 \times 8$$

Product: _____

Divide.

$$24 \div 6$$

Quotient: _____

True or false?

$$9 \times 7 = 56$$

A. True

B. False

$$48 \div 8 = ?$$

A. 8

B. 5

C. 7

D. 6

Multiply.

$$3 \times 9$$

Product: _____

SET 1 • 5.OA.A.1

BACKWARD 3

Adding and subtracting multi-digit numbers • Grade 4 • Number & Operations
in Base Ten • 4.NBT.B.4

Add.

$$348 + 275$$

Sum: _____

Subtract.

$$502 - 168$$

Difference: _____

True or false?

$$1,250 - 375 = 985$$

A. True

B. False

$$609 + 194 = ?$$

A. 793

B. 703

C. 813

D. 803

Subtract.

$$1,000 - 432$$

Difference: _____

SET 1 • 5.OA.A.1

FORWARD 1

Evaluate expressions with variables • Grade 6 • Expressions & Equations •
6.EE.A.2.c

Evaluate $4(x + 3)$ when $x = 5$.

Value: _____

Evaluate $20 - 2y$ when $y = 6$.

Value: _____

Evaluate $(a + b) \div 3$ when $a = 10$ and $b = 8$.

- A.** 4
- B.** 9
- C.** 54
- D.** 6

Evaluate $3m + 2$ when $m = 7$.

Value: _____

True or false?

When $n = 4$, the value of $5(n - 1)$ is 19.

A. True

B. False

SET 1 • 5.OA.A.1

FORWARD 2

Evaluate expressions with negative numbers • Grade 7 • The Number System •
7.NS.A.3

Find the value.

$$-3 \times (4 - 9)$$

Value: _____

Find the value.

$$(-12 + 4) \div 2$$

Value: _____

Find the value.

$$6 - (2 - 10)$$

A. -2

B. 2

C. 14

D. 18

Find the value.

$$(-5) \times (-2) + (-7)$$

Value: _____

True or false?

$$(-20) \div (3 + 2) = 4$$

A. True

B. False

Evaluate expressions with brackets and braces

MAIN	Grade 5 main question	5.OA.A.1
BACKWARD 1	Do the work in parentheses first	3.OA.D.8
BACKWARD 2	Multiplication and division facts	3.OA.C.7
BACKWARD 3	Adding and subtracting within 100	2.NBT.B.5
FORWARD 1	Evaluate expressions with whole-number exponents	6.EE.A.1
FORWARD 2	Grouping symbols with negative numbers	7.NS.A.3

SET 2 • 5.OA.A.1

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$2 \times [5 + (3 \times 4)]$$

Value: _____

Evaluate.

$$[20 - (2 + 6)] \div 3$$

Value: _____

Evaluate.

$$4 \times \{3 + [10 - (2 \times 4)]\}$$

A. 36

B. 20

C. 44

D. 26

Evaluate.

$$60 \div [3 \times (7 - 5)]$$

Value: _____

Evaluate.

$$\{18 - [2 \times (4 + 1)]\} + 9$$

Value: _____

SET 2 • 5.OA.A.1

BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

Evaluate.

$$(12 - 5) \times 3$$

Value: _____

True or false?

In $(8 + 2) \div 5$, you divide $2 \div 5$ first.

A. True

B. False

Evaluate.

$$4 \times (9 - 6)$$

A. 33

B. 10

C. 30

D. 12

Evaluate.

$$30 \div (2 + 4)$$

Value: _____

True or false?

$$(3 + 5) \times 2 = 16$$

A. True

B. False

SET 2 • 5.OA.A.1

BACKWARD 2

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$8 \times 6$$

Product: _____

Divide.

$$56 \div 7$$

Quotient: _____

True or false?

$$36 \div 4 = 8$$

A. True

B. False

$$7 \times 9 = ?$$

A. 63

B. 56

C. 72

D. 49

Divide.

$$81 \div 9$$

Quotient: _____

SET 2 • 5.OA.A.1

BACKWARD 3

Adding and subtracting within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add.

$$46 + 37$$

Sum: _____

Subtract.

$$72 - 28$$

Difference: _____

True or false?

$$55 + 29 = 74$$

A. True

B. False

$$90 - 47 = ?$$

A. 47

B. 43

C. 53

D. 33

Add.

$$38 + 54$$

Sum: _____

SET 2 • 5.OA.A.1

FORWARD 1

Evaluate expressions with whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$2 \times (3 + 1)^2$$

Value: _____

Evaluate.

$$[10 - 2^3] \times 5$$

Value: _____

Evaluate.

$$4^2 - (6 + 2)$$

A. 8

B. 2

C. 0

D. 24

Evaluate.

$$(5^2 - 1) \div 6$$

Value: _____

True or false?

$$3 \times 2^2 = 36$$

A. True

B. False

SET 2 • 5.OA.A.1

FORWARD 2

Grouping symbols with negative numbers • Grade 7 • The Number System •
7.NS.A.3

Find the value.

$$2 \times [-3 + (8 \div (-4))]$$

Value: _____

Find the value.

$$[(-6) + 15] \div (-3)$$

Value: _____

Find the value.

$$-4 \times [5 - (-1)]$$

A. 16

B. -16

C. -24

D. 24

Find the value.

$$\{10 - [2 \times (-3)]\} \div 4$$

Value: _____

True or false?

$$[-8 - (-2)] \times 3 = -30$$

A. True

B. False

Place grouping symbols to make an equation true

MAIN	Grade 5 main question	5.OA.A.1
BACKWARD 1	Do the work in parentheses first	3.OA.D.8
BACKWARD 2	When regrouping keeps the same value	3.OA.B.5
BACKWARD 3	Multiplication and division facts	3.OA.C.7
FORWARD 1	Use parentheses to write equivalent expressions	6.EE.A.3
FORWARD 2	Expand and factor expressions with rational coefficients	7.EE.A.1

SET 3 • 5.OA.A.1

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Insert parentheses to make the equation true.

$$4 + 6 \times 3 = 30$$

Equation: _____

Insert parentheses to make the equation true.

$$20 - 8 - 2 = 14$$

Equation: _____

Which placement of parentheses makes the equation true?

$$2 \times 5 + 3 \times 4 = 52$$

A. $2 \times (5 + 3) \times 4 = 52$

B. $(2 \times 5 + 3) \times 4 = 52$

C. $2 \times (5 + 3 \times 4) = 52$

D. $2 \times 5 + (3 \times 4) = 52$

Insert parentheses to make the equation true.

$$36 \div 6 - 2 = 9$$

Equation: _____

Insert brackets and parentheses to make the equation true.

$$2 \times 9 - 3 + 1 = 10$$

Equation: _____

SET 3 • 5.OA.A.1

BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

Evaluate.

$$(4 + 6) \times 3$$

Value: _____

Evaluate.

$$4 + (6 \times 3)$$

Value: _____

True or false?

$$(20 - 8) - 2 = 20 - (8 - 2)$$

A. True

B. False

Evaluate.

$$36 \div (6 - 2)$$

A. 4

B. 34

C. 9

D. 7

Evaluate.

$$(36 \div 6) - 2$$

Value: _____

SET 3 • 5.OA.A.1

BACKWARD 2

When regrouping keeps the same value • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$(2 \times 3) \times 4 = 2 \times (3 \times 4)$$

A. True

B. False

True or false?

$$(12 \div 6) \div 2 = 12 \div (6 \div 2)$$

A. True

B. False

Which equation is true?

A. $(5 + 2) + 3 = 5 + (2 + 3)$

B. $(9 - 4) - 1 = 9 - (4 - 1)$

C. $(16 \div 4) \div 2 = 16 \div (4 \div 2)$

D. $(7 - 3) - 2 = 7 - (3 - 2)$

True or false?

$$4 \times (5 \times 2) = (4 \times 5) \times 2$$

A. True

B. False

Fill in the blank.

$$(6 \times 2) \times 5 = 6 \times (2 \times \underline{\quad})$$

Blank: _____

SET 3 • 5.OA.A.1

BACKWARD 3

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$9 \times 4$$

Product: _____

Divide.

$$42 \div 6$$

Quotient: _____

True or false?

$$8 \times 7 = 54$$

A. True

B. False

$$64 \div 8 = ?$$

A. 6

B. 7

C. 8

D. 9

Multiply.

$$6 \times 7$$

Product: _____

SET 3 • 5.OA.A.1

FORWARD 1

Use parentheses to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

Expand.

$$3(x + 5)$$

Expression: _____

Expand.

$$4(2y - 1)$$

Expression: _____

Which expression is equivalent to $6(a + 2)$?

- A.** $6a + 2$
- B.** $a + 12$
- C.** $6a + 12$
- D.** $8a$

Factor.

$$5m + 20$$

Expression: _____

True or false?

$$2(x + 7) = 2x + 7$$

A. True

B. False

SET 3 • 5.OA.A.1

FORWARD 2

Expand and factor expressions with rational coefficients • Grade 7 •
Expressions & Equations • 7.EE.A.1

Expand.

$$-2(3x - 4)$$

Expression: _____

Expand.

$$\frac{1}{2}(6y + 10)$$

Expression: _____

Which expression is equivalent to $-5(n + 2)$?

A. $-5n + 2$

B. $-5n - 10$

C. $-5n + 10$

D. $5n - 10$

Factor out 3.

$$3x - 12$$

Expression: _____

True or false?

$$0.5(4a - 8) = 2a - 8$$

A. True

B. False

Write numerical expressions from words

MAIN	Grade 5 main question	5.OA.A.2
BACKWARD 1	Products as equal groups	3.OA.A.1
BACKWARD 2	Quotients as equal shares	3.OA.A.2
BACKWARD 3	Sums and differences	2.OA.A.1
FORWARD 1	Write expressions with variables	6.EE.A.2.a
FORWARD 2	Write and solve equations from words	7.EE.B.4.a

SET 4 • 5.OA.A.2

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Write an expression for "add 9 and 6, then multiply by 4."

Expression: _____

Write an expression for "subtract 5 from 30, then divide by 5."

Expression: _____

Which expression shows "triple the sum of 12 and 8"?

A. $12 + 8 \times 3$

B. $3 \times 12 + 8$

C. $3 \times (12 + 8)$

D. $(3 + 12) \times 8$

Write an expression for "divide 40 by 8, then add 7."

Expression: _____

Write an expression for "the product of 6 and the difference of 15 and 9."

Expression: _____

SET 4 • 5.OA.A.2

BACKWARD 1

Products as equal groups • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.1

True or false?

5×7 can mean 5 groups of 7.

A. True

B. False

Which expression shows 4 groups of 6?

A. 46

B. $6 - 4$

C. $4 + 6$

D. 4×6

Write a multiplication expression for 8 groups of 3.

Expression: _____

True or false?

$$3 + 3 + 3 + 3 = 3 \times 3$$

A. True

B. False

What is the product of 7 and 5?

Product: _____

SET 4 • 5.OA.A.2

BACKWARD 2

Quotients as equal shares • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

True or false?

$15 \div 3$ can mean 15 shared equally into 3 groups.

A. True

B. False

Which expression shows 20 apples shared equally among 4 bags?

A. 20×4

B. $4 \div 20$

C. $20 \div 4$

D. $20 - 4$

What is the quotient of 18 and 6?

Quotient: _____

Write a division expression for 32 split into groups of 8.

Expression: _____

True or false?

The quotient of 12 and 2 is 24.

A. True

B. False

SET 4 • 5.OA.A.2

BACKWARD 3

Sums and differences • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

What is the sum of 14 and 9?

Sum: _____

What is the difference of 20 and 7?

Difference: _____

True or false?

The difference of 15 and 6 is 21.

A. True

B. False

Which word names the answer to an addition problem?

- A.** Difference
- B.** Quotient
- C.** Product
- D.** Sum

Write an expression for "7 more than 25."

Expression: _____

SET 4 • 5.OA.A.2

FORWARD 1

Write expressions with variables • Grade 6 • Expressions & Equations •
6.EE.A.2.a

Write an expression for "the sum of n and 8, times 3."

Expression: _____

Write an expression for "y divided by 5, then minus 2."

Expression: _____

Which expression means "4 less than the product of 2 and x"?

A. $4 - 2x$

B. $2x - 4$

C. $2(x - 4)$

D. $2 + x - 4$

Write an expression for "twice the difference of m and 6."

Expression: _____

True or false?

"9 more than k " can be written as $k + 9$.

A. True

B. False

SET 4 • 5.OA.A.2

FORWARD 2

Write and solve equations from words • Grade 7 • Expressions & Equations •
7.EE.B.4.a

Three times the sum of a number and 4 is 27.

Write and solve an equation to find the number.

Equation and solution: _____

Half of a number, minus 6, is 10.

Write and solve an equation to find the number.

Equation and solution: _____

Which equation represents "2 less than 5 times a number is 33"?

A. $5n + 2 = 33$

B. $5n - 2 = 33$

C. $2 - 5n = 33$

D. $5(n - 2) = 33$

A number divided by 4, plus 3, is 9.

Write and solve an equation to find the number.

Equation and solution: _____

True or false?

The solution of $2(x + 3) = 18$ is $x = 7$.

A. True

B. False

Interpret numerical expressions without evaluating them

MAIN	Grade 5 main question	5.OA.A.2
BACKWARD 1	Multiplicative comparison	4.OA.A.1
BACKWARD 2	Groups of a sum	3.OA.A.1
BACKWARD 3	Equal shares	3.OA.A.2
FORWARD 1	Identify parts of an expression	6.EE.A.2.b
FORWARD 2	Rewrite expressions to show how quantities are related	7.EE.A.2

SET 5 • 5.OA.A.2

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

How does $4 \times (563 + 218)$ compare to $563 + 218$?

- A.** It is 4 more.
- B.** It is 4 times as large.
- C.** It is 4 less.
- D.** It is one-fourth as large.

True or false?

$(7,215 - 1,005) \div 5$ is one-fifth as large as $7,215 - 1,005$.

A. True

B. False

Without calculating, how many times as large is
 $9 \times (48 + 27)$ as $48 + 27$?

Times as large: _____

Which expression is twice as large as 356×12 ?

A. $(356 + 2) \times 12$

B. $356 \times 12 + 2$

C. 356×14

D. $2 \times (356 \times 12)$

True or false?

$6 \times 25 + 4$ is 6 times as large as $25 + 4$.

A. True

B. False

SET 5 • 5.OA.A.2

BACKWARD 1

Multiplicative comparison • Grade 4 • Operations & Algebraic Thinking •
4.OA.A.1

True or false?

35 is 5 times as many as 7.

A. True

B. False

Fill in the blank.

48 is ____ times as many as 6.

Blank: _____

Which equation shows "36 is 4 times as many as 9"?

A. $9 = 4 \times 36$

B. $36 = 4 + 9$

C. $36 - 9 = 4$

D. $36 = 4 \times 9$

True or false?

30 is 3 times as many as 27.

A. True

B. False

Tom has 6 cards. Ana has 3 times as many cards.
How many cards does Ana have?

Cards: _____

SET 5 • 5.OA.A.2

BACKWARD 2

Groups of a sum • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$3 \times (4 + 1)$ means 3 groups of $(4 + 1)$.

A. True

B. False

Which expression means 6 groups of $(10 + 2)$?

A. $6 \times (10 + 2)$

B. $(6 + 10) \times 2$

C. $6 \times 10 + 2$

D. $6 + (10 + 2)$

How many groups of $(5 + 4)$ are in $7 \times (5 + 4)$?

Groups: _____

True or false?

$$2 \times (8 + 1) = (8 + 1) + 2$$

A. True

B. False

Write 4 groups of 9 as a multiplication expression.

Expression: _____

SET 5 • 5.OA.A.2

BACKWARD 3

Equal shares • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

True or false?

12 shared equally among 3 people is 4 each.

A. True

B. False

A 20-inch ribbon is cut into 5 equal pieces.

How long is each piece?

Length: _____

Which expression finds one of 4 equal shares of 28?

A. $28 - 4$

B. $28 \div 4$

C. 28×4

D. $4 \div 28$

True or false?

Dividing a number by 2 splits it into 2 equal shares.

A. True

B. False

30 crayons are shared equally among 6 children.

How many crayons does each child get?

Crayons: _____

SET 5 • 5.OA.A.2

FORWARD 1

Identify parts of an expression • Grade 6 • Expressions & Equations •
6.EE.A.2.b

In $5(x + 2)$, what is 5?

- A.** A term of $x + 2$
- B.** A quotient
- C.** A sum
- D.** A factor

How many terms are in $4a + 7b - 1$?

Terms: _____

What is the coefficient of m in $9m + 4$?

Coefficient: _____

True or false?

In $3(y + 6)$, the quantity $(y + 6)$ is a sum of two terms.

A. True

B. False

What is the expression $12 \div (n + 1)$ called?

- A.** A coefficient
- B.** A quotient
- C.** A sum
- D.** A product

SET 5 • 5.OA.A.2

FORWARD 2

Rewrite expressions to show how quantities are related • Grade 7 •
Expressions & Equations • 7.EE.A.2

A price p is increased by 5%.

Which expression shows the new price?

A. $0.05p$

B. $1.05p$

C. $5p$

D. $p + 5$

True or false?

$$x + 0.3x = 0.4x$$

A. True

B. False

Write $n - 0.4n$ as one term.

Expression: _____

Which expression is equivalent to $2(3t)$?

- A.** $5t$
- B.** $2t + 3t$
- C.** $6t$
- D.** $3t + 2$

A sale price is $0.75p$, where p is the original price.
What percent of the original price is taken off?

Percent: _____

Generate two numerical patterns and compare their terms

MAIN	Grade 5 main question	5.OA.B.3
BACKWARD 1	Patterns that follow a rule	4.OA.C.5
BACKWARD 2	Multiplication facts	3.OA.C.7
FORWARD 1	Equations relating dependent and independent variables	6.EE.C.9
FORWARD 2	Equations for proportional relationships	7.RP.A.2.c

SET 6 • 5.OA.B.3

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 3.

Pattern B starts at 0 and adds 6.

Write the first 5 terms of each pattern.

Pattern A: _____

Pattern B: _____

Pattern A: 0, 4, 8, 12, 16

Pattern B: 0, 12, 24, 36, 48

How does each term in Pattern B compare to the matching term in Pattern A?

A. 8 more

B. 12 more

C. 4 times as large

D. 3 times as large

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 10.

What is the 6th term of each pattern?

Pattern A: _____

Pattern B: _____

True or false?

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 8. Each term in Pattern B is 4 times the matching term in Pattern A.

- A.** True
- B.** False

Pattern A starts at 0 and adds 7. Pattern B starts at 0 and adds 21.

When Pattern A is 28, what is the matching term in Pattern B?

Term: _____

SET 6 • 5.OA.B.3

BACKWARD 1

Patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking •
4.OA.C.5

The rule is "add 6." The pattern starts at 0.
Write the first five terms.

Terms: _____

True or false?

The rule "add 4," starting at 0, gives 0, 4, 8, 16.

A. True

B. False

What is the rule for the pattern?

0, 9, 18, 27, 36

- A.** Add 9
- B.** Add 18
- C.** Multiply by 9
- D.** Add 8

The rule is "add 11." The pattern starts at 0.
What is the 4th term?

Term: _____

What comes next?

0, 15, 30, 45, ____

Next term: _____

SET 6 • 5.OA.B.3

BACKWARD 2

Multiplication facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

$$4 \times 7$$

Product: _____

True or false?

$$6 \times 8 = 42$$

A. True

B. False

What number times 3 equals 24?

A. 8

B. 7

C. 21

D. 6

Multiply.

$$5 \times 9$$

Product: _____

What number times 6 equals 42?

Number: _____

SET 6 • 5.OA.B.3

FORWARD 1

Equations relating dependent and independent variables • Grade 6 •
Expressions & Equations • 6.EE.C.9

Each box holds 6 cans.

Write an equation for the number of cans c in b boxes.

Equation: _____

In $c = 6b$, which variable is the dependent variable?

- A.** b
- B.** c
- C.** 6
- D.** Neither

Write an equation for y in terms of x .

x	1	2	3	4
y	5	10	15	20

Equation: _____

True or false?

In $y = x + 4$, y is the independent variable.

A. True

B. False

A plant grows 2 cm each week.

Write an equation for its growth g after w weeks.

Equation: _____

SET 6 • 5.OA.B.3

FORWARD 2

Equations for proportional relationships • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Pencils cost \$0.35 each.

Write an equation for the cost c of n pencils.

Equation: _____

The table shows a proportional relationship.

Write an equation for y in terms of x .

x	4	6	10
y	10	15	25

Equation: _____

Which equation represents a proportional relationship?

A. $y = 1.5$

B. $y = x + 1.5$

C. $y = 1.5x + 2$

D. $y = 1.5x$

A train travels 150 miles in 2 hours at a constant speed.
Write an equation for the distance d after t hours.

Equation: _____

True or false?

$y = \frac{3}{4}x$ represents a proportional relationship.

A. True

B. False

Form ordered pairs from two patterns and graph them

MAIN	Grade 5 main question	5.OA.B.3
BACKWARD 1	Patterns that follow a rule	4.OA.C.5
BACKWARD 2	Ordered pairs and the coordinate plane	5.G.A.1
BACKWARD 3	Reading the scale on an axis	3.MD.B.3
FORWARD 1	Plot pairs of values from a ratio table	6.RP.A.3.a
FORWARD 2	Use a graph to decide whether a relationship is proportional	7.RP.A.2.a

SET 7 • 5.OA.B.3

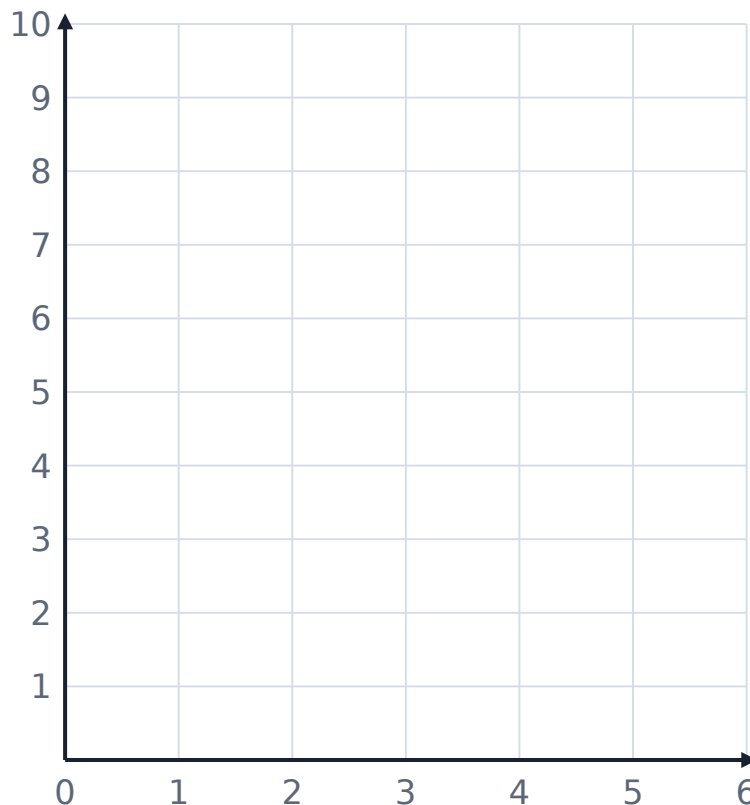
MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A (x): start at 0 and add 1.

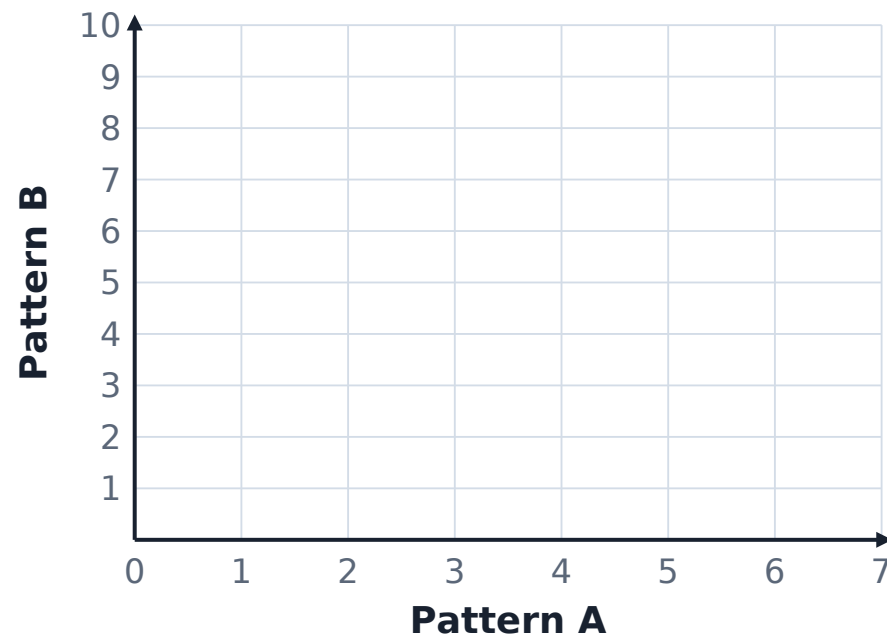
Pattern B (y): start at 0 and add 2.

Graph the ordered pairs for the first 5 terms.



Graph the ordered pairs from the two patterns.

Pattern A	0	2	4	6
Pattern B	0	3	6	9



Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 9.

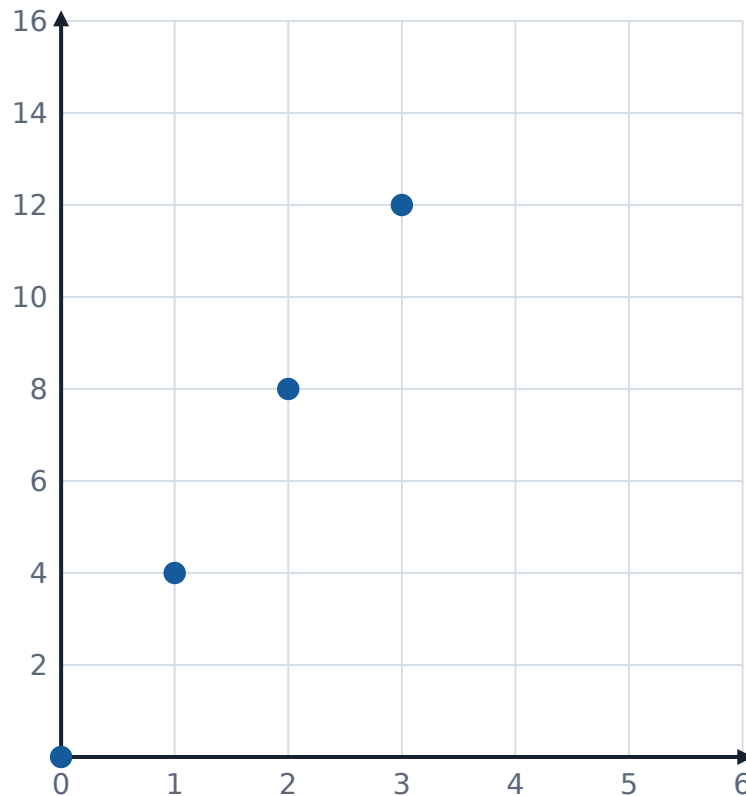
Write the first four ordered pairs (A, B).

Ordered pairs: _____

The points come from two patterns.

Which statement is true?

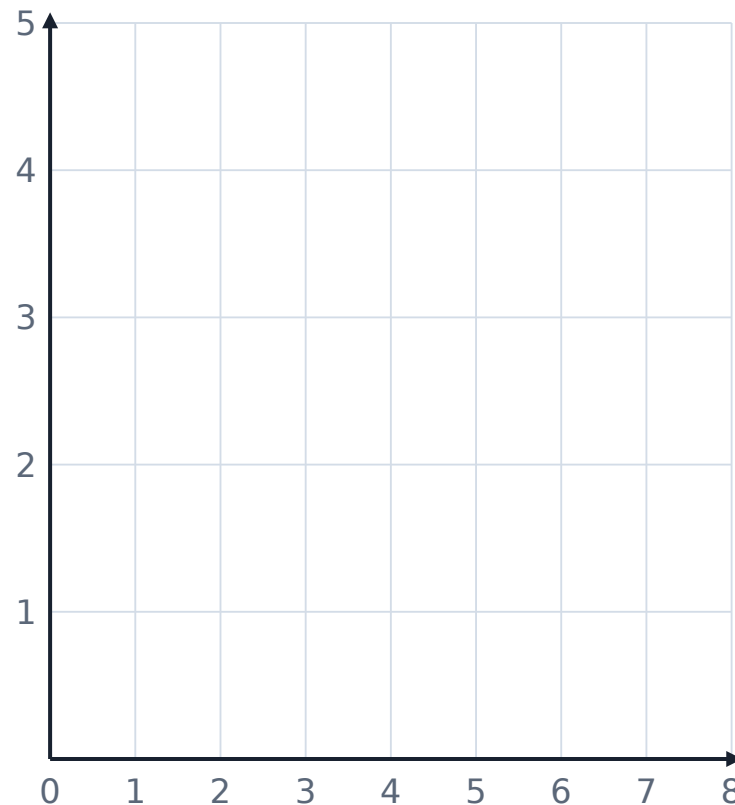
- A.** The y-values increase by 1 each time.
- B.** Each y-value is 4 times its x-value.
- C.** Each x-value is 4 times its y-value.
- D.** Each y-value is 4 more than its x-value.



Pattern A (x): start at 0 and add 2.

Pattern B (y): start at 0 and add 1.

Graph the ordered pairs for the first 4 terms.



SET 7 • 5.OA.B.3

BACKWARD 1

Patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking •
4.OA.C.5

The rule is "add 3." The pattern starts at 0.
What is the 5th term?

Term: _____

True or false?

The pattern 0, 5, 10, 15 follows the rule "add 5."

A. True

B. False

Which pattern follows the rule "add 2," starting at 0?

A. 0, 2, 4, 6

B. 2, 4, 6, 8

C. 0, 1, 2, 3

D. 0, 2, 4, 8

What is the rule for the pattern?

0, 8, 16, 24

Rule: _____

The rule is "add 10." The pattern starts at 0.
What is the 6th term?

Term: _____

SET 7 • 5.OA.B.3

BACKWARD 2

Ordered pairs and the coordinate plane • Grade 5 • Geometry • 5.G.A.1

True or false?

In $(3, 6)$, the 3 tells how far to move along the x-axis.

A. True

B. False

In the ordered pair (4, 9), which number is the y-coordinate?

- A.** 9
- B.** 4
- C.** 13
- D.** 5

Which ordered pair names the origin?

Ordered pair: _____

True or false?

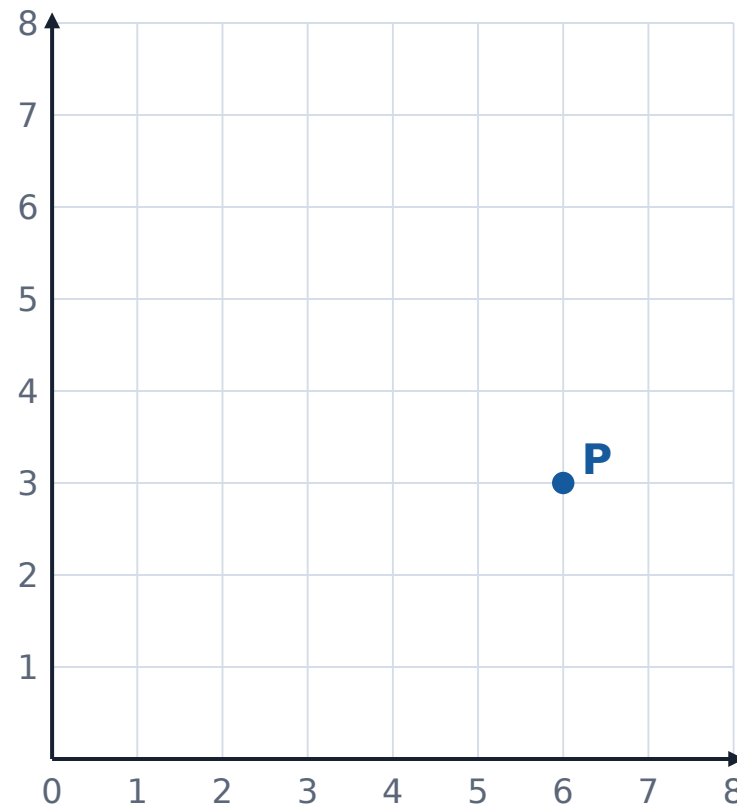
$(2, 5)$ and $(5, 2)$ name the same point.

A. True

B. False

What are the coordinates of point P?

P = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • 5.OA.B.3

BACKWARD 3

Reading the scale on an axis • Grade 3 • Measurement & Data • 3.MD.B.3

An axis is numbered 0, 2, 4, 6, 8.

How much does each grid line stand for?

A. 4

B. 1

C. 8

D. 2

True or false?

On an axis numbered 0, 4, 8, 12, the value 10 is halfway between 4 and 8.

- A. True
- B. False

An axis is numbered 0, 5, 10, 15.

Where does 20 go?

Answer: _____

On an axis numbered by 3s, which value is two grid lines above 0?

A. 6

B. 3

C. 9

D. 2

True or false?

On an axis numbered by 10s, the value 25 is on a grid line.

- A.** True
- B.** False

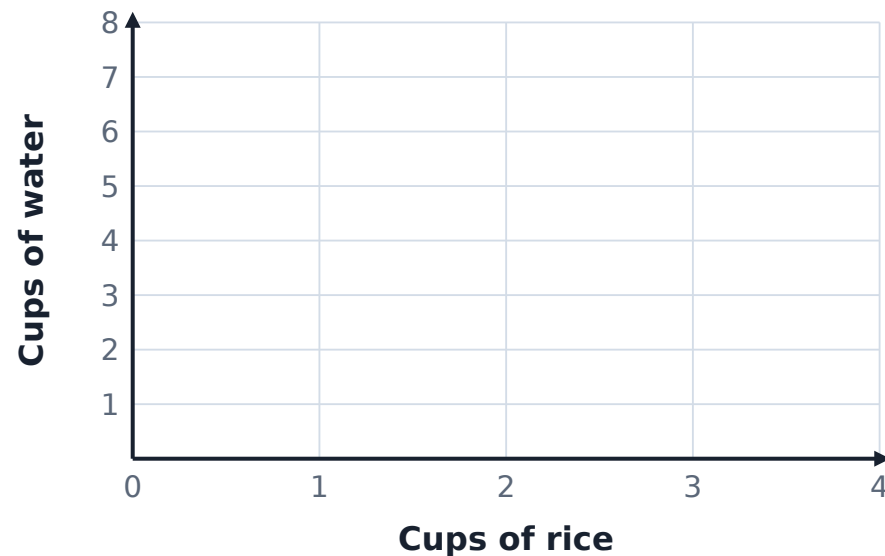
SET 7 • 5.OA.B.3

FORWARD 1

Plot pairs of values from a ratio table • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

Plot the pairs of values from the ratio table on the coordinate plane.

Cups of rice	1	2	3
Cups of water	2	4	6



The table shows equivalent ratios.

Find the missing value.

Laps	2	4	6	8
Minutes	5	10	?	20

Missing value: _____

Which point comes from the ratio table?

Boxes	1	2	3
Pens	6	12	18

A. (2, 6)

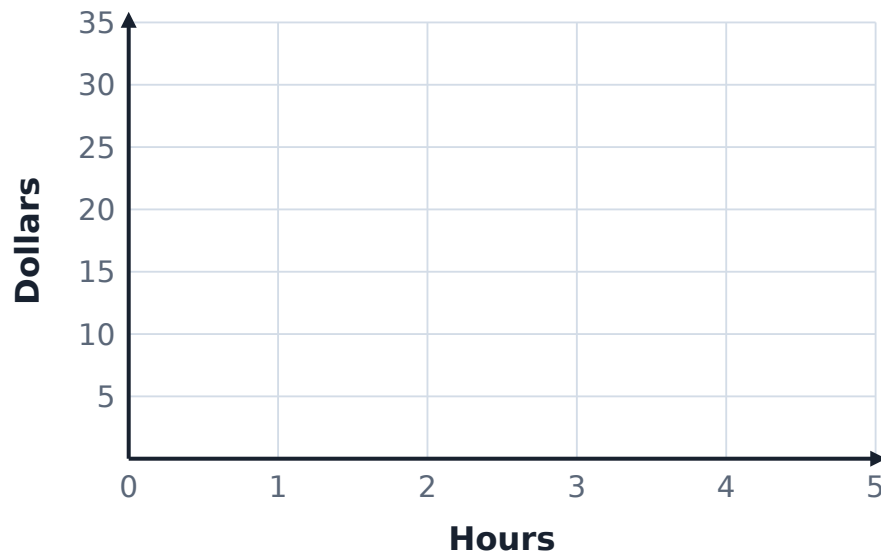
B. (12, 2)

C. (2, 12)

D. (3, 12)

Plot the pairs of values from the ratio table on the coordinate plane.

Hours	1	2	3	4
Dollars	8	16	24	32



True or false?

The points (1, 3), (2, 6), and (3, 9) come from a ratio table with the ratio 1 : 3.

A. True

B. False

SET 7 • 5.OA.B.3

FORWARD 2

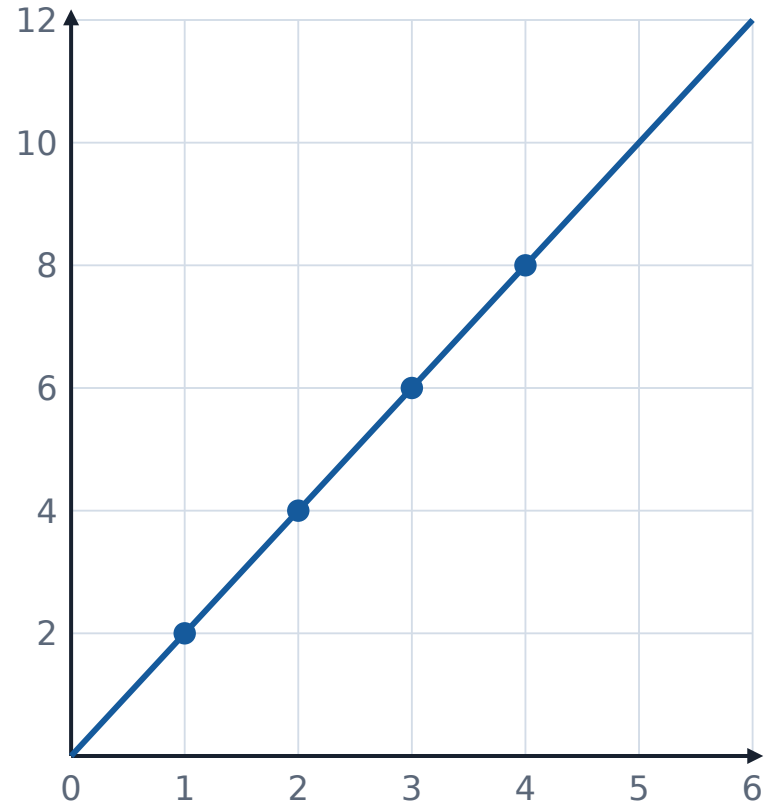
Use a graph to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The points lie on a straight line through the origin, so the relationship is proportional.

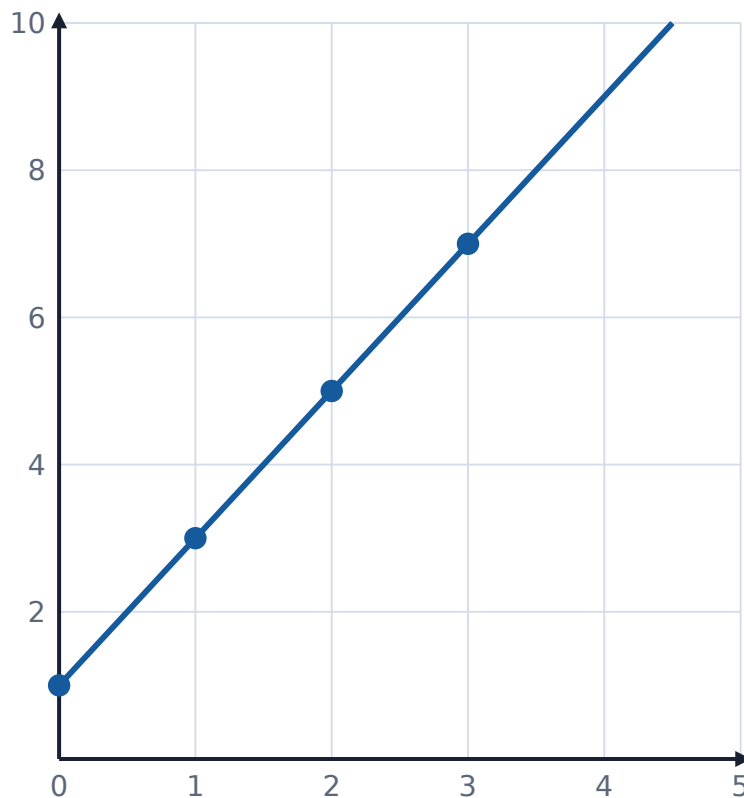
A. True

B. False



Which statement about the graph is true?

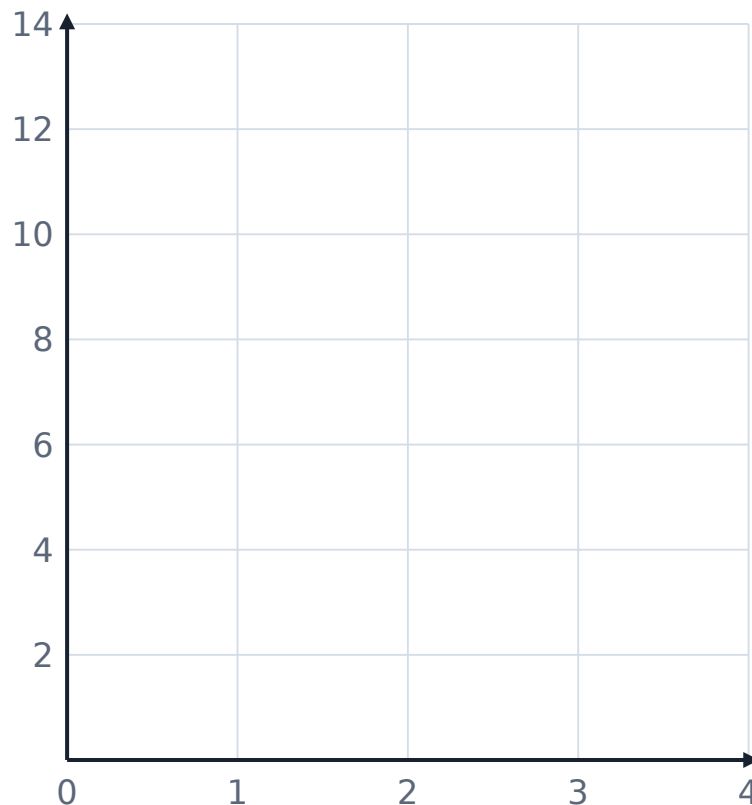
- A.** It is not proportional because the line does not pass through $(0, 0)$.
- B.** It is not proportional because the points are not on a line.
- C.** It is proportional because the points are on a line.
- D.** It is proportional because y increases by 2.



Graph the pairs $(1, 4)$, $(2, 8)$,
and $(3, 12)$.

Is the relationship
proportional?

Answer: _____



True or false?

The points (1, 2), (2, 5), and (3, 8) show a proportional relationship.

- A. True
- B. False

Which set of points shows a proportional relationship?

- A.** (1, 1), (2, 4), (3, 9)
- B.** (1, 2), (2, 3), (3, 4)
- C.** (0, 3), (1, 6), (2, 9)
- D.** (2, 5), (4, 10), (6, 15)

A digit is worth 10 times as much as the same digit one place to its right

MAIN	Grade 5 main question	5.NBT.A.1
BACKWARD 1	Each place is worth 10 times the place to its right	4.NBT.A.1
BACKWARD 2	Tenths and hundredths as decimals	4.NF.C.6
BACKWARD 3	Tenths as hundredths	4.NF.C.5
FORWARD 1	Compute with multi-digit decimals	6.NS.B.3
FORWARD 2	Real-world problems with rational numbers	7.NS.A.3

SET 8 • 5.NBT.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.1

In 5,550, how does the value of the 5 in the hundreds place compare to the value of the 5 in the tens place?

- A.** It is 100 times as much.
- B.** It is 10 times as much.
- C.** It is the same.
- D.** It is $\frac{1}{10}$ as much.

In 3.33, the first 3 is in the ones place and the second 3 is in the tenths place.

How many times as much is the first 3 worth as the second 3?

Times as much: _____

True or false?

In 0.77, the 7 in the tenths place is worth 10 times as much as the 7 in the hundredths place.

- A. True
- B. False

Find the value of the 4 in 642.18 and the value of the 4 in 6,421.8.

How many times as much is the second 4 worth?

4 in 642.18: _____

4 in 6,421.8: _____

Times as much: _____

In which number is the 6 worth 10 times as much as the 6 in 3.6?

- A.** 0.36
- B.** 3.06
- C.** 46.1
- D.** 60.3

SET 8 • 5.NBT.A.1

BACKWARD 1

Each place is worth 10 times the place to its right • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1

True or false?

In 880, the 8 in the hundreds place is worth 10 times as much as the 8 in the tens place.

- A. True
- B. False

What is the value of the 7 in 2,740?

Value: _____

Which number is 10 times as much as 60?

- A.** 600
- B.** 6,000
- C.** 70
- D.** 6

How many tens are in 1 hundred?

Tens: _____

True or false?

70 is 10 times as much as 7.

A. True

B. False

SET 8 • 5.NBT.A.1

BACKWARD 2

Tenths and hundredths as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{3}{10}$ as a decimal.

Decimal: _____

Write $\frac{47}{100}$ as a decimal.

Decimal: _____

True or false?

$$0.6 = \frac{6}{100}$$

A. True

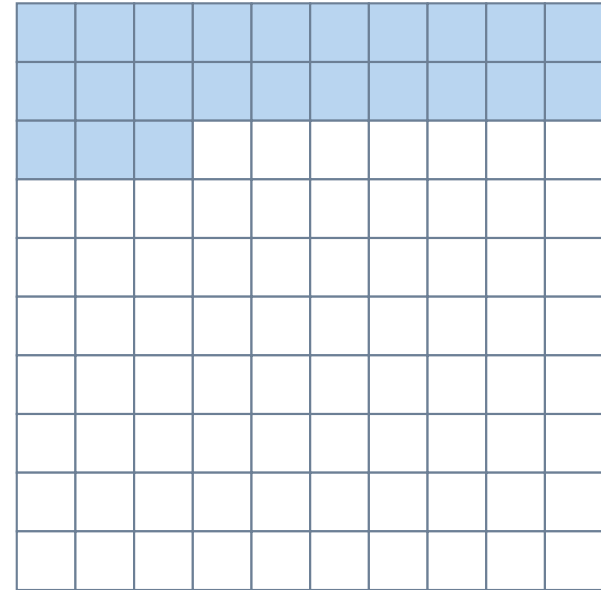
B. False

Which decimal shows 5 hundredths?

- A.** 0.5
- B.** 0.05
- C.** 0.005
- D.** 5.0

What decimal does the shaded part of the grid show?

Decimal: _____



SET 8 • 5.NBT.A.1

BACKWARD 3

Tenths as hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.5

True or false?

$$\frac{1}{10} = \frac{10}{100}$$

A. True

B. False

How many hundredths are in 1 tenth?

Hundredths: _____

Which fraction is equal to $\frac{4}{10}$?

A. $\frac{4}{100}$

B. $\frac{400}{100}$

C. $\frac{14}{100}$

D. $\frac{40}{100}$

Find the missing number.

$$\frac{7}{10} = \frac{?}{100}$$

Missing number: _____

True or false?

$$\frac{3}{10} + \frac{5}{100} = \frac{35}{100}$$

A. True

B. False

SET 8 • 5.NBT.A.1

FORWARD 1

Compute with multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Add.

$$27.6 + 4.38$$

Sum: _____

Subtract.

$$50.2 - 7.65$$

Difference: _____

Multiply.

$$1.2 \times 0.4$$

A. 4.8

B. 0.48

C. 0.048

D. 1.6

Divide.

$$7.5 \div 2.5$$

Quotient: _____

True or false?

$$0.3 \times 0.3 = 0.9$$

A. True

B. False

SET 8 • 5.NBT.A.1

FORWARD 2

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A diver at -4.5 m descends 2.25 m.

What is the diver's new position?

Position: _____

The temperature was 3.5°F . It fell 6.2°F .
What is the new temperature?

Temperature: _____

A stock's value changed by $-\$0.75$ each day for 4 days.
What was the total change?

- A.** $-\$0.30$
- B.** $-\$4.75$
- C.** $\$3.00$
- D.** $-\$3.00$

A bank balance of $-\$12.40$ increases by $\$20.00$.

What is the new balance?

Balance: _____

True or false?

$$-1.5 \times 4 = 6$$

A. True

B. False

A digit is worth $\frac{1}{10}$ of the same digit one place to its left

MAIN	Grade 5 main question	5.NBT.A.1
BACKWARD 1	One place to the right is worth $\frac{1}{10}$ as much	4.NBT.A.1
BACKWARD 2	The unit fraction $\frac{1}{10}$	3.NF.A.1
BACKWARD 3	Decimals for tenths and hundredths	4.NF.C.6
FORWARD 1	Divide multi-digit decimals	6.NS.B.3
FORWARD 2	Write fractions as decimals using long division	7.NS.A.2.d

SET 9 • 5.NBT.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.1

True or false?

In 44.4, the 4 in the tenths place is worth $\frac{1}{10}$ of the 4 in the ones place.

- A. True
- B. False

In 2.22, the 2 in the hundredths place is worth what part of the 2 in the tenths place?

A. 10 times as much

B. $\frac{1}{10}$

C. The same amount

D. $\frac{1}{100}$

What is $\frac{1}{10}$ of 50?

Answer: _____

What is $\frac{1}{10}$ of 0.8?

Answer: _____

In which number is the 9 worth $\frac{1}{10}$ of the 9 in 19.4?

- A.** 5.92
- B.** 3.009
- C.** 9.01
- D.** 90.1

SET 9 • 5.NBT.A.1

BACKWARD 1

One place to the right is worth $\frac{1}{10}$ as much • Grade 4 • Number & Operations
in Base Ten • 4.NBT.A.1

True or false?

In 330, the 3 in the tens place is worth $\frac{1}{10}$ of the 3 in the hundreds place.

- A.** True
- B.** False

Divide.

$$400 \div 10$$

Quotient: _____

Which number is $\frac{1}{10}$ of 3,000?

- A.** 3
- B.** 300
- C.** 30,000
- D.** 30

How many hundreds are in 1 thousand?

Hundreds: _____

True or false?

$$60 \div 10 = 600$$

A. True

B. False

SET 9 • 5.NBT.A.1

BACKWARD 2

The unit fraction $\frac{1}{10}$ • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

$\frac{1}{10}$ means 1 of 10 equal parts.

A. True

B. False

A bar is cut into 10 equal parts.

What fraction of the bar is one part?

Fraction: _____

Which fraction is greater?

A. $\frac{1}{10}$

B. $\frac{1}{2}$

True or false?

A whole cut into 10 equal parts has parts that are each $\frac{1}{10}$ of the whole.

- A.** True
- B.** False

What fraction of the bar is shaded?



Fraction: _____

SET 9 • 5.NBT.A.1

BACKWARD 3

Decimals for tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{9}{100}$ as a decimal.

Decimal: _____

Write 0.8 as a fraction.

Fraction: _____

True or false?

0.4 and 0.04 have the same value.

A. True

B. False

Which decimal is equal to $\frac{6}{10}$?

A. 6.0

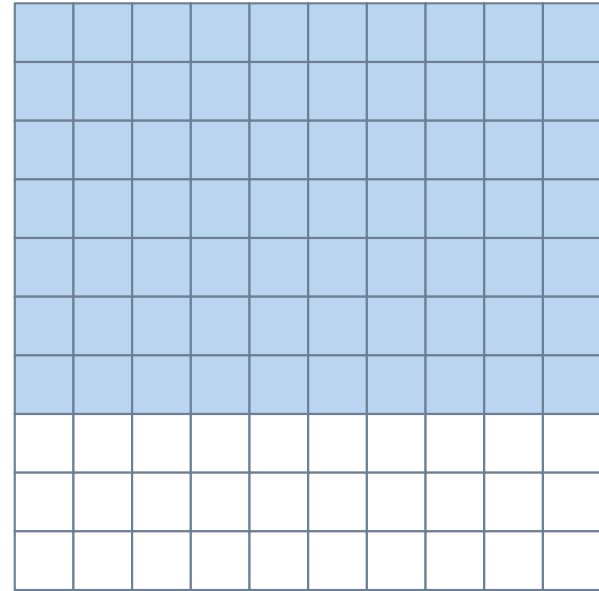
B. 0.06

C. 6.10

D. 0.6

What decimal does the shaded part of the grid show?

Decimal: _____



SET 9 • 5.NBT.A.1

FORWARD 1

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$4.8 \div 0.4$$

Quotient: _____

Divide.

$$9.36 \div 3$$

Quotient: _____

Divide.

$$0.56 \div 0.7$$

A. 80

B. 0.08

C. 8

D. 0.8

Divide.

$$15 \div 0.25$$

Quotient: _____

True or false?

$$6.3 \div 0.9 = 0.7$$

A. True

B. False

SET 9 • 5.NBT.A.1

FORWARD 2

Write fractions as decimals using long division • Grade 7 • The Number System
• 7.NS.A.2.d

Use long division to write $\frac{7}{8}$ as a decimal.

Decimal: _____

Use long division to write $\frac{2}{9}$ as a decimal.

Decimal: _____

Which fraction has a terminating decimal?

A. $\frac{5}{6}$

B. $\frac{4}{7}$

C. $\frac{1}{3}$

D. $\frac{3}{20}$

Use long division to write $\frac{9}{20}$ as a decimal.

Decimal: _____

True or false?

The decimal form of $\frac{1}{11}$ repeats.

A. True

B. False

Multiply and divide whole numbers by powers of 10; write powers of 10 with exponents

MAIN	Grade 5 main question	5.NBT.A.2
BACKWARD 1	Multiplying and dividing by 10	4.NBT.A.1
BACKWARD 2	Multiplying by multiples of 10	3.NBT.A.3
FORWARD 1	Write and evaluate whole-number exponents	6.EE.A.1
FORWARD 2	Multiply and divide negative numbers by powers of 10 — nearest...	7.NS.A.2.c

SET 10 • 5.NBT.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Multiply.

$$37 \times 1,000$$

Product: _____

Divide.

$$45,000 \div 100$$

Quotient: _____

Write $10 \times 10 \times 10 \times 10$ as a power of 10.

Power of 10: _____

$$6 \times 10^3 = ?$$

A. 60,000

B. 6,000

C. 18

D. 600

True or false?

$$820 \div 10^2 = 82$$

A. True

B. False

SET 10 • 5.NBT.A.2

BACKWARD 1

Multiplying and dividing by 10 • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.1

Multiply.

$$25 \times 10$$

Product: _____

True or false?

300 is 10 times as much as 30.

A. True

B. False

Which number is 10 times as much as 450?

- A.** 4,500
- B.** 45,000
- C.** 460
- D.** 45

How many tens are in 800?

Tens: _____

Divide.

$$7,000 \div 10$$

Quotient: _____

SET 10 • 5.NBT.A.2

BACKWARD 2

Multiplying by multiples of 10 • Grade 3 • Number & Operations in Base Ten •
3.NBT.A.3

Multiply.

$$4 \times 60$$

Product: _____

Multiply.

$$9 \times 80$$

Product: _____

True or false?

$$7 \times 30 = 2,100$$

A. True

B. False

$$5 \times 90 = ?$$

A. 4,500

B. 450

C. 45

D. 95

Multiply.

$$8 \times 70$$

Product: _____

SET 10 • 5.NBT.A.2

FORWARD 1

Write and evaluate whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$3^4$$

Value: _____

Write $5 \times 5 \times 5$ using an exponent.

Exponent form: _____

$$2^5 = ?$$

A. 25

B. 10

C. 64

D. 32

Evaluate.

$$10^3 + 2^2$$

Value: _____

True or false?

$$4^3 = 12$$

A. True

B. False

FORWARD 2

Multiply and divide negative numbers by powers of 10 • Grade 7 • The Number System • 7.NS.A.2.c

▮ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Multiply.

$$-3.5 \times 100$$

Product: _____

Divide.

$$-420 \div 10$$

Quotient: _____

$$-0.06 \times 1,000 = ?$$

- A.** -60
- B.** -0.006
- C.** -6
- D.** 60

Divide.

$$7.2 \div (-10)$$

Quotient: _____

True or false?

$$(-10) \times (-10) \times (-10) = -1,000$$

A. True

B. False

Place the decimal point when multiplying or dividing a decimal by a power of 10

MAIN	Grade 5 main question	5.NBT.A.2
BACKWARD 1	Decimal notation for fractions	4.NF.C.6
BACKWARD 2	Place value when multiplying by 10 or 100	4.NBT.A.1
FORWARD 1	Compute fluently with multi-digit decimals	6.NS.B.3
FORWARD 2	Powers of 10 with negative rational numbers — nearest related (n...	7.NS.A.2.c

SET 11 • 5.NBT.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Multiply.

$$3.82 \times 100$$

Product: _____

Divide.

$$56.4 \div 10$$

Quotient: _____

Multiply.

$$0.075 \times 10^2$$

Product: _____

Divide.

$$2.9 \div 10^3$$

A. 0.029

B. 0.29

C. 0.0029

D. 2,900

True or false?

When you multiply a number by 1,000, each digit moves 3 places to the left.

- A. True
- B. False

SET 11 • 5.NBT.A.2

BACKWARD 1

Decimal notation for fractions • Grade 4 • Number & Operations–Fractions •
4.NF.C.6

Write $\frac{35}{100}$ as a decimal.

Decimal: _____

True or false?

$$0.7 = \frac{7}{100}$$

A. True

B. False

Which decimal is $2\frac{4}{10}$?

A. 24

B. 2.4

C. 0.24

D. 2.04

Write 0.09 as a fraction.

Fraction: _____

True or false?

$$1.5 = 1\frac{5}{100}$$

A. True

B. False

SET 11 • 5.NBT.A.2

BACKWARD 2

Place value when multiplying by 10 or 100 • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1

True or false?

When a whole number is multiplied by 10, each digit moves one place to the left.

A. True

B. False

Multiply.

$$64 \times 100$$

Product: _____

$$3,500 \div 10 = ?$$

- A.** 350
- B.** 3,510
- C.** 35
- D.** 35,000

Divide.

$$9,000 \div 100$$

Quotient: _____

True or false?

$$500 \div 10 = 5$$

A. True

B. False

SET 11 • 5.NBT.A.2

FORWARD 1

Compute fluently with multi-digit decimals • Grade 6 • The Number System •
6.NS.B.3

Multiply.

$$2.4 \times 0.35$$

Product: _____

Divide.

$$1.44 \div 0.12$$

Quotient: _____

Multiply.

$$12.5 \times 0.8$$

A. 100

B. 10

C. 0.1

D. 1

Add.

$$0.875 + 12.4$$

Sum: _____

True or false?

$$4.5 \div 0.05 = 9$$

A. True

B. False

FORWARD 2

Powers of 10 with negative rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Find the value.

$$-0.45 \times 100$$

Value: _____

Find the value.

$$-63.5 \div 1,000$$

Value: _____

Find the value.

$$(-2.7) \times (-10)$$

A. -0.27

B. -27

C. 0.27

D. 27

Find the value.

$$0.8 \times (-100)$$

Value: _____

True or false?

$$-5.2 \div 100 = -0.052$$

A. True

B. False

Read and write decimals to thousandths using numerals and words

MAIN	Grade 5 main question	5.NBT.A.3.a
BACKWARD 1	Decimals for tenths and hundredths	4.NF.C.6
BACKWARD 2	Reading and writing whole numbers	4.NBT.A.2
BACKWARD 3	Naming tenths and hundredths	4.NF.C.5
FORWARD 1	Find and position rational numbers on a number line	6.NS.C.6.c
FORWARD 2	Write fractions as decimals	7.NS.A.2.d

SET 12 • 5.NBT.A.3.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

Write the number in standard form.

four and twenty-six hundredths

Number: _____

Write 3.508 in words.

Words: _____

Which number is nine and forty-one thousandths?

- A.** 9.410
- B.** 9.41
- C.** 9.0041
- D.** 9.041

Write the number in standard form.

seventeen and three tenths

Number: _____

True or false?

0.072 is read "seventy-two hundredths."

A. True

B. False

SET 12 • 5.NBT.A.3.a

BACKWARD 1

Decimals for tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{6}{100}$ as a decimal.

Decimal: _____

True or false?

$$0.5 = \frac{5}{10}$$

A. True

B. False

Which decimal is equal to $\frac{12}{100}$?

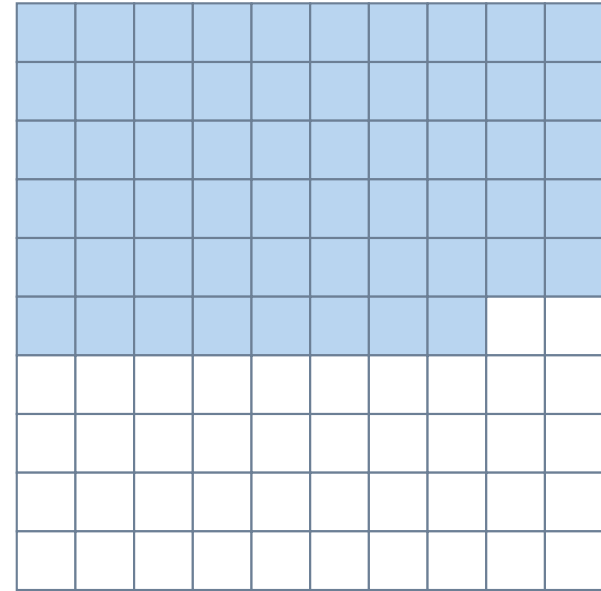
- A.** 1.2
- B.** 12.0
- C.** 0.12
- D.** 0.012

Write $3\frac{7}{10}$ as a decimal.

Decimal: _____

What decimal does the shaded part of the grid show?

Decimal: _____



SET 12 • 5.NBT.A.3.a

BACKWARD 2

Reading and writing whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2

Write the number in standard form.

two thousand, four hundred six

Number: _____

True or false?

5,030 is read "five thousand three hundred."

A. True

B. False

Which number is eighty thousand, nine hundred?

A. 80,900

B. 8,900

C. 89,000

D. 80,090

Write 607 in words.

Words: _____

Write the number in standard form.

twelve thousand, fifty

Number: _____

SET 12 • 5.NBT.A.3.a

BACKWARD 3

Naming tenths and hundredths • Grade 4 • Number & Operations–Fractions •
4.NF.C.5

True or false?

Three tenths is equal to thirty hundredths.

A. True

B. False

How many hundredths are in 4 tenths?

Hundredths: _____

Which fraction is "five hundredths"?

A. $\frac{100}{5}$

B. $\frac{5}{10}$

C. $\frac{50}{100}$

D. $\frac{5}{100}$

Write "nine tenths" as a fraction.

Fraction: _____

True or false?

$$\frac{60}{100} = \frac{6}{10}$$

A. True

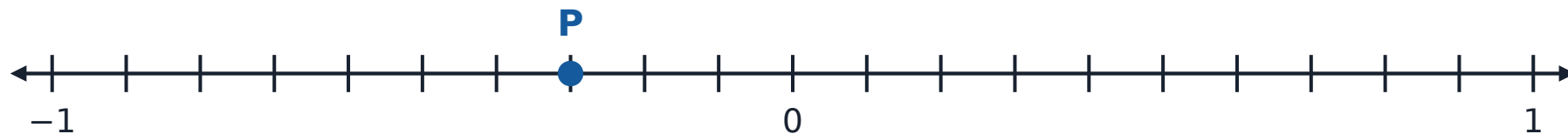
B. False

SET 12 • 5.NBT.A.3.a

FORWARD 1

Find and position rational numbers on a number line • Grade 6 • The Number System • 6.NS.C.6.c

What number does point P represent?

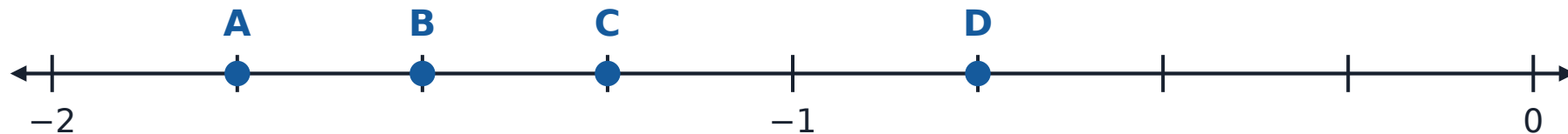


P = _____

Plot and label point Q at 0.45 on the number line.



Which point is at -1.25 ?



A. Point A

B. Point B

C. Point C

D. Point D

What number does point R represent?



R = _____

True or false?

On a number line, -0.6 is to the left of -0.5 .

A. True

B. False

SET 12 • 5.NBT.A.3.a

FORWARD 2

Write fractions as decimals • Grade 7 • The Number System • 7.NS.A.2.d

Use long division to write $\frac{3}{8}$ as a decimal.

Decimal: _____

Use long division to write $\frac{5}{12}$ as a decimal.

Decimal: _____

$$\frac{11}{20} = ?$$

A. 0.55

B. 5.5

C. 0.2

D. 0.11

Use long division to write $\frac{4}{9}$ as a decimal.

Decimal: _____

True or false?

$$\frac{7}{25} = 0.32$$

A. True

B. False

Write decimals to thousandths in expanded form

MAIN	Grade 5 main question	5.NBT.A.3.a
BACKWARD 1	Expanded form of whole numbers	4.NBT.A.2
BACKWARD 2	Decimals and fractions with denominators 10 and 100	4.NF.C.6
FORWARD 1	Write expanded form with exponents	6.EE.A.1
FORWARD 2	Decimal expansions of fractions — nearest related (no direct Gra...	7.NS.A.2.d

SET 13 • 5.NBT.A.3.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

Write 352.68 in expanded form.

Expanded form: _____

Write the number in standard form.

$$4 \times 10 + 7 \times 1 + 3 \times \frac{1}{10} + 9 \times \frac{1}{1000}$$

Number: _____

Which is the expanded form of 6.045?

A. $6 + 45$

B. $6 \times 10 + 4 \times \frac{1}{100} + 5 \times \frac{1}{1000}$

C. $6 \times 1 + 4 \times \frac{1}{100} + 5 \times \frac{1}{1000}$

D. $6 \times 1 + 4 \times \frac{1}{10} + 5 \times \frac{1}{100}$

Write the number in standard form.

$$8 \times 100 + 2 \times \frac{1}{10} + 6 \times \frac{1}{100}$$

Number: _____

True or false?

$$0.908 = 9 \times \frac{1}{10} + 8 \times \frac{1}{1000}$$

A. True

B. False

SET 13 • 5.NBT.A.3.a

BACKWARD 1

Expanded form of whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2

Write 4,371 in expanded form.

Expanded form: _____

True or false?

$$6,052 = 6,000 + 500 + 2$$

A. True

B. False

Which number is $7,000 + 400 + 9$?

- A.** 7,409
- B.** 7,490
- C.** 74,009
- D.** 7,049

Write $3 \times 1,000 + 8 \times 100 + 5 \times 1$ in standard form.

Number: _____

True or false?

$$900 + 20 + 4 = 9,024$$

A. True

B. False

SET 13 • 5.NBT.A.3.a

BACKWARD 2

Decimals and fractions with denominators 10 and 100 • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write 0.6 as a fraction.

Fraction: _____

True or false?

$$\frac{4}{100} = 0.4$$

A. True

B. False

Which fraction is equal to 0.25?

A. $\frac{1}{25}$

B. $\frac{25}{100}$

C. $\frac{25}{10}$

D. $\frac{2}{5}$

Write $2 + \frac{3}{10} + \frac{7}{100}$ as a decimal.

Decimal: _____

True or false?

$$\frac{9}{10} = 0.09$$

A. True

B. False

SET 13 • 5.NBT.A.3.a

FORWARD 1

Write expanded form with exponents • Grade 6 • Expressions & Equations •
6.EE.A.1

Write $5,000 + 300 + 20 + 7$ using powers of 10.

Expression: _____

Evaluate.

$$4 \times 10^3 + 6 \times 10^1$$

Value: _____

Which number is equal to 10^4 ?

- A.** 40
- B.** 100,000
- C.** 1,000
- D.** 10,000

Write $600 + 90 + 1$ using powers of 10.

Expression: _____

True or false?

$$2 \times 10^2 + 5 = 250$$

A. True

B. False

FORWARD 2

Decimal expansions of fractions • Grade 7 • The Number System • 7.NS.A.2.d

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Write $\frac{3}{4}$ as a decimal.

Decimal: _____

Use long division to write $\frac{1}{6}$ as a decimal.

Decimal: _____

Which fraction is equal to 0.666... (the 6 repeats)?

A. $\frac{3}{5}$

B. $\frac{6}{10}$

C. $\frac{2}{3}$

D. $\frac{1}{6}$

Use long division to write $\frac{5}{8}$ as a decimal.

Decimal: _____

True or false?

$$\frac{2}{5} = 0.4$$

A. True

B. False

Compare decimals to thousandths

MAIN	Grade 5 main question	5.NBT.A.3.b
BACKWARD 1	Comparing decimals to hundredths	4.NF.C.7
BACKWARD 2	The symbols $>$, $<$, and $=$	1.NBT.B.3
FORWARD 1	Order rational numbers	6.NS.C.7.b
FORWARD 2	Compare fractions and decimals using decimal expansions	7.NS.A.2.d

SET 14 • 5.NBT.A.3.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.b

Write $>$, $<$, or $=$ to compare.

0.47 0.405

Symbol: _____

Write $>$, $<$, or $=$ to compare.

3.06 ____ 3.060

Symbol: _____

Which number is greatest?

- A.** 5.2
- B.** 5.19
- C.** 5.093
- D.** 5.199

Order from least to greatest.

2.31, 2.301, 2.13, 2.3

Order: _____

True or false?

$$0.8 < 0.79$$

A. True

B. False

SET 14 • 5.NBT.A.3.b

BACKWARD 1

Comparing decimals to hundredths • Grade 4 • Number & Operations–Fractions
• 4.NF.C.7

Write $>$, $<$, or $=$ to compare.

0.6 0.58

Symbol: _____

True or false?

$$0.3 = 0.03$$

A. True

B. False

Which number is less?

A. 0.09

B. 0.1

Write $>$, $<$, or $=$ to compare.

0.75 0.8

Symbol: _____

True or false?

$$0.42 > 0.5$$

A. True

B. False

SET 14 • 5.NBT.A.3.b

BACKWARD 2

The symbols $>$, $<$, and $=$ • Grade 1 • Number & Operations in Base Ten •
1.NBT.B.3

Which symbol means "is less than"?

A. $<$

B. $>$

C. $=$

True or false?

$$9 > 6$$

A. True

B. False

Write $>$, $<$, or $=$ to compare.

52 25

Symbol: _____

True or false?

The symbol $>$ means "is less than."

- A. True
- B. False

Which statement is true?

A. $14 < 41$

B. $14 = 41$

C. $14 > 41$

SET 14 • 5.NBT.A.3.b

FORWARD 1

Order rational numbers • Grade 6 • The Number System • 6.NS.C.7.b

Write $>$, $<$, or $=$ to compare.

-0.4 ____ -0.45

Symbol: _____

Which number is least?

- A.** -2
- B.** -2.5
- C.** 0.5
- D.** -2.05

Order from least to greatest.

-1.2 , 0.3 , -1.25 , 0

Order: _____

True or false?

$$-3.7 > -3.6$$

A. True

B. False

A temperature of -2.5°C is colder than a temperature of -1.8°C .

Write an inequality to show this.

Inequality: _____

SET 14 • 5.NBT.A.3.b

FORWARD 2

Compare fractions and decimals using decimal expansions • Grade 7 • The Number System • 7.NS.A.2.d

Write $\frac{5}{8}$ as a decimal.

Is $\frac{5}{8}$ greater than or less than 0.6?

Decimal: _____

Answer: _____

Which number is greatest?

A. $\frac{4}{5}$

B. 0.7

C. $\frac{3}{4}$

D. 0.78

Order from least to greatest.

$$\frac{2}{3}, 0.6, \frac{5}{8}$$

Order: _____

True or false?

$$\frac{1}{3} > 0.33$$

A. True

B. False

Write $>$, $<$, or $=$ to compare.

$$\frac{7}{20} \text{ ____ } 0.4$$

Symbol: _____

Round decimals to any place

MAIN

Grade 5 main question

5.NBT.A.4

BACKWARD 1

Rounding whole numbers

4.NBT.A.3

BACKWARD 2

Decimals on a number line

4.NF.C.6

FORWARD 1

Estimate to check computations with decimals

6.NS.B.3

FORWARD 2

Use estimation to judge whether an answer is reasonable

7.EE.B.3

SET 15 • 5.NBT.A.4

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.4

Round 6.483 to the nearest tenth.

Rounded: _____

Round 12.749 to the nearest hundredth.

Rounded: _____

Round 0.865 to the nearest whole number.

- A.** 0.9
- B.** 1
- C.** 0
- D.** 0.87

Round 39.96 to the nearest tenth.

Rounded: _____

True or false?

4.351 rounded to the nearest hundredth is 4.36.

A. True

B. False

SET 15 • 5.NBT.A.4

BACKWARD 1

Rounding whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.3

Round 4,682 to the nearest hundred.

Rounded: _____

True or false?

350 rounded to the nearest hundred is 300.

A. True

B. False

Round 7,249 to the nearest thousand.

- A.** 7,200
- B.** 7,000
- C.** 7,250
- D.** 8,000

Round 85 to the nearest ten.

Rounded: _____

True or false?

1,349 rounded to the nearest hundred is 1,400.

A. True

B. False

SET 15 • 5.NBT.A.4

BACKWARD 2

Decimals on a number line • Grade 4 • Number & Operations–Fractions •
4.NF.C.6

What decimal is at point A?



A = _____

True or false?

0.7 is closer to 1 than to 0.

A. True

B. False

Is 0.43 closer to 0.4 or to 0.5?

A. 0.4

B. 0.5

What decimal is at point B?



B = _____

True or false?

2.5 is halfway between 2 and 3.

A. True

B. False

SET 15 • 5.NBT.A.4

FORWARD 1

Estimate to check computations with decimals • Grade 6 • The Number System
• 6.NS.B.3

Round each number to the nearest whole number. Then add to estimate.

$$18.7 + 6.25$$

Estimate: _____

Which is the best estimate of 4.9×6.1 ?

A. 36

B. 30

C. 24

D. 11

Estimate $31.8 \div 7.9$ by rounding each number to the nearest whole number.

Estimate: _____

True or false?

29.6 – 10.2 is about 40.

A. True

B. False

Kim says $3.9 \times 2.1 = 81.9$.

Use an estimate to decide whether her answer is reasonable. Write yes or no.

Answer: _____

SET 15 • 5.NBT.A.4

FORWARD 2

Use estimation to judge whether an answer is reasonable • Grade 7 •
Expressions & Equations • 7.EE.B.3

A \$24.75 shirt is 20% off.

Estimate the sale price.

Estimate: _____

A worker earns \$19.85 per hour for 9.75 hours.
Which is the best estimate of her pay?

- A.** \$20
- B.** \$2,000
- C.** \$200
- D.** \$100

True or false?

A 6.2% tax on \$49.50 is about \$3.

A. True

B. False

Nina says $\frac{3}{4} \times 39.8$ is about 40.

Is her estimate reasonable? Give a better estimate.

Reasonable? _____

Better estimate: _____

A board is $8\frac{1}{4}$ feet long. It is cut into pieces that are $1\frac{7}{8}$ feet long.

Estimate how many whole pieces can be cut.

Estimate: _____

Multiply multi-digit whole numbers using the standard algorithm

MAIN	Grade 5 main question	5.NBT.B.5
BACKWARD 1	Multiplying by a one-digit or two-digit number	4.NBT.B.5
BACKWARD 2	Multiplication facts	3.OA.C.7
BACKWARD 3	Adding partial products	4.NBT.B.4
FORWARD 1	Multiply multi-digit decimals	6.NS.B.3
FORWARD 2	Multiply integers	7.NS.A.2.c

SET 16 • 5.NBT.B.5

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.5

Multiply.

$$246 \times 37$$

Product: _____

Multiply.

$$1,385 \times 24$$

Product: _____

Multiply.

$$508 \times 63$$

- A.** 3,556
- B.** 32,104
- C.** 32,004
- D.** 31,904

Multiply.

$$79 \times 86$$

Product: _____

A theater has 48 rows with 125 seats in each row.
How many seats are in the theater?

Seats: _____

SET 16 • 5.NBT.B.5

BACKWARD 1

Multiplying by a one-digit or two-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

Multiply.

$$346 \times 7$$

Product: _____

Multiply.

$$23 \times 45$$

Product: _____

True or false?

$$1,203 \times 4 = 4,082$$

A. True

B. False

$$62 \times 18 = ?$$

A. 1,126

B. 496

C. 1,116

D. 1,006

Multiply.

$$58 \times 30$$

Product: _____

SET 16 • 5.NBT.B.5

BACKWARD 2

Multiplication facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

$$7 \times 6$$

Product: _____

Multiply.

$$8 \times 9$$

Product: _____

True or false?

$$6 \times 9 = 56$$

A. True

B. False

$$4 \times 8 = ?$$

A. 24

B. 36

C. 32

D. 28

Multiply.

$$9 \times 9$$

Product: _____

SET 16 • 5.NBT.B.5

BACKWARD 3

Adding partial products • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Add.

$$7,380 + 1,722$$

Sum: _____

Add.

$$27,700 + 5,540$$

Sum: _____

True or false?

$$30,480 + 1,524 = 31,904$$

A. True

B. False

$$6,400 + 394 = ?$$

A. 6,784

B. 6,794

C. 7,794

D. 6,694

Add.

$$4,800 + 1,200$$

Sum: _____

SET 16 • 5.NBT.B.5

FORWARD 1

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$2.46 \times 3.7$$

Product: _____

Multiply.

$$13.85 \times 2.4$$

Product: _____

Multiply.

$$5.08 \times 6.3$$

A. 31.904

B. 320.04

C. 3.2004

D. 32.004

Multiply.

$$0.79 \times 8.6$$

Product: _____

True or false?

$$1.25 \times 4.8 = 60$$

A. True

B. False

SET 16 • 5.NBT.B.5

FORWARD 2

Multiply integers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$(-24) \times 15$$

Product: _____

Find the product.

$$(-12) \times (-35)$$

Product: _____

Find the product.

$$18 \times (-21)$$

- A.** 378
- B.** -378
- C.** -39
- D.** -388

Find the product.

$$(-3) \times (-4) \times (-25)$$

Product: _____

True or false?

$$(-6)(-7) = -42$$

A. True

B. False

Divide with up to four-digit dividends and two-digit divisors

MAIN	Grade 5 main question	5.NBT.B.6
BACKWARD 1	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Multiplying to check a quotient	4.NBT.B.5
BACKWARD 3	Division facts	3.OA.C.7
FORWARD 1	Divide multi-digit numbers using the standard algorithm	6.NS.B.2
FORWARD 2	Divide integers	7.NS.A.2.b

SET 17 • 5.NBT.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

Divide.

$$2,016 \div 24$$

Quotient: _____

Divide.

$$7,830 \div 45$$

Quotient: _____

Divide.

$$952 \div 17$$

A. 54

B. 46

C. 56

D. 57

Divide.

$$3,724 \div 38$$

Quotient: _____

Divide. Write the quotient and the remainder.

$$1,000 \div 32$$

Quotient and remainder: _____

SET 17 • 5.NBT.B.6

BACKWARD 1

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$852 \div 6$$

Quotient: _____

Divide.

$$1,935 \div 5$$

Quotient: _____

True or false?

$$728 \div 8 = 901$$

A. True

B. False

$$504 \div 7 = ?$$

A. 82

B. 71

C. 62

D. 72

Divide. Write the quotient and the remainder.

$$95 \div 4$$

Quotient and remainder: _____

SET 17 • 5.NBT.B.6

BACKWARD 2

Multiplying to check a quotient • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

True or false?

To check $2,016 \div 24 = 84$, you can multiply 24×84 .

- A. True
- B. False

Multiply.

$$45 \times 174$$

Product: _____

Which product checks $952 \div 17 = 56$?

- A.** $17 + 56$
- B.** 17×56
- C.** $56 - 17$
- D.** 952×17

Multiply.

$$38 \times 98$$

Product: _____

True or false?

$$32 \times 31 + 8 = 1,000$$

A. True

B. False

SET 17 • 5.NBT.B.6

BACKWARD 3

Division facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Divide.

$$72 \div 8$$

Quotient: _____

Divide.

$$54 \div 9$$

Quotient: _____

True or false?

$$63 \div 7 = 8$$

A. True

B. False

$$40 \div 5 = ?$$

A. 6

B. 7

C. 8

D. 9

Divide.

$$35 \div 7$$

Quotient: _____

SET 17 • 5.NBT.B.6

FORWARD 1

Divide multi-digit numbers using the standard algorithm • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide.

$$6,432 \div 48$$

Quotient: _____

Use the standard algorithm to divide.

$$25,875 \div 69$$

Quotient: _____

Use the standard algorithm to divide.

$$14,040 \div 52$$

- A.** 207
- B.** 2,700
- C.** 270
- D.** 27

Use the standard algorithm to divide.

$$9,594 \div 41$$

Quotient: _____

True or false?

$$11,700 \div 45 = 2,600$$

A. True

B. False

SET 17 • 5.NBT.B.6

FORWARD 2

Divide integers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-2,016 \div 24$$

Quotient: _____

Find the quotient.

$$(-952) \div (-17)$$

Quotient: _____

Find the quotient.

$$780 \div (-15)$$

- A.** -52
- B.** -53
- C.** -765
- D.** 52

Find the quotient.

$$-(144 \div 12)$$

Quotient: _____

True or false?

$$(-90) \div 18 = 90 \div (-18)$$

A. True

B. False

Explain division using area models and equations

MAIN	Grade 5 main question	5.NBT.B.6
BACKWARD 1	Area models for dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Area as length times width	3.MD.C.7.b
BACKWARD 3	Splitting a rectangle to find its area	3.MD.C.7.c
FORWARD 1	Use the distributive property with the greatest common factor	6.NS.B.4
FORWARD 2	Factor and expand linear expressions	7.EE.A.1

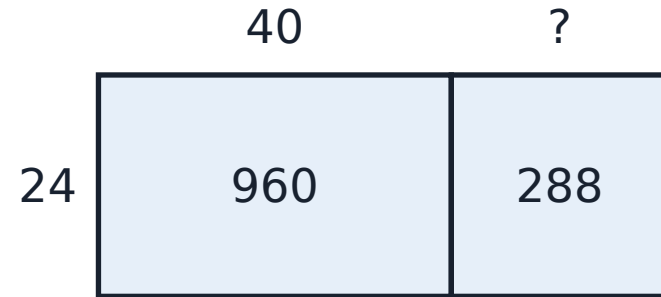
SET 18 • 5.NBT.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

The area model shows
 $1,248 \div 24$.

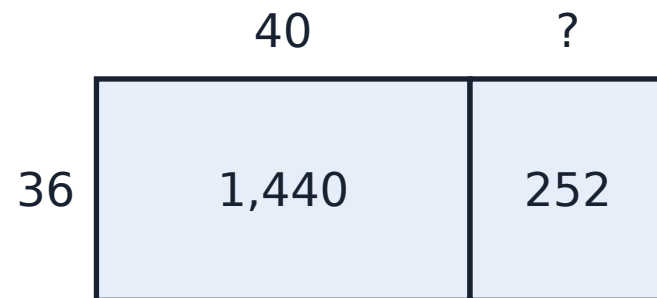
What is the quotient?



Quotient: _____

The area model shows
 $1,692 \div 36$.

What is the quotient?



Quotient: _____

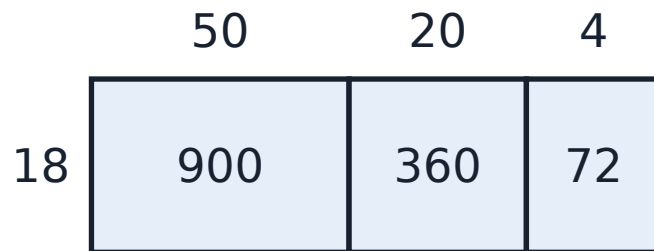
Which equation matches the area model?

A. $1,332 \div 74 = 20$

B. $1,332 \div 18 = 74$

C. $18 \times 50 = 1,332$

D. $900 \div 18 = 74$



Use an area model or equations to divide.

$$2,184 \div 28$$

Quotient: _____

True or false?

$$1,248 \div 24 = (960 \div 24) + (288 \div 24)$$

A. True

B. False

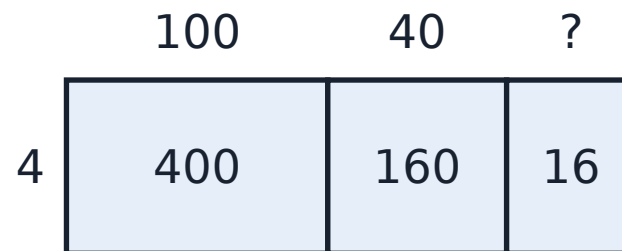
SET 18 • 5.NBT.B.6

BACKWARD 1

Area models for dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

The area model shows
 $576 \div 4$.

What is the quotient?



Quotient: _____

True or false?

$$468 \div 3 = (300 \div 3) + (150 \div 3) + (18 \div 3)$$

A. True

B. False

Which shows $245 \div 5$ split into easier parts?

- A.** $(200 \div 5) + (45 \div 5)$
- B.** $(245 \div 2) + (245 \div 3)$
- C.** $200 \div (5 + 45)$
- D.** $(200 \div 5) \times (45 \div 5)$

Divide.

$$735 \div 7$$

Quotient: _____

A rectangle has an area of 312 square units and a width of 6 units.

What is its length?

Length: _____

SET 18 • 5.NBT.B.6

BACKWARD 2

Area as length times width • Grade 3 • Measurement & Data • 3.MD.C.7.b

A rectangle is 9 units long and 7 units wide.
What is its area?

Area: _____

True or false?

A rectangle with an area of 40 square units and a side of 5 units has another side of 8 units.

A. True

B. False

Which equation gives the area of a 6-by-8 rectangle?

A. $8 - 6 = 2$

B. $6 + 8 = 14$

C. $6 \times 8 = 48$

D. $2 \times (6 + 8) = 28$

A rectangle has an area of 36 square units. One side is 4 units long.

How long is the other side?

Length: _____

Find the area of the rectangle.

Area: _____



SET 18 • 5.NBT.B.6

BACKWARD 3

Splitting a rectangle to find its area • Grade 3 • Measurement & Data •
3.MD.C.7.c

True or false?

The area of a rectangle that is 4 by $(10 + 3)$ equals $4 \times 10 + 4 \times 3$.

A. True

B. False

A 5-by-12 rectangle is split into a 5-by-10 rectangle and a 5-by-2 rectangle.

What is the total area?

Area: _____

A 7-by-13 rectangle is split into a 7-by-10 part and a 7-by-3 part.

Which expression gives its area?

A. $7 \times 10 + 7 \times 3$

B. $7 \times 10 + 3$

C. $7 + 10 + 3$

D. $7 \times 10 \times 3$

Fill in the blank.

$$6 \times 14 = 6 \times 10 + 6 \times \underline{\quad}$$

Blank: _____

True or false?

$$8 \times 12 = 8 \times 10 + 2$$

A. True

B. False

SET 18 • 5.NBT.B.6

FORWARD 1

Use the distributive property with the greatest common factor • Grade 6 • The Number System • 6.NS.B.4

Use the greatest common factor to write $36 + 48$ as a product.

Expression: _____

Find the greatest common factor of 32 and 56.

GCF: _____

Which expression is equal to $45 + 60$ and uses the greatest common factor?

- A.** $15(3 + 6)$
- B.** $15(3 + 4)$
- C.** $3(15 + 20)$
- D.** $5(9 + 12)$

Complete the equation.

$$24 + 40 = 8(\underline{\quad} + \underline{\quad})$$

First blank: _____

Second blank: _____

True or false?

$$18 + 27 = 9(2 + 3)$$

A. True

B. False

SET 18 • 5.NBT.B.6

FORWARD 2

Factor and expand linear expressions • Grade 7 • Expressions & Equations •
7.EE.A.1

Factor using the greatest common factor.

$$12x + 30$$

Expression: _____

Expand.

$$7(4y - 3)$$

Expression: _____

Which expression is equivalent to $18a - 24$?

- A.** $3(6a - 4)$
- B.** $6(3a - 24)$
- C.** $18(a - 6)$
- D.** $6(3a - 4)$

Factor out -4 .

$$-8n - 20$$

Expression: _____

True or false?

$$4(2.5x + 3) = 10x + 3$$

A. True

B. False

Add and subtract decimals to hundredths

MAIN

Grade 5 main question

5.NBT.B.7

BACKWARD 1

Adding and subtracting multi-digit whole numbers

4.NBT.B.4

BACKWARD 2

Adding tenths and hundredths

4.NF.C.5

BACKWARD 3

Writing decimals with the same number of places

4.NF.C.6

FORWARD 1

Add and subtract multi-digit decimals

6.NS.B.3

FORWARD 2

Add and subtract rational numbers

7.NS.A.1.d

SET 19 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Add.

$$23.45 + 8.7$$

Sum: _____

Subtract.

$$15.3 - 6.48$$

Difference: _____

Add.

$$4.06 + 12.9$$

- A.** 16.15
- B.** 16.069
- C.** 5.35
- D.** 16.96

Subtract.

$$20 - 7.35$$

Difference: _____

Maya had \$18.50. She spent \$6.75.

How much money does she have left?

Money left: _____

SET 19 • 5.NBT.B.7

BACKWARD 1

Adding and subtracting multi-digit whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

Add.

$$2,345 + 870$$

Sum: _____

Subtract.

$$1,530 - 648$$

Difference: _____

True or false?

$$406 + 1,290 = 1,596$$

A. True

B. False

$$2,000 - 735 = ?$$

A. 2,735

B. 1,265

C. 1,275

D. 1,365

Subtract.

$$1,850 - 675$$

Difference: _____

SET 19 • 5.NBT.B.7

BACKWARD 2

Adding tenths and hundredths • Grade 4 • Number & Operations–Fractions •
4.NF.C.5

Add.

$$\frac{3}{10} + \frac{45}{100}$$

Sum: _____

True or false?

$$\frac{6}{10} + \frac{2}{100} = \frac{8}{100}$$

A. True

B. False

$$\frac{7}{10} + \frac{15}{100} = ?$$

A. $\frac{85}{100}$

B. $\frac{22}{100}$

C. $\frac{22}{110}$

D. $\frac{85}{10}$

Write $\frac{4}{10}$ as hundredths.

Fraction: _____

True or false?

$$\frac{5}{10} + \frac{5}{100} = \frac{10}{100}$$

A. True

B. False

SET 19 • 5.NBT.B.7

BACKWARD 3

Writing decimals with the same number of places • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

True or false?

$$8.7 = 8.70$$

A. True

B. False

Write 15.3 with two decimal places.

Decimal: _____

Which decimal is equal to 20?

- A.** 20.00
- B.** 0.20
- C.** 2.00
- D.** 200.0

True or false?

$$6.4 = 6.04$$

A. True

B. False

Write 12.9 with a digit in the hundredths place, without changing its value.

Decimal: _____

SET 19 • 5.NBT.B.7

FORWARD 1

Add and subtract multi-digit decimals • Grade 6 • The Number System •
6.NS.B.3

Add.

$$128.47 + 9.605$$

Sum: _____

Subtract.

$$54.2 - 18.376$$

Difference: _____

Add.

$$0.952 + 3.48 + 11.6$$

A. 16.132

B. 4.53

C. 15.032

D. 16.032

Subtract.

$$100 - 0.485$$

Difference: _____

True or false?

$$7.3 - 2.145 = 5.245$$

A. True

B. False

SET 19 • 5.NBT.B.7

FORWARD 2

Add and subtract rational numbers • Grade 7 • The Number System •
7.NS.A.1.d

Find the value.

$$-12.5 + 4.75$$

Value: _____

Find the value.

$$3.2 - 8.45$$

Value: _____

Find the value.

$$-6.3 + (-2.85)$$

- A.** 9.15
- B.** -9.15
- C.** -3.45
- D.** 3.45

Find the value.

$$-4.6 - (-10.25)$$

Value: _____

True or false?

$$-1.5 + 1.5 = 0$$

A. True

B. False

Multiply decimals to hundredths

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Multiplying whole numbers	4.NBT.B.5
BACKWARD 2	Decimals as tenths and hundredths	4.NF.C.6
BACKWARD 3	Dividing by 10 and 100 to place the decimal point	5.NBT.A.2
FORWARD 1	Multiply multi-digit decimals	6.NS.B.3
FORWARD 2	Multiply rational numbers	7.NS.A.2.c

SET 20 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Multiply.

$$3.4 \times 6$$

Product: _____

Multiply.

$$0.7 \times 0.5$$

Product: _____

Multiply.

$$2.5 \times 1.2$$

A. 3

B. 30

C. 3.7

D. 0.3

Multiply.

$$0.06 \times 8$$

Product: _____

A pen costs \$1.25.

How much do 7 pens cost?

Cost: _____

SET 20 • 5.NBT.B.7

BACKWARD 1

Multiplying whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Multiply.

$$34 \times 6$$

Product: _____

Multiply.

$$25 \times 12$$

Product: _____

True or false?

$$125 \times 7 = 785$$

A. True

B. False

$$18 \times 4 = ?$$

A. 72

B. 82

C. 48

D. 64

Multiply.

$$106 \times 8$$

Product: _____

SET 20 • 5.NBT.B.7

BACKWARD 2

Decimals as tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write 35 hundredths as a decimal.

Decimal: _____

True or false?

0.4 is 4 tenths.

A. True

B. False

Which decimal is 204 tenths?

- A.** 204
- B.** 20.4
- C.** 2.04
- D.** 0.204

How many hundredths are in 0.48?

Hundredths: _____

True or false?

0.06 is 6 tenths.

A. True

B. False

SET 20 • 5.NBT.B.7

BACKWARD 3

Dividing by 10 and 100 to place the decimal point • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Divide.

$$204 \div 10$$

Quotient: _____

Divide.

$$35 \div 100$$

Quotient: _____

True or false?

$$300 \div 100 = 30$$

A. True

B. False

$$875 \div 100 = ?$$

- A.** 875
- B.** 8.75
- C.** 0.875
- D.** 87.5

Divide.

$$48 \div 100$$

Quotient: _____

SET 20 • 5.NBT.B.7

FORWARD 1

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$3.45 \times 0.6$$

Product: _____

Multiply.

$$12.08 \times 2.5$$

Product: _____

Multiply.

$$0.125 \times 0.4$$

- A.** 0.5
- B.** 0.05
- C.** 5
- D.** 0.005

Multiply.

$$7.2 \times 1.15$$

Product: _____

True or false?

$$0.03 \times 0.03 = 0.09$$

A. True

B. False

SET 20 • 5.NBT.B.7

FORWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$-3.4 \times 6$$

Product: _____

Find the product.

$$(-0.7) \times (-0.5)$$

Product: _____

Find the product.

$$2.5 \times (-1.2)$$

A. -0.3

B. -30

C. 3

D. -3

Find the product.

$$(-0.06) \times 8$$

Product: _____

True or false?

$$(-1.25) \times (-4) = -5$$

A. True

B. False

Divide decimals to hundredths

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Dividing whole numbers	4.NBT.B.6
BACKWARD 2	Division as an unknown-factor problem	3.OA.B.6
BACKWARD 3	Multiplying both numbers by a power of 10 keeps the quotient	5.NBT.A.2
FORWARD 1	Divide multi-digit decimals	6.NS.B.3
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 21 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Divide.

$$8.4 \div 4$$

Quotient: _____

Divide.

$$3.6 \div 0.6$$

Quotient: _____

Divide.

$$0.45 \div 0.05$$

A. 0.09

B. 0.9

C. 90

D. 9

Divide.

$$7.5 \div 0.3$$

Quotient: _____

A rope is 4.8 meters long. It is cut into pieces that are each 0.6 meter long.

How many pieces are there?

Pieces: _____

SET 21 • 5.NBT.B.7

BACKWARD 1

Dividing whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$84 \div 4$$

Quotient: _____

Divide.

$$360 \div 6$$

Quotient: _____

True or false?

$$450 \div 5 = 9$$

A. True

B. False

$$750 \div 3 = ?$$

- A.** 25
- B.** 250
- C.** 2,500
- D.** 260

Divide.

$$48 \div 6$$

Quotient: _____

SET 21 • 5.NBT.B.7

BACKWARD 2

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $36 \div 6$ by thinking $6 \times \square = 36$.

Quotient: _____

True or false?

$7 \times 9 = 63$, so $63 \div 7 = 9$.

A. True

B. False

Which fact helps you find $56 \div 8$?

A. $56 \times 8 = 448$

B. $8 \times 7 = 56$

C. $8 + 48 = 56$

D. $8 - 7 = 1$

What number goes in the box?

$$\square \times 5 = 45$$

Number: _____

True or false?

$24 \div 3 = 6$ because $3 \times 6 = 24$.

A. True

B. False

SET 21 • 5.NBT.B.7

BACKWARD 3

Multiplying both numbers by a power of 10 keeps the quotient • Grade 5 •
Number & Operations in Base Ten • 5.NBT.A.2

True or false?

$3.6 \div 0.6$ has the same quotient as $36 \div 6$.

A. True

B. False

Which has the same quotient as $0.45 \div 0.05$?

A. $4.5 \div 5$

B. $0.45 \div 5$

C. $45 \div 5$

D. $45 \div 0.5$

Multiply.

$$0.6 \times 10$$

Product: _____

Multiply.

$$0.05 \times 100$$

Product: _____

True or false?

$7.5 \div 0.3$ has the same quotient as $75 \div 30$.

A. True

B. False

SET 21 • 5.NBT.B.7

FORWARD 1

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$13.65 \div 0.35$$

Quotient: _____

Divide.

$$9.072 \div 2.4$$

Quotient: _____

Divide.

$$0.081 \div 0.09$$

A. 90

B. 0.09

C. 0.9

D. 9

Divide.

$$52.5 \div 1.25$$

Quotient: _____

True or false?

$$1.8 \div 0.006 = 3$$

A. True

B. False

SET 21 • 5.NBT.B.7

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-8.4 \div 4$$

Quotient: _____

Find the quotient.

$$(-3.6) \div (-0.6)$$

Quotient: _____

Find the quotient.

$$0.45 \div (-0.05)$$

A. -9

B. -0.9

C. 0.9

D. 9

Find the quotient.

$$-7.5 \div 0.3$$

Quotient: _____

True or false?

$$-(4.8 \div 0.6) = (-4.8) \div 0.6$$

A. True

B. False

Solve real-world problems with decimal operations

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Word problems with money and measurement	4.MD.A.2
BACKWARD 2	Hundredths on a grid	4.NF.C.6
FORWARD 1	Real-world problems with multi-digit decimals	6.NS.B.3
FORWARD 2	Real-world problems with rational numbers	7.NS.A.3

SET 22 • 5.NBT.B.7

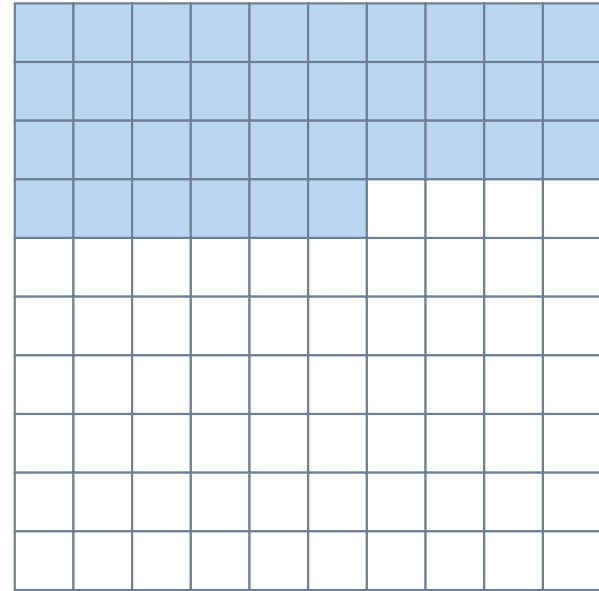
MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

The grid shows 0.36 shaded.

How much more must be shaded to show 0.5?

Amount: _____



Juice costs \$2.35 a bottle. Ben buys 4 bottles and pays with a \$20 bill.

How much change does he get?

Change: _____

A 2.5-pound bag of nuts is shared equally among 5 friends.

How many pounds does each friend get?

- A.** 0.05 pound
- B.** 0.5 pound
- C.** 2 pounds
- D.** 12.5 pounds

A runner ran 3.75 km on Monday and 4.6 km on Tuesday.

How much farther did she run on Tuesday?

Distance: _____

Tiles cost \$0.45 each.

How much do 12 tiles cost?

Cost: _____

SET 22 • 5.NBT.B.7

BACKWARD 1

Word problems with money and measurement • Grade 4 • Measurement & Data • 4.MD.A.2

Pens cost 75 cents each.

How much do 4 pens cost, in dollars?

Cost: _____

A board is 120 cm long. It is cut into 4 equal pieces.
How long is each piece?

Length: _____

True or false?

Two quarters and three dimes are worth 65 cents.

A. True

B. False

Ann has \$5. She spends \$2.25.

How much money is left?

A. \$7.25

B. \$3.25

C. \$2.75

D. \$2.25

A jug holds 2 liters.

How many milliliters is that?

Milliliters: _____

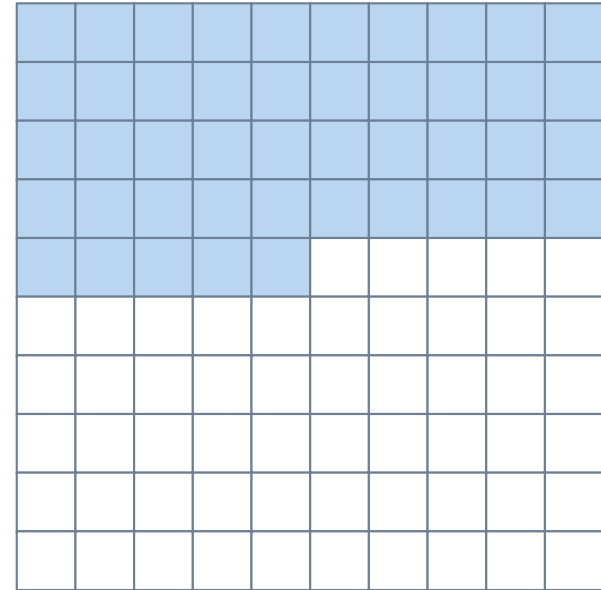
SET 22 • 5.NBT.B.7

BACKWARD 2

Hundredths on a grid • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

What decimal does the shaded part of the grid show?

Decimal: _____

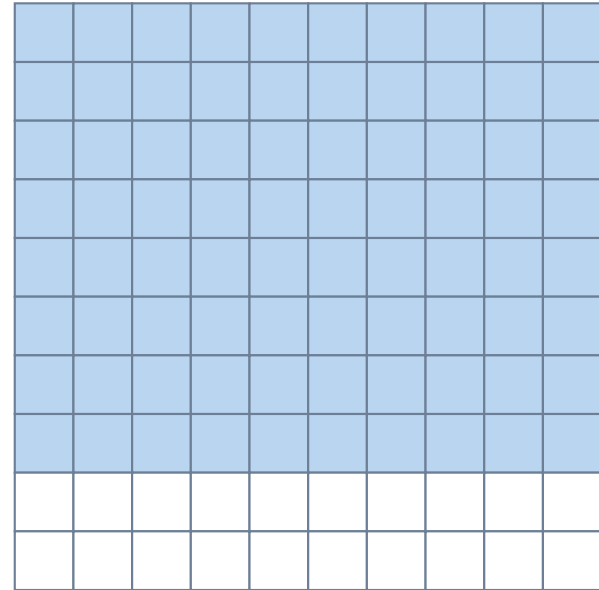


True or false?

The shaded part of the grid shows 0.8.

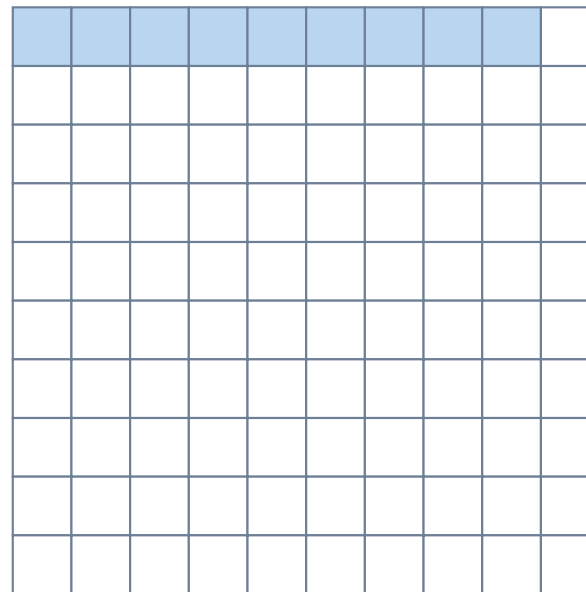
A. True

B. False



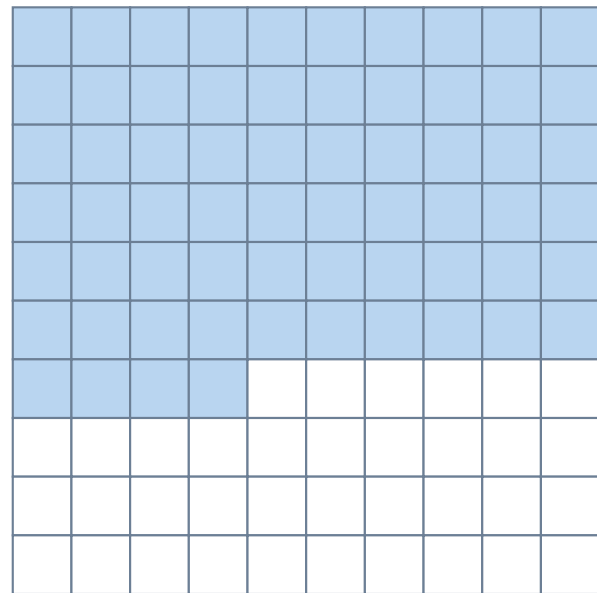
Which decimal does the shaded part of the grid show?

- A.** 0.9
- B.** 9
- C.** 0.009
- D.** 0.09



How many more squares
must be shaded to fill the
whole grid?

Squares: _____



True or false?

0.5 of a hundredths grid is 5 squares.

A. True

B. False

SET 22 • 5.NBT.B.7

FORWARD 1

Real-world problems with multi-digit decimals • Grade 6 • The Number System
• 6.NS.B.3

Gas costs \$3.85 per gallon.

How much do 12 gallons cost?

Cost: _____

A board is 6.75 m long. It is cut into pieces that are 0.45 m long.

How many pieces are there?

Pieces: _____

Rice costs \$1.20 per pound.

How much do 2.5 pounds cost?

- A.** \$3.00
- B.** \$30.00
- C.** \$2.70
- D.** \$3.70

Lee buys items that cost \$12.49, \$3.05, and \$7.60.

What is the total cost?

Total: _____

True or false?

A 2.4-pound bag split into 0.3-pound portions makes 8 portions.

A. True

B. False

SET 22 • 5.NBT.B.7

FORWARD 2

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A submarine at -45.5 m rises 12.75 m and then dives 20 m.

What is its final position?

Position: _____

A store's profit was $-\$125.40$ in May and $\$310.15$ in June.

What was the total profit for the two months?

Total: _____

The temperature was 4°F . It dropped 1.5°F each hour for 6 hours.

What was the final temperature?

A. -5°F

B. 13°F

C. 5°F

D. -9°F

An account changes by $-\$2.40$ each week for 5 weeks.
What is the total change?

Total change: _____

True or false?

$$-3.6 \div 0.4 = -0.9$$

A. True

B. False

Add fractions with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Equivalent fractions	4.NF.A.1
BACKWARD 2	Adding fractions with like denominators	4.NF.B.3.a
BACKWARD 3	Common multiples	4.OA.B.4
FORWARD 1	Find the least common multiple	6.NS.B.4
FORWARD 2	Add rational numbers	7.NS.A.1.d

SET 23 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$\frac{2}{3} + \frac{1}{4}$$

Sum: _____

Add.

$$\frac{3}{5} + \frac{1}{10}$$

Sum: _____

Add.

$$\frac{5}{6} + \frac{3}{8}$$

A. $1\frac{5}{24}$

B. $1\frac{1}{24}$

C. $\frac{8}{24}$

D. $\frac{8}{14}$

Add.

$$\frac{1}{2} + \frac{2}{7}$$

Sum: _____

Add.

$$\frac{3}{4} + \frac{5}{12} + \frac{1}{6}$$

Sum: _____

SET 23 • 5.NF.A.1

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{2}{3} = \frac{?}{12}$$

Missing number: _____

True or false?

$$\frac{1}{4} = \frac{4}{12}$$

A. True

B. False

Which fraction is equivalent to $\frac{3}{5}$?

A. $\frac{6}{5}$

B. $\frac{5}{3}$

C. $\frac{3}{10}$

D. $\frac{6}{10}$

Find the missing number.

$$\frac{5}{6} = \frac{20}{?}$$

Missing number: _____

True or false?

$$\frac{2}{7} = \frac{4}{21}$$

A. True

B. False

SET 23 • 5.NF.A.1

BACKWARD 2

Adding fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.a

Add.

$$\frac{3}{8} + \frac{2}{8}$$

Sum: _____

True or false?

$$\frac{4}{12} + \frac{5}{12} = \frac{9}{24}$$

A. True

B. False

$$\frac{7}{10} + \frac{6}{10} = ?$$

A. $1\frac{6}{10}$

B. $1\frac{3}{10}$

C. $\frac{13}{20}$

D. $\frac{1}{10}$

Add.

$$\frac{1}{6} + \frac{4}{6}$$

Sum: _____

True or false?

$$\frac{2}{5} + \frac{3}{5} = 1$$

A. True

B. False

SET 23 • 5.NF.A.1

BACKWARD 3

Common multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first five multiples of 4.

Multiples: _____

True or false?

10 is a multiple of both 3 and 4.

A. True

B. False

Which number is a multiple of both 6 and 8?

A. 16

B. 30

C. 24

D. 14

Name a number that is a multiple of both 5 and 10.

Number: _____

True or false?

18 is a multiple of both 4 and 6.

A. True

B. False

SET 23 • 5.NF.A.1

FORWARD 1

Find the least common multiple • Grade 6 • The Number System • 6.NS.B.4

Find the least common multiple of 6 and 10.

LCM: _____

Find the least common multiple of 8 and 12.

LCM: _____

What is the least common multiple of 4 and 9?

- A.** 72
- B.** 13
- C.** 18
- D.** 36

Find the least common multiple of 5 and 15.

LCM: _____

True or false?

The least common multiple of 6 and 9 is 54.

A. True

B. False

SET 23 • 5.NF.A.1

FORWARD 2

Add rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$-\frac{2}{3} + \frac{1}{4}$$

Value: _____

Find the value.

$$\frac{3}{5} + \left(-\frac{7}{10}\right)$$

Value: _____

Find the value.

$$-\frac{1}{2} + \left(-\frac{3}{8}\right)$$

A. $-\frac{7}{8}$

B. $-\frac{4}{10}$

C. $\frac{1}{8}$

D. $-\frac{1}{8}$

Find the value.

$$-1\frac{1}{3} + \frac{5}{6}$$

Value: _____

True or false?

$$-\frac{3}{4} + \frac{3}{4} = 0$$

A. True

B. False

Subtract fractions with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Equivalent fractions	4.NF.A.1
BACKWARD 2	Subtracting fractions with like denominators	4.NF.B.3.a
BACKWARD 3	Common multiples	4.OA.B.4
FORWARD 1	Solve equations of the form $x + p = q$ with fractions	6.EE.B.7
FORWARD 2	Subtract rational numbers	7.NS.A.1.c

SET 24 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$$\frac{5}{6} - \frac{1}{4}$$

Difference: _____

Subtract.

$$\frac{7}{8} - \frac{1}{2}$$

Difference: _____

Subtract.

$$\frac{4}{5} - \frac{2}{3}$$

A. $\frac{2}{2}$

B. $\frac{1}{5}$

C. $\frac{6}{15}$

D. $\frac{2}{15}$

Subtract.

$$\frac{9}{10} - \frac{2}{5}$$

Difference: _____

Subtract.

$$1 - \frac{3}{7}$$

Difference: _____

SET 24 • 5.NF.A.1

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{1}{4} = \frac{?}{12}$$

Missing number: _____

True or false?

$$\frac{1}{2} = \frac{4}{8}$$

A. True

B. False

Which fraction is equivalent to $\frac{2}{3}$?

A. $\frac{10}{15}$

B. $\frac{4}{9}$

C. $\frac{3}{4}$

D. $\frac{2}{6}$

Find the missing number.

$$\frac{4}{5} = \frac{?}{15}$$

Missing number: _____

True or false?

$$\frac{3}{4} = \frac{9}{16}$$

A. True

B. False

SET 24 • 5.NF.A.1

BACKWARD 2

Subtracting fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.a

Subtract.

$$\frac{7}{9} - \frac{2}{9}$$

Difference: _____

True or false?

$$\frac{5}{8} - \frac{3}{8} = \frac{2}{8}$$

A. True

B. False

$$\frac{11}{12} - \frac{4}{12} = ?$$

A. $\frac{7}{24}$

B. $\frac{7}{12}$

C. $\frac{15}{12}$

D. $\frac{4}{12}$

Subtract.

$$1 - \frac{3}{5}$$

Difference: _____

True or false?

$$\frac{9}{10} - \frac{4}{10} = \frac{5}{20}$$

A. True

B. False

SET 24 • 5.NF.A.1

BACKWARD 3

Common multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first four multiples of 6.

Multiples: _____

True or false?

15 is a multiple of both 3 and 5.

A. True

B. False

Which number is a multiple of both 4 and 5?

A. 15

B. 20

C. 9

D. 25

Name a number that is a multiple of both 3 and 8.

Number: _____

True or false?

16 is a multiple of both 4 and 6.

A. True

B. False

SET 24 • 5.NF.A.1

FORWARD 1

Solve equations of the form $x + p = q$ with fractions • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + \frac{1}{3} = \frac{5}{6}$$

x = _____

Solve for y.

$$y - \frac{2}{5} = \frac{3}{10}$$

y = _____

Solve for n.

$$\frac{3}{4} + n = 1\frac{1}{2}$$

A. $1\frac{1}{4}$

B. $\frac{3}{4}$

C. $2\frac{1}{4}$

D. $\frac{1}{4}$

Solve for m.

$$m + \frac{5}{8} = 2$$

m = _____

True or false?

The solution of $x - \frac{1}{6} = \frac{2}{3}$ is $x = \frac{5}{6}$.

A. True

B. False

SET 24 • 5.NF.A.1

FORWARD 2

Subtract rational numbers • Grade 7 • The Number System • 7.NS.A.1.c

Find the value.

$$\frac{1}{4} - \frac{5}{6}$$

Value: _____

Find the value.

$$-\frac{2}{3} - \frac{1}{6}$$

Value: _____

Find the value.

$$\frac{1}{2} - \left(-\frac{3}{4}\right)$$

A. $-\frac{1}{4}$

B. $\frac{1}{4}$

C. $1\frac{1}{4}$

D. $-1\frac{1}{4}$

Find the value.

$$-\frac{3}{5} - (-\frac{1}{2})$$

Value: _____

True or false?

$$\frac{2}{9} - \frac{5}{9} = \frac{1}{3}$$

A. True

B. False

Add and subtract mixed numbers with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Adding and subtracting mixed numbers with like denominators	4.NF.B.3.c
BACKWARD 2	Mixed numbers and fractions greater than 1	4.NF.B.3.b
BACKWARD 3	Equivalent fractions	4.NF.A.1
FORWARD 1	Solve equations of the form $x + p = q$ with mixed numbers	6.EE.B.7
FORWARD 2	Solve equations of the form $px + q = r$ with fractions	7.EE.B.4.a

SET 25 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$2\frac{1}{3} + 1\frac{1}{2}$$

Sum: _____

Subtract.

$$5\frac{3}{4} - 2\frac{1}{6}$$

Difference: _____

Subtract.

$$4\frac{1}{5} - 1\frac{1}{2}$$

A. $3\frac{3}{10}$

B. $2\frac{3}{10}$

C. $3\frac{7}{10}$

D. $2\frac{7}{10}$

Add.

$$3\frac{5}{8} + 2\frac{3}{4}$$

Sum: _____

Subtract.

$$6 - 2\frac{2}{3}$$

Difference: _____

SET 25 • 5.NF.A.1

BACKWARD 1

Adding and subtracting mixed numbers with like denominators • Grade 4 •
Number & Operations–Fractions • 4.NF.B.3.c

Add.

$$1\frac{2}{5} + 2\frac{1}{5}$$

Sum: _____

Subtract.

$$4\frac{5}{6} - 1\frac{4}{6}$$

Difference: _____

True or false?

$$2\frac{3}{4} + 1\frac{3}{4} = 3\frac{6}{8}$$

A. True

B. False

$$5\frac{1}{3} - 2\frac{2}{3} = ?$$

A. $3\frac{1}{3}$

B. $2\frac{1}{3}$

C. $2\frac{2}{3}$

D. $3\frac{2}{3}$

Add.

$$3\frac{7}{8} + \frac{3}{8}$$

Sum: _____

SET 25 • 5.NF.A.1

BACKWARD 2

Mixed numbers and fractions greater than 1 • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write $2\frac{3}{4}$ as a fraction.

Fraction: _____

Write $\frac{17}{5}$ as a mixed number.

Mixed number: _____

True or false?

$$1\frac{1}{3} = \frac{4}{3}$$

A. True

B. False

Which fraction is equal to $3\frac{1}{2}$?

A. $\frac{31}{2}$

B. $\frac{7}{2}$

C. $\frac{5}{2}$

D. $\frac{6}{2}$

True or false?

$$5\frac{2}{3} = 4\frac{5}{3}$$

A. True

B. False

SET 25 • 5.NF.A.1

BACKWARD 3

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{3}{4} = \frac{?}{8}$$

Missing number: _____

True or false?

$$\frac{1}{6} = \frac{2}{12}$$

A. True

B. False

Which fraction is equivalent to $\frac{1}{2}$?

A. $\frac{5}{10}$

B. $\frac{2}{3}$

C. $\frac{2}{2}$

D. $\frac{1}{4}$

Find the missing number.

$$\frac{2}{5} = \frac{?}{10}$$

Missing number: _____

True or false?

$$\frac{5}{6} = \frac{10}{18}$$

A. True

B. False

SET 25 • 5.NF.A.1

FORWARD 1

Solve equations of the form $x + p = q$ with mixed numbers • Grade 6 •
Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + 2\frac{1}{2} = 6\frac{1}{4}$$

x = _____

Solve for y.

$$y - 1\frac{2}{3} = 3\frac{1}{6}$$

$$y = \underline{\hspace{10cm}}$$

Solve for n.

$$n + 4\frac{3}{8} = 7$$

A. $3\frac{5}{8}$

B. $2\frac{3}{8}$

C. $11\frac{3}{8}$

D. $2\frac{5}{8}$

A board was $8\frac{1}{2}$ feet long. After a piece was cut off, it was $5\frac{3}{4}$ feet long.

Write and solve an equation to find the length p of the piece.

Equation and solution: _____

True or false?

The solution of $x - 1\frac{1}{4} = 2\frac{1}{2}$ is $x = 1\frac{1}{4}$.

A. True

B. False

SET 25 • 5.NF.A.1

FORWARD 2

Solve equations of the form $px + q = r$ with fractions • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$$2x + \frac{1}{2} = 3\frac{1}{2}$$

x = _____

Solve for x.

$$\frac{1}{3}x - 1\frac{1}{4} = 2\frac{3}{4}$$

x = _____

Solve for x.

$$4x - 2\frac{1}{3} = 1$$

A. $1\frac{1}{12}$

B. $\frac{1}{6}$

C. $\frac{5}{6}$

D. $\frac{1}{3}$

Jo bought 3 bags of flour that weigh the same and one $1\frac{1}{2}$ -pound bag of sugar. The total weight was $9\frac{3}{4}$ pounds.

How much does each bag of flour weigh?

Weight: _____

True or false?

The solution of $\frac{1}{2}(x + 3) = 4$ is $x = 5$.

A. True

B. False

Solve word problems by adding and subtracting fractions

MAIN	Grade 5 main question	5.NF.A.2
BACKWARD 1	Word problems with like denominators	4.NF.B.3.d
BACKWARD 2	Choosing addition or subtraction	2.OA.A.1
BACKWARD 3	Adding and subtracting fractions with unlike denominators	5.NF.A.1
FORWARD 1	Write and solve equations of the form $x + p = q$	6.EE.B.7
FORWARD 2	Write and solve equations of the form $px + q = r$	7.EE.B.4.a

SET 26 • 5.NF.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

Mia walked $\frac{3}{4}$ mile to school and then $\frac{2}{3}$ mile to the park.

How far did she walk in all?

Distance: _____

A jug had $\frac{7}{8}$ gallon of milk. Ben drank $\frac{1}{3}$ gallon.

How much milk is left?

Milk left: _____

Ana used $2\frac{1}{2}$ cups of flour and $1\frac{3}{4}$ cups of sugar.

How many cups did she use in all?

A. $4\frac{1}{4}$ cups

B. $3\frac{3}{4}$ cups

C. $4\frac{3}{4}$ cups

D. $3\frac{1}{4}$ cups

A ribbon is $5\frac{1}{6}$ feet long. Lea cuts off $2\frac{1}{2}$ feet.

How long is the ribbon now?

Length: _____

Tom read $\frac{1}{4}$ of a book on Monday and $\frac{3}{8}$ of the book on Tuesday.

What fraction of the book does he still have to read?

Fraction: _____

SET 26 • 5.NF.A.2

BACKWARD 1

Word problems with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.d

Kim ate $\frac{3}{8}$ of a pizza and Joe ate $\frac{2}{8}$ of the pizza.

What fraction of the pizza did they eat in all?

Fraction: _____

A rope is $\frac{9}{10}$ meter long. $\frac{4}{10}$ meter is cut off.

How much rope is left?

Rope left: _____

True or false?

$\frac{2}{6}$ of a garden is corn and $\frac{3}{6}$ is beans. Corn and beans cover $\frac{5}{6}$ of the garden.

A. True

B. False

Lu had $\frac{7}{12}$ of a cake. She gave away $\frac{5}{12}$ of the cake.

How much of the cake is left?

A. $\frac{12}{12}$

B. $\frac{2}{12}$

C. $\frac{35}{12}$

D. $\frac{2}{24}$

A tank is $\frac{3}{5}$ full. Water is added to fill another $\frac{1}{5}$ of the tank.

How full is the tank now?

Fraction: _____

SET 26 • 5.NF.A.2

BACKWARD 2

Choosing addition or subtraction • Grade 2 • Operations & Algebraic Thinking •
2.OA.A.1

A class has 14 boys and 13 girls.

Which operation finds the number of students in all?

- A.** Subtraction
- B.** Division
- C.** Addition
- D.** Multiplication

True or false?

To find how much is left after some is used, you subtract.

- A.** True
- B.** False

Sam had 42 marbles. He lost 15 marbles.
How many marbles are left?

Marbles: _____

A shelf holds 35 books. 18 books are on it now.

Which equation finds how many more books fit?

A. $35 - 18 = \square$

B. $18 - 35 = \square$

C. $35 \times 18 = \square$

D. $35 + 18 = \square$

Liz has 26 stickers. She gets 17 more.

How many stickers does she have now?

Stickers: _____

SET 26 • 5.NF.A.2

BACKWARD 3

Adding and subtracting fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$\frac{1}{3} + \frac{1}{2}$$

Sum: _____

Subtract.

$$\frac{3}{4} - \frac{1}{3}$$

Difference: _____

True or false?

$$\frac{1}{2} + \frac{1}{4} = \frac{2}{6}$$

A. True

B. False

$$\frac{5}{6} - \frac{1}{2} = ?$$

A. $\frac{2}{3}$

B. $\frac{1}{3}$

C. $\frac{4}{4}$

D. $\frac{1}{6}$

Add.

$$\frac{2}{5} + \frac{1}{2}$$

Sum: _____

SET 26 • 5.NF.A.2

FORWARD 1

Write and solve equations of the form $x + p = q$ • Grade 6 • Expressions & Equations • 6.EE.B.7

After using $\frac{2}{3}$ cup of oil, Ray has $\frac{3}{4}$ cup left.

Write and solve an equation to find how much oil c he started with.

Equation and solution: _____

A plant grew $\frac{5}{8}$ inch and is now $4\frac{1}{4}$ inches tall.

Write and solve an equation to find its starting height h .

Equation and solution: _____

$\frac{3}{10}$ inch of rain fell on day 1. In all, $\frac{9}{10}$ inch fell on days 1 and 2.

Which equation finds the rain r on day 2?

A. $\frac{3}{10}r = \frac{9}{10}$

B. $r + \frac{9}{10} = \frac{3}{10}$

C. $\frac{3}{10} + r = \frac{9}{10}$

D. $r - \frac{3}{10} = \frac{9}{10}$

Solve for x.

$$x + \frac{2}{9} = \frac{2}{3}$$

x = _____

True or false?

If $n - \frac{1}{4} = \frac{5}{8}$, then $n = \frac{7}{8}$.

A. True

B. False

SET 26 • 5.NF.A.2

FORWARD 2

Write and solve equations of the form $px + q = r$ • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A recipe uses $\frac{3}{4}$ cup of sugar per batch, plus $\frac{1}{2}$ cup for frosting. Ali used $2\frac{3}{4}$ cups of sugar in all.

How many batches did he make?

Batches: _____

Solve for x.

$$\frac{2}{3}x + \frac{1}{6} = \frac{5}{6}$$

x = _____

A 6-foot board is cut into 4 equal pieces, with $\frac{1}{2}$ foot left over.

How long is each piece?

A. $1\frac{1}{2}$ feet

B. $1\frac{5}{8}$ feet

C. $1\frac{3}{8}$ feet

D. $5\frac{1}{2}$ feet

Solve for x.

$$5x - \frac{3}{4} = 4\frac{1}{4}$$

x = _____

True or false?

The solution of $2x + \frac{1}{3} = 1$ is $x = \frac{2}{3}$.

A. True

B. False

Use benchmark fractions to estimate and to check answers

MAIN	Grade 5 main question	5.NF.A.2
BACKWARD 1	Comparing fractions to $\frac{1}{2}$ and 1	4.NF.A.2
BACKWARD 2	Fractions on a number line	3.NF.A.2
FORWARD 1	Reason about the size of a quotient of fractions	6.NS.A.1
FORWARD 2	Use estimation to judge whether an answer is reasonable	7.EE.B.3

SET 27 • 5.NF.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

True or false?

$\frac{2}{5} + \frac{1}{2} = \frac{3}{7}$ must be wrong, because $\frac{3}{7}$ is less than $\frac{1}{2}$.

A. True

B. False

Which is the best estimate of $\frac{7}{8} + \frac{5}{12}$?

- A.** About $\frac{1}{2}$
- B.** About $2\frac{1}{2}$
- C.** About $1\frac{1}{2}$
- D.** About 3

Is $\frac{3}{5} + \frac{4}{9}$ more or less than 1?

Use benchmark fractions to explain.

Answer: _____

Explanation: _____

Jen ran $\frac{9}{10}$ mile and walked $\frac{1}{8}$ mile.

Estimate the total distance to the nearest whole mile.

Estimate: _____

Leo says $\frac{5}{6} - \frac{1}{3} = \frac{4}{3}$.

Which statement explains why his answer is not reasonable?

- A.** The answer is greater than $\frac{5}{6}$, but subtracting should give less than $\frac{5}{6}$.
- B.** The denominators are different.
- C.** $\frac{4}{3}$ is less than 1.
- D.** The answer should be 0.

SET 27 • 5.NF.A.2

BACKWARD 1

Comparing fractions to $\frac{1}{2}$ and 1 • Grade 4 • Number & Operations–Fractions •
4.NF.A.2

True or false?

$\frac{5}{8}$ is less than $\frac{1}{2}$.

A. True

B. False

Which fraction is greater than $\frac{1}{2}$?

A. $\frac{1}{3}$

B. $\frac{3}{8}$

C. $\frac{2}{5}$

D. $\frac{4}{7}$

Is $\frac{5}{9}$ closest to 0, $\frac{1}{2}$, or 1?

Answer: _____

True or false?

$\frac{6}{10}$ is less than $\frac{1}{2}$.

A. True

B. False

Is $\frac{11}{12}$ closer to $\frac{1}{2}$ or to 1?

Answer: _____

SET 27 • 5.NF.A.2

BACKWARD 2

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction is at point A?



A = _____

True or false?

On a number line, $\frac{1}{2}$ is halfway between 0 and 1.

A. True

B. False

Which point is at $\frac{1}{3}$?



A. Point A

B. Point B

C. Point C

D. Point D

What fraction is at point B?



B = _____

True or false?

$\frac{7}{8}$ is closer to 1 than to $\frac{1}{2}$.

A. True

B. False

SET 27 • 5.NF.A.2

FORWARD 1

Reason about the size of a quotient of fractions • Grade 6 • The Number System • 6.NS.A.1

True or false?

$\frac{3}{4} \div \frac{1}{8}$ is greater than 1.

A. True

B. False

Without dividing, which quotient is less than 1?

A. $\frac{3}{4} \div \frac{1}{2}$

B. $\frac{1}{3} \div \frac{1}{2}$

C. $2 \div \frac{1}{2}$

D. $\frac{5}{6} \div \frac{1}{3}$

Divide.

$$\frac{5}{6} \div \frac{5}{12}$$

Quotient: _____

Estimate $2\frac{7}{8} \div \frac{1}{2}$ by first rounding $2\frac{7}{8}$ to the nearest whole number.

Estimate: _____

True or false?

$\frac{1}{2} \div 4$ is greater than $\frac{1}{2}$.

A. True

B. False

SET 27 • 5.NF.A.2

FORWARD 2

Use estimation to judge whether an answer is reasonable • Grade 7 •
Expressions & Equations • 7.EE.B.3

A board $9\frac{7}{8}$ feet long is cut into 3 equal pieces.

Estimate the length of each piece.

Estimate: _____

Ann earns \$11.85 per hour and works $7\frac{3}{4}$ hours.

Which is the best estimate of her pay?

- A.** \$96
- B.** \$84
- C.** \$120
- D.** \$77

True or false?

$\frac{3}{4}$ of \$79.95 is about \$60.

A. True

B. False

Kai says $4\frac{5}{6} + 3\frac{1}{8}$ is about 9.

Is this reasonable? Give a better estimate.

Reasonable? _____

Better estimate: _____

A recipe needs $2\frac{1}{3}$ cups of flour per batch.

Estimate the flour needed for 4 batches.

Estimate: _____

Interpret a fraction as division of the numerator by the denominator

MAIN	Grade 5 main question	5.NF.B.3
BACKWARD 1	Unit fractions	3.NF.A.1
BACKWARD 2	Division with remainders	4.NBT.B.6
BACKWARD 3	Mixed numbers and fractions greater than 1	4.NF.B.3.b
FORWARD 1	Unit rates written as fractions	6.RP.A.2
FORWARD 2	Write fractions as decimals using division	7.NS.A.2.d

SET 28 • 5.NF.B.3

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

Write $3 \div 5$ as a fraction.

Fraction: _____

4 friends share 3 pizzas equally.

How much pizza does each friend get?

Each friend gets: _____

Which division expression is equal to $\frac{7}{4}$?

A. $7 \div 4$

B. $7 - 4$

C. $4 \div 7$

D. 7×4

Write $\frac{13}{6}$ as a division expression. Then write the quotient as a mixed number.

Division: _____

Mixed number: _____

True or false?

When 2 pounds of rice are shared equally among 3 bags, each bag gets $\frac{2}{3}$ pound.

- A. True
- B. False

SET 28 • 5.NF.B.3

BACKWARD 1

Unit fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

$\frac{1}{4}$ is one of 4 equal parts of a whole.

A. True

B. False

A sandwich is cut into 3 equal parts.

What fraction of the sandwich is one part?

Fraction: _____

What fraction of the bar is shaded?



A. $\frac{5}{3}$

B. $\frac{3}{5}$

C. $\frac{2}{5}$

D. $\frac{3}{2}$

How many $\frac{1}{6}$ s make $\frac{5}{6}$?

Answer: _____

True or false?

$\frac{3}{8}$ is 3 parts of size $\frac{1}{8}$.

A. True

B. False

SET 28 • 5.NF.B.3

BACKWARD 2

Division with remainders • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide. Write the quotient and the remainder.

$$17 \div 5$$

Quotient and remainder: _____

True or false?

$$23 \div 4 = 6 \text{ R } 1$$

A. True

B. False

$$29 \div 6 = ?$$

A. 5 R 5

B. 5 R 1

C. 4 R 3

D. 4 R 5

Divide. Write the quotient and the remainder.

$$50 \div 8$$

Quotient and remainder: _____

True or false?

$$15 \div 2 = 7 \text{ R } 2$$

A. True

B. False

SET 28 • 5.NF.B.3

BACKWARD 3

Mixed numbers and fractions greater than 1 • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write $\frac{9}{4}$ as a mixed number.

Mixed number: _____

Write $1\frac{2}{3}$ as a fraction.

Fraction: _____

True or false?

$$\frac{11}{5} = 1\frac{1}{5}$$

A. True

B. False

Which mixed number is equal to $\frac{7}{3}$?

A. $2\frac{2}{3}$

B. $3\frac{1}{3}$

C. $2\frac{1}{3}$

D. $1\frac{1}{3}$

Write $\frac{15}{4}$ as a mixed number.

Mixed number: _____

SET 28 • 5.NF.B.3

FORWARD 1

Unit rates written as fractions • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.2

A recipe uses 3 cups of flour for every 4 cups of water.
How many cups of flour are used per cup of water?

Cups of flour: _____

8 apples cost \$6.

What is the cost per apple?

Cost per apple: _____

A car goes 150 miles on 4 gallons of gas.

What is the rate in miles per gallon?

A. $\frac{4}{150}$

B. 600

C. 154

D. 37.5

There are 5 cups of juice for every 2 pitchers.

How many cups of juice are there per pitcher?

Cups per pitcher: _____

True or false?

The ratio 3 : 8 has the unit rate $\frac{3}{8}$.

A. True

B. False

SET 28 • 5.NF.B.3

FORWARD 2

Write fractions as decimals using division • Grade 7 • The Number System •
7.NS.A.2.d

Write $5 \div 8$ as a decimal.

Decimal: _____

Use long division to write $\frac{7}{12}$ as a decimal.

Decimal: _____

Which fraction has a decimal form that repeats?

A. $\frac{7}{20}$

B. $\frac{5}{6}$

C. $\frac{9}{25}$

D. $\frac{3}{8}$

Write $11 \div 4$ as a decimal.

Decimal: _____

True or false?

$$\frac{1}{3} = 0.3$$

A. True

B. False

Solve sharing problems whose answers are fractions or mixed numbers

MAIN	Grade 5 main question	5.NF.B.3
BACKWARD 1	Division word problems	3.OA.A.3
BACKWARD 2	Whole numbers as fractions	3.NF.A.3.c
FORWARD 1	Solve unit rate problems	6.RP.A.3.b
FORWARD 2	Unit rates with fractional quantities	7.RP.A.1

SET 29 • 5.NF.B.3

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

6 friends share 10 granola bars equally.

How many bars does each friend get? Write a mixed number.

Bars: _____

A 50-pound bag of rice is shared equally among 8 families.

How many pounds does each family get?

Pounds: _____

9 yards of ribbon are cut into 4 equal pieces.

How long is each piece?

A. $4\frac{1}{2}$ yards

B. $\frac{4}{9}$ yard

C. $2\frac{1}{9}$ yards

D. $2\frac{1}{4}$ yards

7 liters of juice are poured equally into 3 jugs.

How many liters are in each jug? Between which two whole numbers is your answer?

Liters: _____

Between: _____

True or false?

When 5 cookies are shared equally by 2 children, each child gets $2\frac{1}{2}$ cookies.

A. True

B. False

SET 29 • 5.NF.B.3

BACKWARD 1

Division word problems • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

24 stickers are shared equally by 6 friends.

How many stickers does each friend get?

Stickers: _____

35 chairs are set out in 5 equal rows.

How many chairs are in each row?

A. 30

B. 6

C. 40

D. 7

True or false?

If 4 boxes hold 28 toys, with the same number in each box, each box holds 7 toys.

A. True

B. False

A baker puts 45 muffins into bags of 9.
How many bags does she fill?

Bags: _____

18 pencils are shared equally by 3 students.

How many pencils does each student get?

Pencils: _____

SET 29 • 5.NF.B.3

BACKWARD 2

Whole numbers as fractions • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.c

True or false?

$$\frac{6}{3} = 3$$

A. True

B. False

Write 4 as a fraction with a denominator of 1.

Fraction: _____

Which fraction is equal to 3?

A. $\frac{3}{3}$

B. $\frac{3}{4}$

C. $\frac{4}{12}$

D. $\frac{12}{4}$

How many halves are in 5 wholes?

Halves: _____

True or false?

$$\frac{8}{8} = 8$$

A. True

B. False

SET 29 • 5.NF.B.3

FORWARD 1

Solve unit rate problems • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

3 pounds of grapes cost \$7.

What is the cost per pound?

Cost per pound: _____

A car travels 100 miles in 3 hours at a constant speed.
At this rate, how far does it travel in 6 hours?

Distance: _____

8 cans of soup cost \$6.

At this rate, how much do 12 cans cost?

A. \$16

B. \$8

C. \$9

D. \$4

A runner runs 5 miles in 40 minutes at a constant pace.
At this pace, how long does she take to run 8 miles?

Time: _____

True or false?

If 4 pens cost \$3, then 10 pens cost \$7.00.

- A. True
- B. False

SET 29 • 5.NF.B.3

FORWARD 2

Unit rates with fractional quantities • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A snail crawls $\frac{2}{3}$ foot in $\frac{1}{4}$ hour.

What is its speed in feet per hour?

Speed: _____

A recipe uses $\frac{3}{4}$ cup of sugar for every $\frac{1}{2}$ cup of milk.

How many cups of sugar are used per cup of milk?

Cups of sugar: _____

Jo walks $\frac{1}{2}$ mile in $\frac{1}{6}$ hour.

What is her speed?

A. 2 miles per hour

B. 3 miles per hour

C. $\frac{1}{3}$ mile per hour

D. $\frac{1}{12}$ mile per hour

A pump fills $\frac{5}{8}$ of a tank in $\frac{1}{2}$ hour.

How much of the tank does it fill per hour?

Tanks per hour: _____

True or false?

$\frac{1}{2}$ cup of juice for every $\frac{1}{4}$ cup of water is a rate of 2 cups of juice per cup of water.

- A. True
- B. False

Multiply a fraction by a whole number

MAIN	Grade 5 main question	5.NF.B.4.a
BACKWARD 1	A fraction times a whole number as repeated unit fractions	4.NF.B.4.b
BACKWARD 2	Parts of a whole	3.NF.A.1
BACKWARD 3	Splitting a number into equal parts	3.OA.A.2
FORWARD 1	Find a percent of a quantity	6.RP.A.3.c
FORWARD 2	Multistep percent problems	7.RP.A.3

SET 30 • 5.NF.B.4.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{3}{4} \times 12$$

Product: _____

Multiply.

$$5 \times \frac{2}{3}$$

Product: _____

What is $\frac{2}{5}$ of 20?

A. 10

B. 8

C. 4

D. 50

Multiply.

$$\frac{5}{8} \times 6$$

Product: _____

True or false?

$\frac{1}{3} \times 15$ means 15 split into 3 equal parts, with 1 of the parts taken.

- A.** True
- B.** False

SET 30 • 5.NF.B.4.a

BACKWARD 1

A fraction times a whole number as repeated unit fractions • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.b

Write $\frac{4}{5}$ as a whole number times $\frac{1}{5}$.

Expression: _____

True or false?

$$3 \times \frac{2}{7} = \frac{6}{21}$$

A. True

B. False

$$3 \times \frac{3}{8} = ?$$

A. $3\frac{3}{8}$

B. $\frac{6}{8}$

C. $\frac{9}{8}$

D. $\frac{9}{24}$

Multiply.

$$4 \times \frac{2}{9}$$

Product: _____

True or false?

$$2 \times \frac{3}{5} = \frac{6}{10}$$

A. True

B. False

SET 30 • 5.NF.B.4.a

BACKWARD 2

Parts of a whole • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

$\frac{1}{4}$ of a bar means 1 of 4 equal parts.

A. True

B. False

What fraction of the bar is shaded?



Fraction: _____

What fraction is 2 parts of size $\frac{1}{3}$?

A. $\frac{2}{3}$

B. $\frac{1}{6}$

C. $\frac{1}{3}$

D. $\frac{3}{2}$

How many $\frac{1}{8}$ s are in $\frac{5}{8}$?

Answer: _____

True or false?

$\frac{3}{5}$ means 3 parts of size $\frac{1}{5}$.

A. True

B. False

SET 30 • 5.NF.B.4.a

BACKWARD 3

Splitting a number into equal parts • Grade 3 • Operations & Algebraic Thinking
• 3.OA.A.2

12 is split into 4 equal parts.

How much is in each part?

Answer: _____

True or false?

20 split into 5 equal parts gives 4 in each part.

A. True

B. False

Which expression shows 18 split into 3 equal parts?

A. $3 \div 18$

B. $18 - 3$

C. $18 \div 3$

D. 18×3

Divide.

$$24 \div 8$$

Quotient: _____

30 is split into 6 equal parts.

How much is in each part?

Answer: _____

SET 30 • 5.NF.B.4.a

FORWARD 1

Find a percent of a quantity • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

What is 25% of 48?

Answer: _____

What is 60% of 35?

Answer: _____

A \$40 game is 15% off.

How many dollars are taken off the price?

- A.** \$6
- B.** \$15
- C.** \$4
- D.** \$34

A class has 30 students. 70% of them ride the bus.
How many students ride the bus?

Students: _____

True or false?

50% of 18 is 36.

A. True

B. False

SET 30 • 5.NF.B.4.a

FORWARD 2

Multistep percent problems • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.3

A \$60 coat is 30% off.

What is the sale price?

Sale price: _____

A meal costs \$25. A 20% tip is added.

What is the total cost?

Total: _____

A \$80 bike has a 6% sales tax added.

What is the total cost?

A. \$86.00

B. \$4.80

C. \$85.60

D. \$84.80

A town of 500 people grew by 12%.

What is the new population?

Population: _____

True or false?

A 25% markup on a \$40 item makes the price \$50.

A. True

B. False

Multiply a fraction by a fraction

MAIN	Grade 5 main question	5.NF.B.4.a
BACKWARD 1	A whole number times a fraction	4.NF.B.4.b
BACKWARD 2	Equivalent fractions and simplest form	4.NF.A.1
FORWARD 1	Divide fractions by fractions	6.NS.A.1
FORWARD 2	Multiply rational numbers	7.NS.A.2.c

SET 31 • 5.NF.B.4.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{2}{3} \times \frac{4}{5}$$

Product: _____

Multiply.

$$\frac{3}{4} \times \frac{1}{6}$$

Product: _____

Multiply.

$$\frac{5}{6} \times \frac{3}{10}$$

A. $\frac{15}{6}$

B. $\frac{1}{2}$

C. $\frac{1}{4}$

D. $\frac{8}{16}$

What is $\frac{1}{2}$ of $\frac{3}{8}$?

Answer: _____

True or false?

$$\frac{2}{5} \times \frac{2}{3} = \frac{4}{15}$$

A. True

B. False

SET 31 • 5.NF.B.4.a

BACKWARD 1

A whole number times a fraction • Grade 4 • Number & Operations–Fractions •
4.NF.B.4.b

Multiply.

$$2 \times \frac{3}{10}$$

Product: _____

True or false?

$$5 \times \frac{1}{4} = \frac{5}{20}$$

A. True

B. False

$$4 \times \frac{2}{3} = ?$$

A. $\frac{8}{12}$

B. $\frac{8}{3}$

C. $\frac{6}{3}$

D. $\frac{2}{12}$

Multiply.

$$3 \times \frac{5}{12}$$

Product: _____

True or false?

$$6 \times \frac{1}{6} = 1$$

A. True

B. False

SET 31 • 5.NF.B.4.a

BACKWARD 2

Equivalent fractions and simplest form • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Write $\frac{6}{8}$ as an equivalent fraction with a denominator of 4.

Fraction: _____

True or false?

$$\frac{10}{15} = \frac{3}{4}$$

A. True

B. False

Which fraction is equivalent to $\frac{4}{12}$?

A. $\frac{4}{3}$

B. $\frac{1}{4}$

C. $\frac{2}{4}$

D. $\frac{1}{3}$

Find the missing number.

$$\frac{9}{12} = \frac{?}{4}$$

Missing number: _____

True or false?

$$\frac{5}{20} = \frac{1}{5}$$

A. True

B. False

SET 31 • 5.NF.B.4.a

FORWARD 1

Divide fractions by fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{2}{3} \div \frac{4}{5}$$

Quotient: _____

Divide.

$$\frac{3}{4} \div \frac{1}{6}$$

Quotient: _____

Divide.

$$\frac{5}{8} \div \frac{1}{4}$$

A. $\frac{5}{32}$

B. $\frac{1}{2}$

C. $2\frac{1}{2}$

D. $1\frac{1}{4}$

How many $\frac{3}{8}$ -cup servings are in $\frac{3}{4}$ cup?

Servings: _____

True or false?

$$\frac{1}{2} \div \frac{1}{4} = \frac{1}{8}$$

A. True

B. False

SET 31 • 5.NF.B.4.a

FORWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$-\frac{2}{3} \times \frac{3}{4}$$

Product: _____

Find the product.

$$\left(-\frac{4}{5}\right) \times \left(-\frac{5}{6}\right)$$

Product: _____

Find the product.

$$\frac{3}{7} \times \left(-\frac{7}{9}\right)$$

A. $-\frac{3}{7}$

B. $\frac{1}{3}$

C. $-\frac{1}{3}$

D. $\frac{10}{16}$

Find the product.

$$\left(-\frac{1}{2}\right) \times \left(-\frac{1}{2}\right) \times \left(-\frac{1}{2}\right)$$

Product: _____

True or false?

$$\left(-\frac{2}{5}\right) \times \frac{5}{2} = 1$$

A. True

B. False

Find the area of a rectangle with fractional side lengths

MAIN

Grade 5 main question

5.NF.B.4.b

BACKWARD 1

Area by tiling with unit squares

3.MD.C.7.a

BACKWARD 2

The area formula for rectangles

4.MD.A.3

BACKWARD 3

Multiplying fractions

5.NF.B.4.a

FORWARD 1

Area of triangles and other polygons with fractional measures

6.G.A.1

FORWARD 2

Area of composite figures

7.G.B.6

SET 32 • 5.NF.B.4.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.b

Find the area of the rectangle.

Area: _____



A rectangle is $2\frac{1}{2}$ inches long and $\frac{4}{5}$ inch wide.

What is its area?

Area: _____

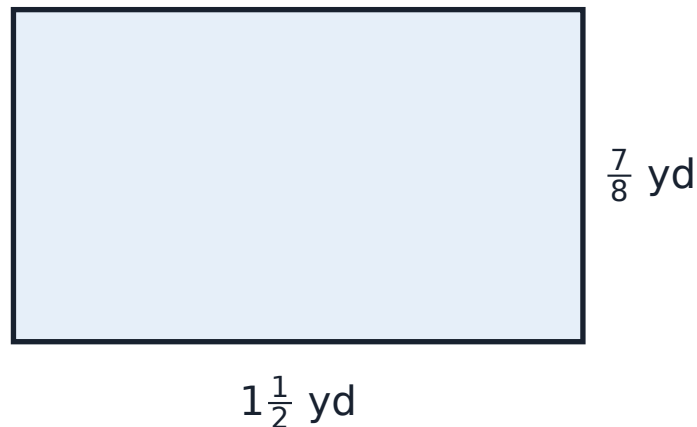
A rectangle that is $\frac{2}{3}$ unit by $\frac{3}{4}$ unit is tiled with small rectangles that are each $\frac{1}{3}$ unit by $\frac{1}{4}$ unit.

What is the area of the large rectangle?

- A.** $\frac{6}{12}$ square unit
- B.** $\frac{6}{7}$ square unit
- C.** $\frac{5}{7}$ square unit
- D.** $1\frac{5}{12}$ square units

Find the area of the rectangle.

Area: _____



True or false?

A rectangle that is $\frac{1}{2}$ foot by $\frac{1}{2}$ foot has an area of 1 square foot.

- A. True
- B. False

SET 32 • 5.NF.B.4.b

BACKWARD 1

Area by tiling with unit squares • Grade 3 • Measurement & Data • 3.MD.C.7.a

A rectangle is covered by 4 rows of 6 unit squares, with no gaps.

What is its area?

Area: _____

True or false?

Tiling a 3-by-5 rectangle takes 8 unit squares.

A. True

B. False

How many unit squares cover a 7-by-2 rectangle?

A. 72

B. 9

C. 14

D. 18

A rectangle has 5 rows of unit squares with 8 squares in each row.

What is its area?

Area: _____

True or false?

The number of unit squares that cover a rectangle equals its length times its width.

A. True

B. False

SET 32 • 5.NF.B.4.b

BACKWARD 2

The area formula for rectangles • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 12 m long and 5 m wide.

What is its area?

Area: _____

A rectangle has an area of 54 square feet and a length of 9 feet.

What is its width?

Width: _____

True or false?

The formula $A = l \times w$ gives the area of a rectangle.

- A. True
- B. False

A rectangle is 15 cm long and 4 cm wide.

What is its area?

- A.** 60 square cm
- B.** 45 square cm
- C.** 38 square cm
- D.** 19 square cm

Find the area of the rectangle.

Area: _____



SET 32 • 5.NF.B.4.b

BACKWARD 3

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{5}{6} \times \frac{3}{5}$$

Product: _____

Multiply.

$$\frac{5}{2} \times \frac{4}{5}$$

Product: _____

True or false?

$$\frac{2}{3} \times \frac{3}{4} = \frac{6}{12}$$

A. True

B. False

$$\frac{3}{2} \times \frac{7}{8} = ?$$

A. $\frac{10}{10}$

B. $\frac{21}{16}$

C. $\frac{21}{8}$

D. $\frac{3}{7}$

Multiply.

$$\frac{1}{2} \times \frac{1}{2}$$

Product: _____

SET 32 • 5.NF.B.4.b

FORWARD 1

Area of triangles and other polygons with fractional measures • Grade 6 •
Geometry • 6.G.A.1

A triangle has a base of $4\frac{1}{2}$ inches and a height of 2 inches.

What is its area?

Area: _____

A parallelogram has a base of $3\frac{1}{4}$ feet and a height of 4 feet.

What is its area?

Area: _____

A triangle has a base of $\frac{3}{4}$ m and a height of $\frac{2}{3}$ m.

What is its area?

A. $\frac{1}{2}$ square meter

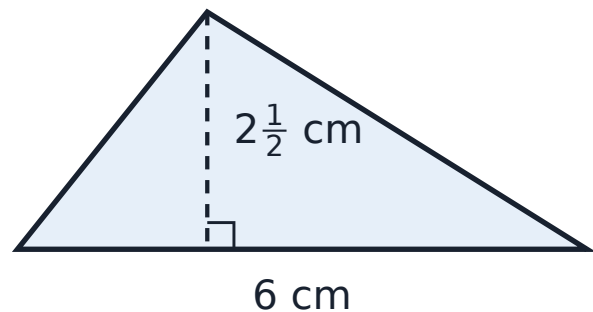
B. $\frac{5}{7}$ square meter

C. $\frac{1}{4}$ square meter

D. $\frac{6}{7}$ square meter

Find the area of the triangle.

Area: _____



True or false?

A trapezoid with bases of $2\frac{1}{2}$ m and $3\frac{1}{2}$ m and a height of 2 m has an area of 6 square meters.

- A. True
- B. False

SET 32 • 5.NF.B.4.b

FORWARD 2

Area of composite figures • Grade 7 • Geometry • 7.G.B.6

A figure is made of a rectangle that is $6\frac{1}{2}$ ft by 4 ft and a triangle with a base of 4 ft and a height of 3 ft.

What is the total area?

Area: _____

A rectangular garden is 10 m by $7\frac{1}{2}$ m. A square patio with $2\frac{1}{2}$ -m sides is inside the garden.

What is the area of the garden that is not patio?

Area: _____

A $\frac{1}{2}$ -by-2 rectangle is cut out of a 5-by-4 rectangle.

What is the area of the part that is left?

- A.** 21 square units
- B.** 18 square units
- C.** 19 square units
- D.** 20 square units

A kite is made of two triangles. Each triangle has a base of 8 in. and a height of $3\frac{1}{2}$ in.

What is the area of the kite?

Area: _____

True or false?

A figure made of two $\frac{1}{2}$ -by-3 rectangles that do not overlap has an area of 6 square units.

- A. True
- B. False

Compare the size of a product to the size of one factor without multiplying

MAIN	Grade 5 main question	5.NF.B.5.a
BACKWARD 1	Multiplicative comparison	4.OA.A.2
BACKWARD 2	Comparing fractions to 1	4.NF.A.2
FORWARD 1	Scale quantities in ratio tables	6.RP.A.3.a
FORWARD 2	Use a scale to find lengths in scale drawings	7.G.A.1

SET 33 • 5.NF.B.5.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.5.a

Without multiplying, how does $\frac{3}{4} \times 20$ compare to 20?

- A.** It is less than 20.
- B.** It is equal to 20.
- C.** It is 4 times 20.
- D.** It is greater than 20.

True or false?

$5 \times 3\frac{1}{2}$ is greater than 5.

A. True

B. False

Without multiplying, is $\frac{2}{3} \times 15$ greater than, less than, or equal to 15?

Answer: _____

Which product is greater than 8?

A. $\frac{9}{8} \times 8$

B. $\frac{1}{2} \times 8$

C. $\frac{7}{8} \times 8$

D. $\frac{8}{8} \times 8$

A photo is 6 inches wide. It is resized to $\frac{5}{4}$ of its width.

Is the new width greater than or less than 6 inches?

Answer: _____

SET 33 • 5.NF.B.5.a

BACKWARD 1

Multiplicative comparison • Grade 4 • Operations & Algebraic Thinking •
4.OA.A.2

True or false?

3×12 is 3 more than 12.

A. True

B. False

A dog weighs 4 times as much as a 6-pound cat.
How much does the dog weigh?

Weight: _____

Which number is 5 times as much as 7?

A. 57

B. 12

C. 2

D. 35

True or false?

2×9 is less than 9.

A. True

B. False

Fill in the blank.

42 is ____ times as much as 7.

Blank: _____

SET 33 • 5.NF.B.5.a

BACKWARD 2

Comparing fractions to 1 • Grade 4 • Number & Operations–Fractions •
4.NF.A.2

True or false?

$\frac{4}{5}$ is greater than 1.

A. True

B. False

Which fraction is less than 1?

A. $\frac{6}{7}$

B. $\frac{9}{9}$

C. $\frac{7}{6}$

D. $\frac{3}{2}$

Write $>$, $<$, or $=$ to compare.

$$\frac{8}{8} \text{ ____ } 1$$

Symbol: _____

True or false?

$$\frac{3}{5} > 1$$

A. True

B. False

Is $\frac{11}{10}$ greater than, less than, or equal to 1?

Answer: _____

SET 33 • 5.NF.B.5.a

FORWARD 1

Scale quantities in ratio tables • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

A recipe uses 2 cups of rice for every 3 cups of water.
How many cups of water are needed for 8 cups of rice?

Cups of water: _____

A ratio table starts with 4 : 10.

Which ratio could also be in the table?

- A.** 4 : 5
- B.** 2 : 5
- C.** 8 : 15
- D.** 10 : 4

The table shows equivalent ratios.

Find the missing value.

Red	3	6	?
Blue	5	10	25

Missing value: _____

True or false?

Multiplying both numbers in $6 : 9$ by $\frac{1}{3}$ gives $2 : 3$, an equivalent ratio.

- A.** True
- B.** False

A map shows 5 km for every 2 cm.

How many kilometers do 7 cm show?

Kilometers: _____

SET 33 • 5.NF.B.5.a

FORWARD 2

Use a scale to find lengths in scale drawings • Grade 7 • Geometry • 7.G.A.1

A scale drawing uses 1 inch = 4 feet. A wall is $3\frac{1}{2}$ inches long in the drawing.

How long is the real wall?

Length: _____

A room is 15 feet long.

How long is it on a drawing with a scale of 1 inch = 3 feet?

Length: _____

A drawing is reduced to $\frac{1}{2}$ of its size.

A 10-cm side becomes how long?

- A.** 2 cm
- B.** 20 cm
- C.** 10 cm
- D.** 5 cm

A drawing that is 4 cm by 6 cm is enlarged by a scale factor of $\frac{3}{2}$.

What are the new dimensions?

Dimensions: _____

True or false?

Using a scale factor less than 1 makes a drawing smaller.

A. True

B. False

Explain the effect of multiplying by a fraction greater than, less than, or equal to 1

MAIN	Grade 5 main question	5.NF.B.5.b
BACKWARD 1	Equivalent fractions	4.NF.A.1
BACKWARD 2	Fractions equal to 1	3.NF.A.3.c
FORWARD 1	Percents greater than and less than 100%	6.RP.A.3.c
FORWARD 2	Use a multiplier for percent increase and decrease	7.RP.A.3

SET 34 • 5.NF.B.5.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.5.b

True or false?

Multiplying 12 by $\frac{7}{6}$ gives a product greater than 12.

- A. True
- B. False

Which statement explains why $\frac{4}{5} \times 30$ is less than 30?

- A.** $\frac{4}{5}$ is less than 1.
- B.** $\frac{4}{5}$ is greater than 1.
- C.** 30 is greater than 1.
- D.** The product is a fraction.

Use the fact that $\frac{4}{4} = 1$ to explain why $\frac{2}{3} = \frac{8}{12}$.

Explanation: _____

Which expression has a value equal to 9?

A. $\frac{6}{5} \times 9$

B. $\frac{5}{6} \times 9$

C. $\frac{1}{5} \times 9$

D. $\frac{5}{5} \times 9$

Without multiplying, order the products from least to greatest.

$$\frac{3}{4} \times 10, \frac{4}{3} \times 10, \frac{4}{4} \times 10$$

Order: _____

SET 34 • 5.NF.B.5.b

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

True or false?

To write a fraction equal to $\frac{3}{5}$, you can multiply its numerator and denominator by the same number.

- A.** True
- B.** False

Find the missing number.

$$\frac{3}{5} = \frac{?}{20}$$

Missing number: _____

Which fraction is equivalent to $\frac{7}{8}$?

A. $\frac{14}{24}$

B. $\frac{21}{24}$

C. $\frac{7}{16}$

D. $\frac{8}{7}$

Find the missing number.

$$\frac{1}{3} = \frac{4}{?}$$

Missing number: _____

True or false?

$$\frac{2}{9} = \frac{4}{27}$$

A. True

B. False

SET 34 • 5.NF.B.5.b

BACKWARD 2

Fractions equal to 1 • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.c

True or false?

$$\frac{4}{4} = 4$$

A. True

B. False

Write 1 as a fraction with a denominator of 6.

Fraction: _____

Which fraction is equal to 1?

A. $\frac{0}{9}$

B. $\frac{9}{1}$

C. $\frac{1}{9}$

D. $\frac{9}{9}$

True or false?

$\frac{5}{5}$ is greater than 1.

A. True

B. False

How many eighths make 1 whole?

Eighths: _____

SET 34 • 5.NF.B.5.b

FORWARD 1

Percents greater than and less than 100% • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

True or false?

150% of 40 is less than 40.

A. True

B. False

What is 120% of 50?

Answer: _____

Which value is less than 80?

- A.** 200% of 80
- B.** 125% of 80
- C.** 75% of 80
- D.** 100% of 80

What is 100% of 37?

Answer: _____

What is 250% of 12?

Answer: _____

SET 34 • 5.NF.B.5.b

FORWARD 2

Use a multiplier for percent increase and decrease • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A \$60 price increases by 15%.

Which expression gives the new price?

A. 1.15×60

B. $60 + 15$

C. 0.85×60

D. 0.15×60

A \$90 jacket is 20% off.

Multiply by a decimal to find the sale price.

Sale price: _____

True or false?

Multiplying a price by 0.7 decreases it by 30%.

A. True

B. False

A population of 2,000 grows by 5%.

What is the new population?

Population: _____

Which multiplier shows a 40% decrease?

A. 0.6

B. 0.4

C. 1.4

D. 1.04

Solve real-world problems by multiplying fractions and mixed numbers

MAIN	Grade 5 main question	5.NF.B.6
BACKWARD 1	Word problems: a whole number times a fraction	4.NF.B.4.c
BACKWARD 2	Mixed numbers as fractions	4.NF.B.3.b
BACKWARD 3	Multiplying fractions	5.NF.B.4.a
FORWARD 1	Word problems with division of fractions	6.NS.A.1
FORWARD 2	Real-world problems with rational numbers	7.NS.A.3

SET 35 • 5.NF.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A recipe uses $\frac{3}{4}$ cup of oats per batch.

How many cups of oats are in $2\frac{1}{2}$ batches?

Cups: _____

A garden is 6 m long. $\frac{2}{3}$ of its length is planted with beans.

How many meters are planted with beans?

Meters: _____

Sam walks $\frac{5}{8}$ mile each day.

How far does he walk in 4 days?

A. $\frac{5}{32}$ mile

B. $\frac{9}{8}$ miles

C. $2\frac{1}{2}$ miles

D. $4\frac{5}{8}$ miles

A board is $3\frac{1}{3}$ feet long. Jo uses $\frac{3}{5}$ of the board.

How many feet does she use?

Feet: _____

A pitcher holds $1\frac{1}{2}$ liters. It is $\frac{2}{3}$ full.

How many liters are in the pitcher?

Liters: _____

SET 35 • 5.NF.B.6

BACKWARD 1

Word problems: a whole number times a fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.c

Each serving is $\frac{2}{3}$ cup.

How many cups are in 5 servings?

Cups: _____

True or false?

4 children each drink $\frac{3}{8}$ liter of milk. They drink $1\frac{1}{2}$ liters in all.

- A. True
- B. False

Each piece of ribbon is $\frac{5}{6}$ yard long.

How long are 3 pieces in all?

A. $\frac{5}{18}$ yard

B. $2\frac{1}{2}$ yards

C. $3\frac{5}{6}$ yards

D. $\frac{15}{18}$ yard

A bug crawls $\frac{3}{10}$ meter each minute.

How far does it crawl in 7 minutes?

Distance: _____

Each bag holds $\frac{1}{4}$ pound of nuts.

How many pounds are in 6 bags?

Pounds: _____

SET 35 • 5.NF.B.6

BACKWARD 2

Mixed numbers as fractions • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.b

Write $2\frac{1}{2}$ as a fraction.

Fraction: _____

Write $3\frac{1}{3}$ as a fraction.

Fraction: _____

True or false?

$$1\frac{1}{2} = \frac{11}{2}$$

A. True

B. False

Which fraction is equal to $4\frac{2}{5}$?

A. $\frac{22}{5}$

B. $\frac{42}{5}$

C. $\frac{11}{5}$

D. $\frac{24}{5}$

Write $\frac{15}{8}$ as a mixed number.

Mixed number: _____

SET 35 • 5.NF.B.6

BACKWARD 3

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{3}{4} \times \frac{5}{2}$$

Product: _____

Multiply.

$$\frac{2}{3} \times \frac{3}{2}$$

Product: _____

True or false?

$$\frac{10}{3} \times \frac{3}{5} = 2$$

A. True

B. False

$$\frac{3}{2} \times \frac{2}{3} = ?$$

A. $\frac{5}{6}$

B. 1

C. $\frac{6}{5}$

D. $\frac{9}{4}$

Multiply.

$$\frac{5}{8} \times 4$$

Product: _____

SET 35 • 5.NF.B.6

FORWARD 1

Word problems with division of fractions • Grade 6 • The Number System •
6.NS.A.1

How many $\frac{3}{4}$ -cup servings are in 6 cups of soup?

Servings: _____

A $\frac{7}{8}$ -mile trail is split into $\frac{1}{8}$ -mile sections.

How many sections are there?

Sections: _____

A rectangle has an area of $\frac{3}{4}$ square foot and a width of $\frac{1}{2}$ foot.

What is its length?

A. $\frac{3}{8}$ foot

B. $1\frac{1}{4}$ feet

C. $1\frac{1}{2}$ feet

D. $\frac{2}{3}$ foot

Ann has $2\frac{1}{2}$ pounds of clay. Each pot uses $\frac{5}{8}$ pound.

How many pots can she make?

Pots: _____

True or false?

$$\frac{2}{3} \div \frac{1}{6} = \frac{1}{9}$$

A. True

B. False

SET 35 • 5.NF.B.6

FORWARD 2

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A diver descends $\frac{3}{4}$ foot per second for $6\frac{1}{2}$ seconds.

What is the diver's change in depth?

Change: _____

The temperature was 2°F . It dropped $\frac{1}{2}^{\circ}\text{F}$ each hour for 9 hours.

What was the final temperature?

Temperature: _____

A stock loses $\frac{3}{8}$ dollar each day for 4 days.

What is the total change in value?

A. $-4\frac{3}{8}$ dollars

B. $-\frac{3}{32}$ dollar

C. $-1\frac{1}{2}$ dollars

D. $1\frac{1}{2}$ dollars

A tank holds $12\frac{1}{2}$ gallons. $\frac{2}{5}$ of the water drains out.

How many gallons drain out?

Gallons: _____

True or false?

$$-\frac{2}{3} \times 4\frac{1}{2} = -3$$

A. True

B. False

Multiply mixed numbers

MAIN

Grade 5 main question

5.NF.B.6

BACKWARD 1

Mixed numbers and fractions greater than 1

4.NF.B.3.b

BACKWARD 2

Multiplying fractions

5.NF.B.4.a

BACKWARD 3

Breaking apart a factor to multiply

3.OA.B.5

FORWARD 1

Volume of prisms with fractional edge lengths

6.G.A.2

FORWARD 2

Surface area and volume of prisms

7.G.B.6

SET 36 • 5.NF.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.6

Multiply.

$$2\frac{1}{2} \times 3$$

Product: _____

Multiply.

$$1\frac{1}{3} \times 1\frac{1}{2}$$

Product: _____

Multiply.

$$2\frac{1}{4} \times 1\frac{2}{3}$$

A. $3\frac{1}{6}$

B. $4\frac{1}{2}$

C. $2\frac{2}{12}$

D. $3\frac{3}{4}$

Multiply.

$$3\frac{1}{5} \times 2\frac{1}{2}$$

Product: _____

True or false?

$$1\frac{1}{2} \times 1\frac{1}{2} = 1\frac{1}{4}$$

A. True

B. False

SET 36 • 5.NF.B.6

BACKWARD 1

Mixed numbers and fractions greater than 1 • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write $1\frac{3}{4}$ as a fraction.

Fraction: _____

Write $\frac{13}{3}$ as a mixed number.

Mixed number: _____

True or false?

$$2\frac{2}{3} = \frac{22}{3}$$

A. True

B. False

Which mixed number is equal to $\frac{9}{2}$?

A. $2\frac{1}{9}$

B. $4\frac{1}{9}$

C. $9\frac{1}{2}$

D. $4\frac{1}{2}$

Write $5\frac{1}{2}$ as a fraction.

Fraction: _____

SET 36 • 5.NF.B.6

BACKWARD 2

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{7}{4} \times \frac{2}{3}$$

Product: _____

True or false?

$$\frac{5}{2} \times \frac{3}{2} = \frac{15}{2}$$

A. True

B. False

$$\frac{4}{3} \times \frac{9}{4} = ?$$

A. 3

B. 2

C. $\frac{36}{7}$

D. $\frac{13}{7}$

Multiply.

$$\frac{5}{3} \times 6$$

Product: _____

Multiply.

$$\frac{3}{2} \times \frac{3}{2}$$

Product: _____

SET 36 • 5.NF.B.6

BACKWARD 3

Breaking apart a factor to multiply • Grade 3 • Operations & Algebraic Thinking
• 3.OA.B.5

True or false?

$$4 \times (5 + 2) = 4 \times 5 + 4 \times 2$$

A. True

B. False

Fill in the blank.

$$6 \times 12 = 6 \times 10 + 6 \times \underline{\quad}$$

Blank: _____

Which expression is equal to 3×15 ?

A. $3 \times 10 + 5$

B. $3 + 10 \times 5$

C. $3 \times 10 \times 5$

D. $3 \times 10 + 3 \times 5$

True or false?

$$7 \times 13 = 7 \times 10 + 3$$

A. True

B. False

Fill in the blank.

$$8 \times (4 + 1) = 8 \times 4 + 8 \times \underline{\hspace{1cm}}$$

Blank: _____

SET 36 • 5.NF.B.6

FORWARD 1

Volume of prisms with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

A box is $2\frac{1}{2}$ ft by 2 ft by $1\frac{1}{2}$ ft.

What is its volume?

Volume: _____

A cube has edges that are $2\frac{1}{2}$ inches long.

What is its volume?

Volume: _____

A prism has a base area of $3\frac{1}{3}$ square cm and a height of $1\frac{1}{2}$ cm.

What is its volume?

A. $3\frac{1}{2}$ cubic cm

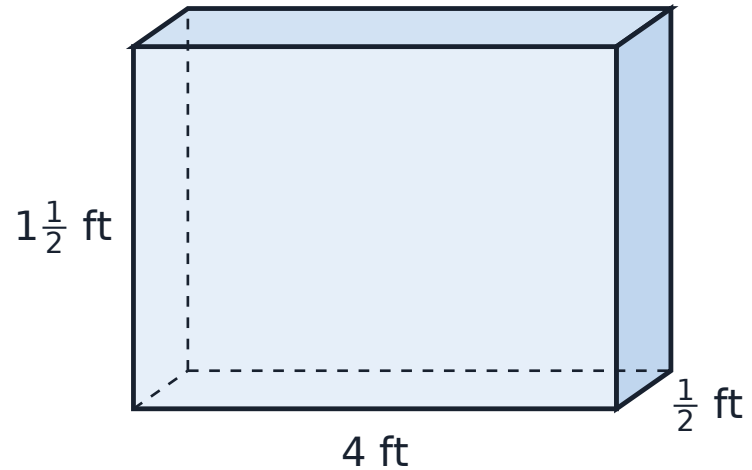
B. 10 cubic cm

C. $4\frac{5}{6}$ cubic cm

D. 5 cubic cm

Find the volume of the prism.

Volume: _____



True or false?

A box that is $\frac{1}{2}$ m by $\frac{1}{2}$ m by $\frac{1}{2}$ m has a volume of $\frac{1}{8}$ cubic meter.

A. True

B. False

SET 36 • 5.NF.B.6

FORWARD 2

Surface area and volume of prisms • Grade 7 • Geometry • 7.G.B.6

A rectangular prism is 4 cm by 3 cm by $2\frac{1}{2}$ cm.

What is its surface area?

Surface area: _____

A triangular prism has triangle bases with a base of 3 in. and a height of $2\frac{1}{2}$ in. The prism is 6 in. long.

What is its volume?

Volume: _____

A box is 5 in. by 4 in. by $1\frac{1}{2}$ in.

What is its volume?

- A.** 20 cubic inches
- B.** $10\frac{1}{2}$ cubic inches
- C.** 30 cubic inches
- D.** 300 cubic inches

A cube has edges that are $1\frac{1}{2}$ m long.

What is its surface area?

Surface area: _____

True or false?

A prism with a base area of $7\frac{1}{2}$ square feet and a height of 4 feet has a volume of $11\frac{1}{2}$ cubic feet.

A. True

B. False

Divide a unit fraction by a whole number

MAIN	Grade 5 main question	5.NF.B.7.a
BACKWARD 1	Unit fractions get smaller as the number of parts grows	3.NF.A.1
BACKWARD 2	Sharing equally	3.OA.A.2
BACKWARD 3	Multiplying unit fractions	5.NF.B.4.a
FORWARD 1	Divide fractions	6.NS.A.1
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 37 • 5.NF.B.7.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.a

Divide.

$$\frac{1}{3} \div 4$$

Quotient: _____

Divide.

$$\frac{1}{2} \div 5$$

Quotient: _____

$$\frac{1}{4} \div 2 = ?$$

A. $\frac{1}{2}$

B. $\frac{2}{4}$

C. $\frac{1}{8}$

D. 8

$\frac{1}{5}$ of a pan of brownies is shared equally by 3 friends.

What fraction of the whole pan does each friend get?

Fraction: _____

True or false?

$$\frac{1}{6} \div 3 = \frac{1}{2}$$

A. True

B. False

SET 37 • 5.NF.B.7.a

BACKWARD 1

Unit fractions get smaller as the number of parts grows • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

When a whole is cut into more equal parts, each part is smaller.

A. True

B. False

Which unit fraction is the smallest?

A. $\frac{1}{8}$

B. $\frac{1}{12}$

C. $\frac{1}{2}$

D. $\frac{1}{5}$

A bar is cut into 6 equal parts.

What fraction of the bar is one part?

Fraction: _____

True or false?

$$\frac{1}{3} < \frac{1}{4}$$

A. True

B. False

What fraction of the bar is shaded?



Fraction: _____

SET 37 • 5.NF.B.7.a

BACKWARD 2

Sharing equally • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

15 is shared equally among 5 groups.

How much is in each group?

Answer: _____

True or false?

Sharing 8 equally between 2 groups gives 4 in each group.

A. True

B. False

Which expression shows 21 shared equally among 7 people?

A. $21 \div 7$

B. $21 - 7$

C. $7 \div 21$

D. 21×7

Divide.

$$16 \div 4$$

Quotient: _____

36 is shared equally among 6 people.

How much does each person get?

Answer: _____

SET 37 • 5.NF.B.7.a

BACKWARD 3

Multiplying unit fractions • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{1}{3} \times \frac{1}{4}$$

Product: _____

True or false?

$$\frac{1}{2} \times \frac{1}{5} = \frac{1}{10}$$

A. True

B. False

$$\frac{1}{6} \times \frac{1}{2} = ?$$

A. $\frac{1}{8}$

B. $\frac{1}{3}$

C. $\frac{2}{6}$

D. $\frac{1}{12}$

Multiply.

$$\frac{1}{4} \times \frac{1}{4}$$

Product: _____

True or false?

$$\frac{1}{2} \times \frac{1}{3} = \frac{2}{5}$$

A. True

B. False

SET 37 • 5.NF.B.7.a

FORWARD 1

Divide fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{3}{4} \div 3$$

Quotient: _____

Divide.

$$\frac{2}{5} \div 4$$

Quotient: _____

Divide.

$$\frac{5}{6} \div \frac{1}{3}$$

A. $\frac{5}{18}$

B. $2\frac{1}{2}$

C. $\frac{6}{9}$

D. $\frac{1}{2}$

$\frac{3}{4}$ pound of trail mix is shared equally among 6 bags.

How much trail mix is in each bag?

Pounds: _____

True or false?

$$\frac{2}{3} \div 2 = 1\frac{1}{3}$$

A. True

B. False

SET 37 • 5.NF.B.7.a

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-\frac{1}{3} \div 4$$

Quotient: _____

Find the quotient.

$$\frac{3}{5} \div \left(-\frac{3}{10}\right)$$

Quotient: _____

Find the quotient.

$$\left(-\frac{1}{2}\right) \div \left(-\frac{1}{4}\right)$$

A. $\frac{1}{8}$

B. $-\frac{1}{8}$

C. -2

D. 2

Find the quotient.

$$-\frac{4}{9} \div \frac{2}{3}$$

Quotient: _____

True or false?

$$-\left(\frac{1}{2} \div 5\right) = \left(-\frac{1}{2}\right) \div 5$$

A. True

B. False

Divide a whole number by a unit fraction

MAIN

Grade 5 main question

5.NF.B.7.b

BACKWARD 1

Division as "how many groups?"

3.OA.A.2

BACKWARD 2

Unit fractions that make a whole

3.NF.A.1

FORWARD 1

Divide whole numbers by fractions

6.NS.A.1

FORWARD 2

Unit rates from quantities measured in fractions

7.RP.A.1

SET 38 • 5.NF.B.7.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.b

Divide.

$$4 \div \frac{1}{5}$$

Quotient: _____

Divide.

$$3 \div \frac{1}{6}$$

Quotient: _____

How many $\frac{1}{3}$ s are in 7?

A. 10

B. 21

C. $\frac{7}{3}$

D. $\frac{1}{21}$

Divide.

$$9 \div \frac{1}{4}$$

Quotient: _____

True or false?

$$5 \div \frac{1}{2} = 2\frac{1}{2}$$

A. True

B. False

SET 38 • 5.NF.B.7.b

BACKWARD 1

Division as "how many groups?" • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

How many groups of 3 are in 18?

Groups: _____

True or false?

$20 \div 5$ can mean "How many 5s are in 20?"

A. True

B. False

Pam puts 24 eggs into cartons of 6.

Which equation finds the number of cartons?

A. $6 \div 24 = \square$

B. $24 - 6 = \square$

C. $24 \times 6 = \square$

D. $24 \div 6 = \square$

How many 4s are in 32?

Answer: _____

A 30-inch ribbon is cut into 5-inch pieces.
How many pieces are there?

Pieces: _____

SET 38 • 5.NF.B.7.b

BACKWARD 2

Unit fractions that make a whole • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

How many $\frac{1}{4}$ s make 1 whole?

Answer: _____

True or false?

There are 3 thirds in 2 wholes.

A. True

B. False

How many $\frac{1}{5}$ s make 2 wholes?

- A.** 2
- B.** 7
- C.** 10
- D.** 5

How many $\frac{1}{8}$ s make 1 whole?

Answer: _____

True or false?

There are 2 halves in 2 wholes.

A. True

B. False

SET 38 • 5.NF.B.7.b

FORWARD 1

Divide whole numbers by fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$6 \div \frac{3}{4}$$

Quotient: _____

Divide.

$$4 \div \frac{2}{5}$$

Quotient: _____

How many $\frac{2}{3}$ -cup servings are in 8 cups?

A. 6

B. $\frac{1}{12}$

C. $\frac{16}{3}$

D. 12

Divide.

$$5 \div \frac{5}{6}$$

Quotient: _____

True or false?

$$3 \div \frac{3}{8} = \frac{9}{8}$$

A. True

B. False

SET 38 • 5.NF.B.7.b

FORWARD 2

Unit rates from quantities measured in fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A car uses $\frac{1}{4}$ gallon of gas to go 6 miles.

How many miles does it go per gallon?

Miles per gallon: _____

A machine makes 5 parts every $\frac{1}{3}$ minute.

How many parts does it make per minute?

Parts per minute: _____

Lee runs 2 miles in $\frac{1}{6}$ hour.

What is his speed?

- A.** $\frac{1}{12}$ mile per hour
- B.** 3 miles per hour
- C.** 8 miles per hour
- D.** 12 miles per hour

A hose fills 10 liters in $\frac{1}{2}$ minute.

How many liters does it fill per minute?

Liters per minute: _____

True or false?

Walking 3 miles in $\frac{3}{4}$ hour is a speed of 4 miles per hour.

- A.** True
- B.** False

Solve word problems that divide unit fractions and whole numbers

MAIN	Grade 5 main question	5.NF.B.7.c
BACKWARD 1	Division word problems	3.OA.A.3
BACKWARD 2	A fraction as division	5.NF.B.3
FORWARD 1	Word problems that divide fractions by fractions	6.NS.A.1
FORWARD 2	Real-world problems that divide rational numbers	7.NS.A.3

SET 39 • 5.NF.B.7.c

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.c

How many $\frac{1}{3}$ -cup servings of raisins are in 2 cups?

Servings: _____

3 friends share $\frac{1}{2}$ pound of chocolate equally.

How much chocolate does each friend get?

Pounds: _____

A 5-meter rope is cut into pieces that are $\frac{1}{4}$ meter long.

How many pieces are there?

A. 20

B. $\frac{1}{20}$

C. $\frac{5}{4}$

D. 9

$\frac{1}{8}$ of a pizza is shared equally by 2 people.

What fraction of the whole pizza does each person get?

Fraction: _____

True or false?

A 6-mile trail is split into $\frac{1}{2}$ -mile sections. There are 12 sections.

- A.** True
- B.** False

SET 39 • 5.NF.B.7.c

BACKWARD 1

Division word problems • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

Kim has 28 beads. Each bracelet uses 7 beads.
How many bracelets can she make?

Bracelets: _____

True or false?

If 32 students sit 4 at each table, there are 6 tables.

A. True

B. False

Raj shares 40 cards equally among 5 friends.

How many cards does each friend get?

- A.** 8
- B.** 200
- C.** 35
- D.** 45

A 27-foot fence is split into 9 equal sections.
How long is each section?

Length: _____

48 eggs are packed 6 to a carton.

How many cartons are filled?

Cartons: _____

SET 39 • 5.NF.B.7.c

BACKWARD 2

A fraction as division • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

Write $1 \div 4$ as a fraction.

Fraction: _____

True or false?

$$2 \div 3 = \frac{2}{3}$$

A. True

B. False

4 people share 1 pie equally.

How much pie does each person get?

A. $\frac{1}{4}$ pie

B. $\frac{4}{1}$ pies

C. $\frac{3}{4}$ pie

D. 4 pies

Write $\frac{1}{6}$ as a division expression.

Expression: _____

True or false?

5 people share 2 sandwiches equally, so each person gets $\frac{5}{2}$ sandwiches.

A. True

B. False

SET 39 • 5.NF.B.7.c

FORWARD 1

Word problems that divide fractions by fractions • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{3}{8}$ -pound bags can be filled from 3 pounds of rice?

Bags: _____

$\frac{2}{3}$ of a cake is shared equally by 4 people.

What fraction of the whole cake does each person get?

Fraction: _____

A path is $\frac{9}{10}$ mile long. It is split into sections that are $\frac{3}{10}$ mile long.

How many sections are there?

A. $\frac{27}{100}$

B. 6

C. $\frac{1}{3}$

D. 3

A rectangle has an area of 2 square yards and a length of $2\frac{1}{2}$ yards.

What is its width?

Width: _____

True or false?

$\frac{3}{4}$ cup of oil fills exactly 3 bottles that hold $\frac{1}{4}$ cup each.

- A. True
- B. False

SET 39 • 5.NF.B.7.c

FORWARD 2

Real-world problems that divide rational numbers • Grade 7 • The Number System • 7.NS.A.3

A diver descends 18 feet in $\frac{1}{2}$ minute at a constant rate.

What is the change in depth per minute?

Change per minute: _____

The temperature fell 5°F in $\frac{1}{4}$ hour.

At this rate, what is the change in temperature per hour?

Change per hour: _____

A bank account changes by $-\$45$ in $\frac{1}{3}$ month at a steady rate.

What is the change per month?

A. $\$135$

B. $-\$45$

C. $-\$15$

D. $-\$135$

Find the value.

$$-6 \div \frac{1}{3}$$

Value: _____

True or false?

$$-2 \div \frac{1}{4} = -\frac{1}{2}$$

A. True

B. False

Convert from a larger unit to a smaller unit

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Relative sizes of measurement units	4.MD.A.1
BACKWARD 2	Multiplying whole numbers	4.NBT.B.5
BACKWARD 3	Multiplying decimals by powers of 10	5.NBT.A.2
FORWARD 1	Use ratio reasoning to convert measurement units	6.RP.A.3.d
FORWARD 2	Unit rates with fractional quantities and different units	7.RP.A.1

SET 40 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

How many inches are in $4\frac{1}{2}$ feet?

Inches: _____

How many ounces are in 3 pounds?

Ounces: _____

A track is 2.5 kilometers long.

How many meters is that?

- A.** 250 m
- B.** 25,000 m
- C.** 2,500 m
- D.** 25 m

Convert.

6 yards = ____ feet

Feet: _____

A jug holds 3 gallons.

How many quarts is that?

Quarts: _____

SET 40 • 5.MD.A.1

BACKWARD 1

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many inches are in 1 foot?

Inches: _____

True or false?

1 pound = 16 ounces

A. True

B. False

How many quarts are in 1 gallon?

- A.** 2
- B.** 16
- C.** 8
- D.** 4

How many meters are in 1 kilometer?

Meters: _____

True or false?

1 yard = 12 inches

A. True

B. False

SET 40 • 5.MD.A.1

BACKWARD 2

Multiplying whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Multiply.

$$16 \times 3$$

Product: _____

Multiply.

$$1,000 \times 7$$

Product: _____

True or false?

$$3 \times 36 = 98$$

A. True

B. False

$$12 \times 9 = ?$$

- A.** 96
- B.** 118
- C.** 21
- D.** 108

Multiply.

$$60 \times 5$$

Product: _____

SET 40 • 5.MD.A.1

BACKWARD 3

Multiplying decimals by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

$$2.5 \times 1,000$$

Product: _____

True or false?

$$0.75 \times 100 = 7.5$$

A. True

B. False

$$4.2 \times 10 = ?$$

A. 0.42

B. 420

C. 42

D. 4.12

Multiply.

$$1.6 \times 1,000$$

Product: _____

Multiply.

$$0.3 \times 100$$

Product: _____

SET 40 • 5.MD.A.1

FORWARD 1

Use ratio reasoning to convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

There are 2.54 centimeters in 1 inch.

How many centimeters are in 5 inches?

Centimeters: _____

A car travels 3 miles every 4 minutes.

At this rate, how many miles does it travel in 1 hour?

Miles: _____

There are 3 feet in 1 yard.

How many yards are in 27 feet?

- A.** 81
- B.** 30
- C.** 9
- D.** 24

1 kilogram is about 2.2 pounds.

About how many pounds are in 15 kilograms?

Pounds: _____

True or false?

If 1 cup = 8 fluid ounces, then 5 cups = 13 fluid ounces.

- A. True
- B. False

SET 40 • 5.MD.A.1

FORWARD 2

Unit rates with fractional quantities and different units • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A snail moves 6 feet in $\frac{1}{2}$ hour.

What is its speed in feet per minute?

Speed: _____

A cyclist rides $\frac{3}{4}$ mile in 3 minutes.

What is the speed in miles per hour?

Speed: _____

A faucet drips $\frac{1}{2}$ cup every 5 minutes.

How many cups does it drip per hour?

A. $2\frac{1}{2}$

B. $\frac{1}{10}$

C. 6

D. 10

A plant grows $\frac{1}{4}$ inch per week.

How many inches does it grow in 52 weeks?

Inches: _____

True or false?

Running $1\frac{1}{2}$ miles in 15 minutes is a speed of 6 miles per hour.

- A. True
- B. False

Convert from a smaller unit to a larger unit

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Relative sizes of measurement units	4.MD.A.1
BACKWARD 2	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 3	Dividing by powers of 10	5.NBT.A.2
FORWARD 1	Use ratio reasoning to convert measurement units	6.RP.A.3.d
FORWARD 2	Equations for unit conversions	7.RP.A.2.c

SET 41 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

Convert.

30 inches = ____ feet

Feet: _____

Convert.

750 milliliters = ____ liters

Liters: _____

A rope is 45 feet long.

How many yards is that?

A. 15

B. 5

C. 135

D. 42

Convert.

40 ounces = ____ pounds

Pounds: _____

Convert.

1,350 grams = ____ kilograms

Kilograms: _____

SET 41 • 5.MD.A.1

BACKWARD 1

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many milliliters are in 1 liter?

Milliliters: _____

True or false?

1 kilogram = 1,000 grams

A. True

B. False

How many feet are in 1 yard?

A. 10

B. 3

C. 36

D. 12

How many cups are in 1 pint?

Cups: _____

True or false?

1 foot = 10 inches

A. True

B. False

SET 41 • 5.MD.A.1

BACKWARD 2

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$36 \div 3$$

Quotient: _____

Divide.

$$64 \div 4$$

Quotient: _____

True or false?

$$84 \div 7 = 14$$

A. True

B. False

$$90 \div 3 = ?$$

- A.** 33
- B.** 300
- C.** 27
- D.** 30

Divide.

$$5,000 \div 5$$

Quotient: _____

SET 41 • 5.MD.A.1

BACKWARD 3

Dividing by powers of 10 • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Divide.

$$750 \div 1,000$$

Quotient: _____

True or false?

$$1,350 \div 1,000 = 13.5$$

A. True

B. False

$$45 \div 100 = ?$$

- A.** 0.45
- B.** 4.5
- C.** 0.045
- D.** 4,500

Divide.

$$2,600 \div 1,000$$

Quotient: _____

Divide.

$$80 \div 100$$

Quotient: _____

SET 41 • 5.MD.A.1

FORWARD 1

Use ratio reasoning to convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

There are 12 inches in 1 foot.

How many feet are in 102 inches?

Feet: _____

There are 1,000 meters in 1 kilometer.

How many kilometers are in 4,200 meters?

Kilometers: _____

A recipe needs 10 cups of milk. There are 4 cups in 1 quart.

How many quarts of milk are needed?

A. 40

B. 6

C. $2\frac{1}{2}$

D. 14

A trail is 5 kilometers long. 1 kilometer is about 0.62 mile.

About how many miles long is the trail?

Miles: _____

True or false?

There are 60 minutes in 1 hour, so 150 minutes = $1\frac{1}{2}$ hours.

A. True

B. False

SET 41 • 5.MD.A.1

FORWARD 2

Equations for unit conversions • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.2.c

There are 2.54 centimeters in 1 inch.

Write an equation for the number of centimeters c in i inches.

Equation: _____

Write an equation for the number of feet f in y yards.

Equation: _____

Which equation converts k kilograms to g grams?

A. $k = 1,000g$

B. $g = 1,000k$

C. $g = k \div 1,000$

D. $g = k + 1,000$

Write an equation for the number of pounds p in z ounces.

Equation: _____

True or false?

The equation $m = 60h$ gives the number of minutes m in h hours.

- A. True
- B. False

Solve multistep problems that involve converting units

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Word problems with measurement	4.MD.A.2
BACKWARD 2	Relative sizes of measurement units	4.MD.A.1
FORWARD 1	Convert units to compare rates	6.RP.A.3.d
FORWARD 2	Multistep ratio problems	7.RP.A.3

SET 42 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

Ana has a 3-yard ribbon. She cuts off 4 feet.
How many feet of ribbon are left?

Feet: _____

A recipe needs 2 pounds of flour. Jo has 20 ounces.
How many more ounces does she need?

Ounces: _____

A runner ran 1.2 km and then 850 m.

How many meters did she run in all?

- A.** 1,050 m
- B.** 2,050 m
- C.** 9,700 m
- D.** 862 m

A jug holds 2 liters of water. Six glasses of 250 milliliters each are poured from it.

How many milliliters are left in the jug?

Milliliters: _____

A board is 8 feet long. It is cut into pieces that are 16 inches long.

How many pieces are there?

Pieces: _____

SET 42 • 5.MD.A.1

BACKWARD 1

Word problems with measurement • Grade 4 • Measurement & Data •
4.MD.A.2

A movie starts at 2:15 and lasts 1 hour 40 minutes.
What time does it end?

Time: _____

True or false?

3 quarters and 2 dimes make 85 cents.

A. True

B. False

Jake has 2 meters of string. He uses 75 cm.
How many centimeters are left?

- A.** 1.25 cm
- B.** 75 cm
- C.** 125 cm
- D.** 275 cm

A bag holds 3 kilograms of apples.

How many grams is that?

Grams: _____

Liz runs 400 meters four times.

How many meters does she run in all?

Meters: _____

SET 42 • 5.MD.A.1

BACKWARD 2

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many ounces are in 2 pounds?

Ounces: _____

True or false?

1 gallon = 4 quarts

A. True

B. False

How many centimeters are in 1 meter?

- A.** 1,000
- B.** 100
- C.** 10
- D.** 12

How many feet are in 3 yards?

Feet: _____

True or false?

1 liter = 100 milliliters

A. True

B. False

SET 42 • 5.MD.A.1

FORWARD 1

Convert units to compare rates • Grade 6 • Ratios & Proportional Relationships
• 6.RP.A.3.d

A car uses 2 gallons of gas to drive 64 miles. There are 4 quarts in 1 gallon.

How many miles does the car drive per quart?

Miles: _____

Mo walks 3 kilometers in 40 minutes.

At this rate, how many meters does he walk per minute?

Meters per minute: _____

Fabric costs \$6 per yard.

How much do 12 feet of fabric cost?

A. \$72

B. \$18

C. \$24

D. \$2

A 24-ounce jar of juice costs \$3. There are 16 ounces in 1 pound.

What is the cost per pound?

Cost per pound: _____

True or false?

60 miles per hour is the same speed as 1 mile per minute.

- A. True
- B. False

SET 42 • 5.MD.A.1

FORWARD 2

Multistep ratio problems • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.3

A 2-liter bottle of soda costs \$1.80. A 500-milliliter can costs \$0.60.

Which is the better buy per liter?

Better buy: _____

A map scale is 1 cm : 2 km. Two towns are 7.5 cm apart on the map. A car drives 60 km per hour.

How many minutes does the drive take?

Minutes: _____

One gallon of paint covers 350 square feet.

How many gallons are needed to cover 1,400 square feet?

A. 1,050

B. 3

C. 4

D. 5

A recipe for 4 people uses 600 grams of pasta.
How many kilograms are needed for 10 people?

Kilograms: _____

True or false?

At 5 miles per hour, a 2.5-mile walk takes 50 minutes.

- A. True
- B. False

Make a line plot of measurements in fractions of a unit

MAIN	Grade 5 main question	5.MD.B.2
BACKWARD 1	Reading a line plot with fractions	4.MD.B.4
BACKWARD 2	Fractions on a number line	3.NF.A.2
BACKWARD 3	Line plots of whole-number measurements	2.MD.D.9
FORWARD 1	Display numerical data in a dot plot	6.SP.B.4
FORWARD 2	Compare two distributions shown in dot plots	7.SP.B.3

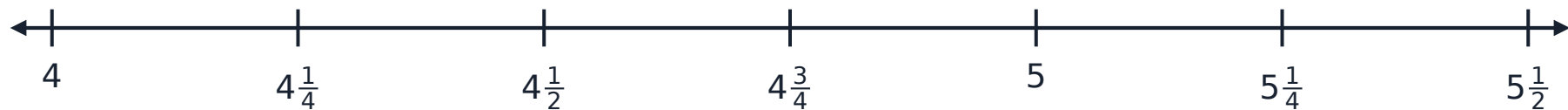
SET 43 • 5.MD.B.2

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.B.2

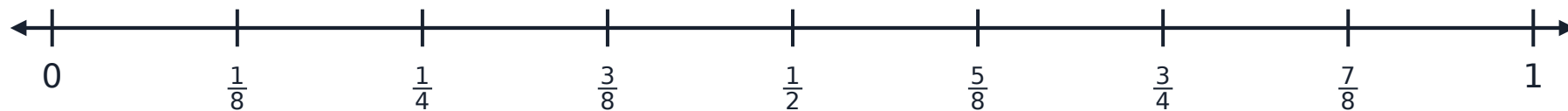
Pencil lengths (inches): $4\frac{1}{2}$, 5, $4\frac{1}{2}$, $5\frac{1}{4}$, $4\frac{3}{4}$, 5, $4\frac{1}{2}$

Make a line plot of the data.



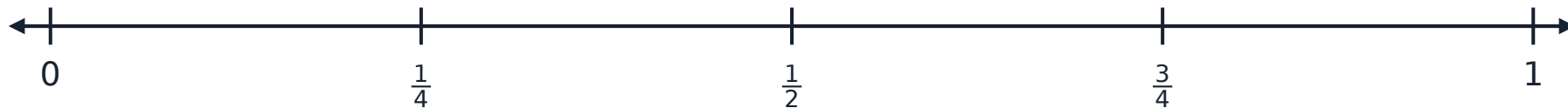
Rainfall (inches): $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{8}$, $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{4}$, $\frac{1}{2}$

Make a line plot of the data.



Apple weights (pounds): $\frac{3}{4}$, $\frac{1}{2}$, $\frac{3}{4}$, 1 , $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{4}$

Make a line plot of the data.



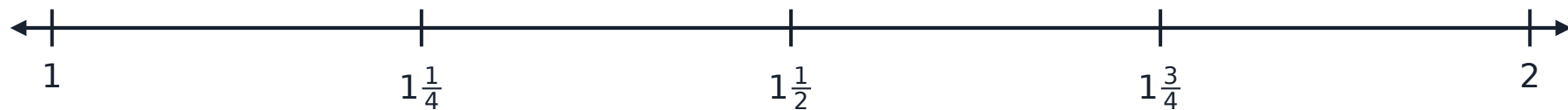
Lengths (inches): $\frac{1}{8}$, $\frac{3}{8}$, $\frac{3}{8}$, $\frac{5}{8}$, $\frac{3}{8}$

In a line plot, how many X's are drawn at $\frac{3}{8}$?

X's: _____

Water in each bottle (liters): $1\frac{1}{2}$, 2, $1\frac{1}{4}$, $1\frac{1}{2}$, $1\frac{3}{4}$, 2, $1\frac{1}{2}$

Make a line plot of the data.



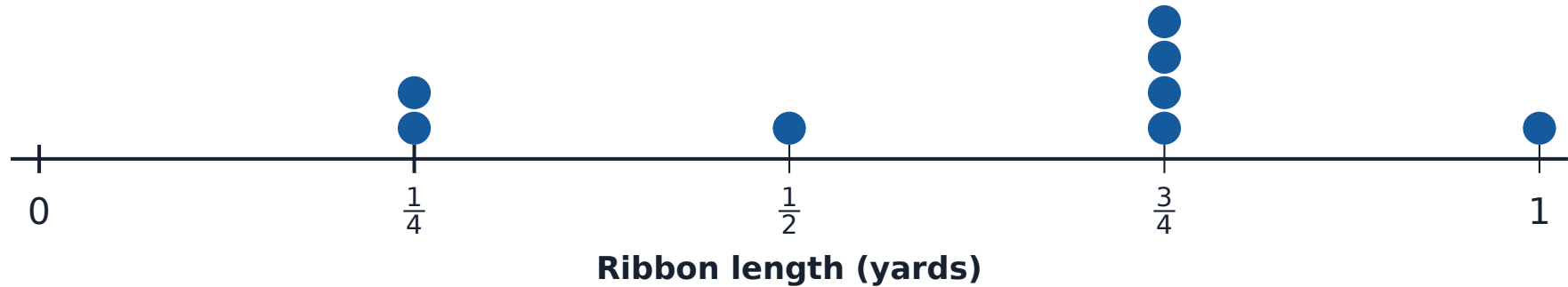
SET 43 • 5.MD.B.2

BACKWARD 1

Reading a line plot with fractions • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows ribbon lengths.

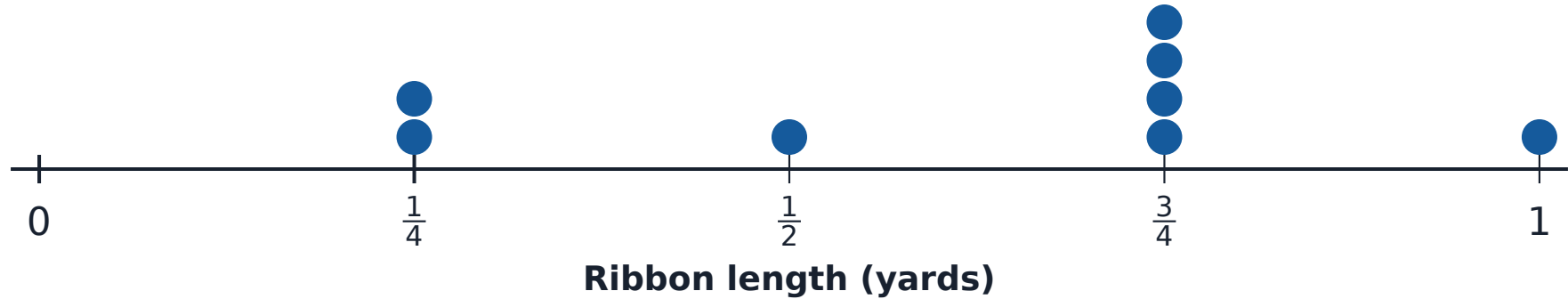
How many ribbons are $\frac{3}{4}$ yard long?



Ribbons: _____

True or false?

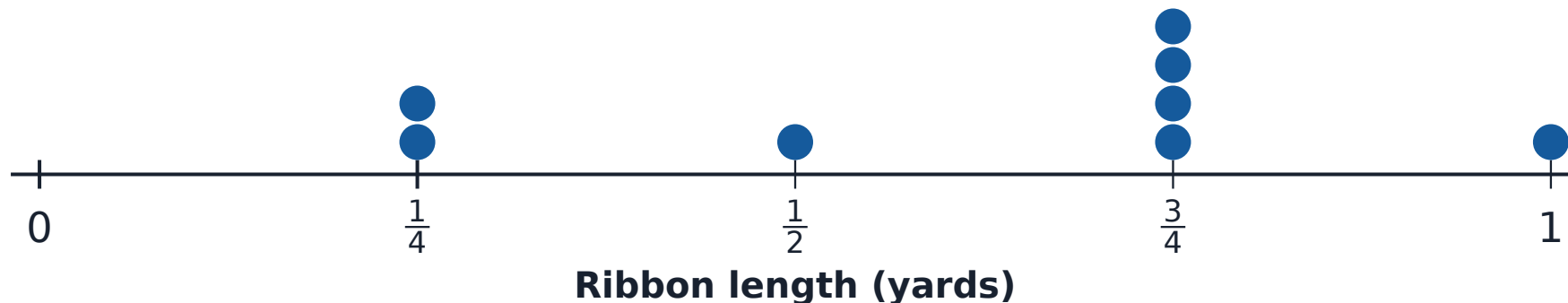
Two ribbons are $\frac{1}{4}$ yard long.



A. True

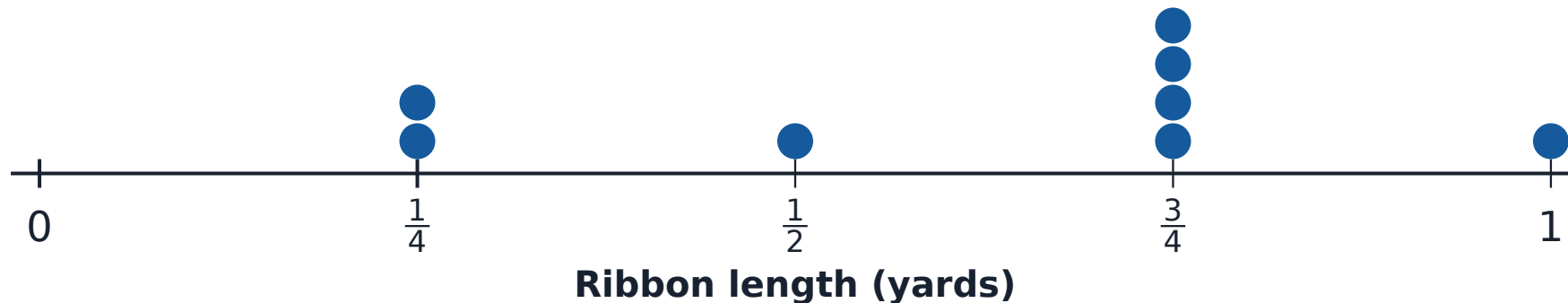
B. False

Which ribbon length is the most common?



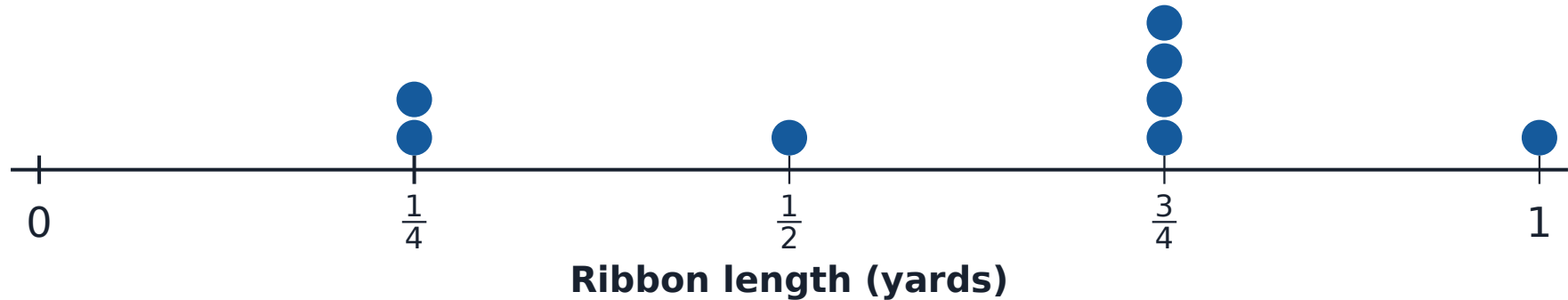
- A.** 1 yard **B.** $\frac{1}{2}$ yard **C.** $\frac{1}{4}$ yard **D.** $\frac{3}{4}$ yard

How many ribbons were measured?



Ribbons: _____

What is the difference between the longest and the shortest ribbon?



Difference: _____

SET 43 • 5.MD.B.2

BACKWARD 2

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction is at point P?



P = _____

True or false?

$\frac{2}{4}$ and $\frac{1}{2}$ are at the same point on a number line.

A. True

B. False

Into how many equal parts is the space from 0 to 1 divided?



A. 5

B. 7

C. 6

D. 1

What number is at point Q?



Q = _____

True or false?

On a number line from 0 to 1 marked in eighths, $\frac{6}{8}$ is one mark to the left of 1.

- A. True
- B. False

SET 43 • 5.MD.B.2

BACKWARD 3

Line plots of whole-number measurements • Grade 2 • Measurement & Data •
2.MD.D.9

Lengths (inches): 5, 6, 6, 8, 6

In a line plot, how many X's are drawn at 6?

X's: _____

True or false?

A line plot shows each measurement as a mark above a number line.

- A.** True
- B.** False

Which number line would you use for a line plot of 3, 4, 4, 7?

- A.** A number line from 0 to 8
- B.** A number line from 0 to 3
- C.** A number line from 5 to 6
- D.** A number line from 10 to 20

Lengths (cm): 2, 4, 4, 5

Make a line plot of the data.



Lengths (feet): 9, 10, 10, 10, 12

Which length gets the most X's?

Length: _____

SET 43 • 5.MD.B.2

FORWARD 1

Display numerical data in a dot plot • Grade 6 • Statistics & Probability •
6.SP.B.4

Hours of sleep: 8, 9, 9, 10, 8, 9, 7

Make a dot plot of the data.



Number of siblings: 0, 1, 1, 2, 3, 1, 0, 2

Make a dot plot of the data.



Which number line fits a dot plot of 12, 15, 15, 18, 20?

- A.** A number line from 0 to 5
- B.** A number line from 20 to 40
- C.** A number line from 10 to 20
- D.** A number line from 15 to 18

Data: 4, 6, 6, 6, 7, 9

In a dot plot, how many dots go at 6?

Dots: _____

True or false?

A dot plot shows every value in a data set.

- A. True
- B. False

SET 43 • 5.MD.B.2

FORWARD 2

Compare two distributions shown in dot plots • Grade 7 • Statistics & Probability • 7.SP.B.3

About how many inches greater is the center of Plot B than the center of Plot A?



Plot A: plant heights (inches)



Plot B: plant heights (inches)

Inches: _____

True or false?

The two plots have about the same spread.



Plot A: plant heights (inches)



Plot B: plant heights (inches)

A. True

B. False

Which statement is true?



Plot A: plant heights (inches)



Plot B: plant heights (inches)

- A.** Plants in Plot A are typically taller. **B.** Plot A has much more spread.
C. The centers are equal. **D.** Plants in Plot B are typically taller.

What is the range of the heights in Plot A?



Plot A: plant heights (inches)



Plot B: plant heights (inches)

Range: _____

True or false?

The two dot plots overlap.



Plot A: plant heights (inches)



Plot B: plant heights (inches)

A. True

B. False

Use operations on fractions to solve problems with line plot data

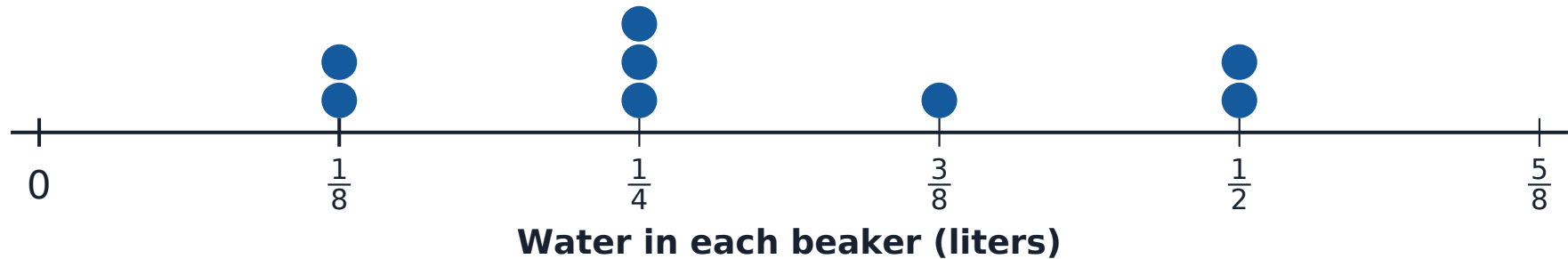
MAIN	Grade 5 main question	5.MD.B.2
BACKWARD 1	Solving problems from a line plot with like denominators	4.MD.B.4
BACKWARD 2	Adding and subtracting fractions with unlike denominators	5.NF.A.1
BACKWARD 3	Sharing a total equally	5.NF.B.3
FORWARD 1	Find the mean of a data set	6.SP.B.5.c
FORWARD 2	Compare two populations using means	7.SP.B.4

SET 44 • 5.MD.B.2

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.B.2

The line plot shows the water in some beakers.
What is the total amount of water?



Total: _____

If all of the water were shared equally among the beakers, how much would each beaker hold?



Liters: _____

How much more water is in the fullest beaker than in the least full beaker?



- A.** $\frac{1}{2}$ liter **B.** $\frac{1}{8}$ liter **C.** $\frac{5}{8}$ liter **D.** $\frac{3}{8}$ liter

What is the total amount of water in the beakers that hold $\frac{1}{4}$ liter each?



Total: _____

True or false?

The beakers that hold $\frac{1}{2}$ liter each contain 1 liter in all.



A. True

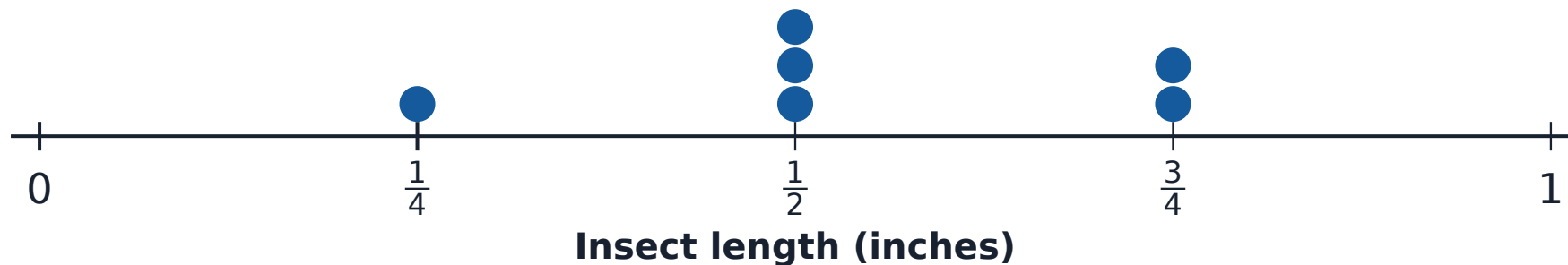
B. False

SET 44 • 5.MD.B.2

BACKWARD 1

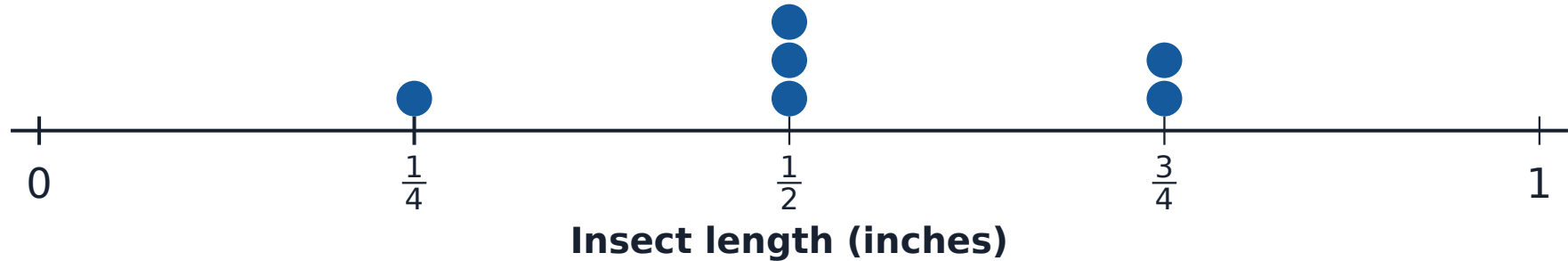
Solving problems from a line plot with like denominators • Grade 4 •
Measurement & Data • 4.MD.B.4

How many insects are $\frac{1}{2}$ inch long?



Insects: _____

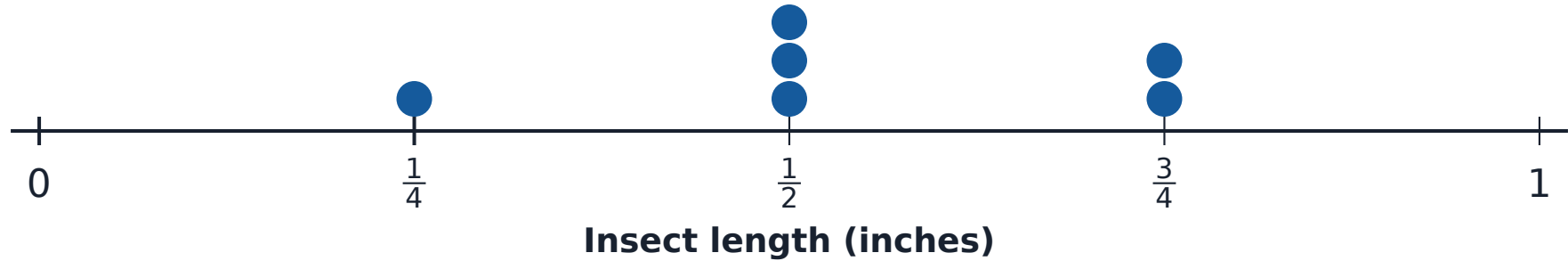
What is the difference between the longest and the shortest insect?



Difference: _____

True or false?

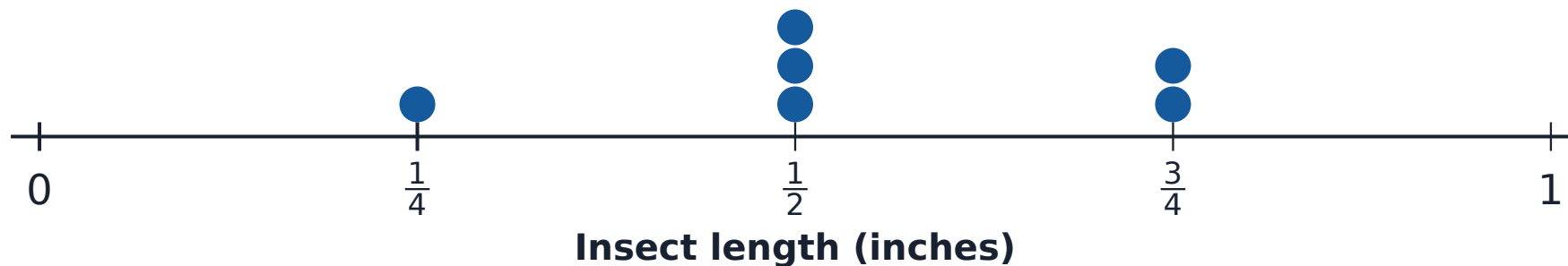
The total length of the $\frac{3}{4}$ -inch insects is $1\frac{1}{2}$ inches.



A. True

B. False

How many insects were measured?



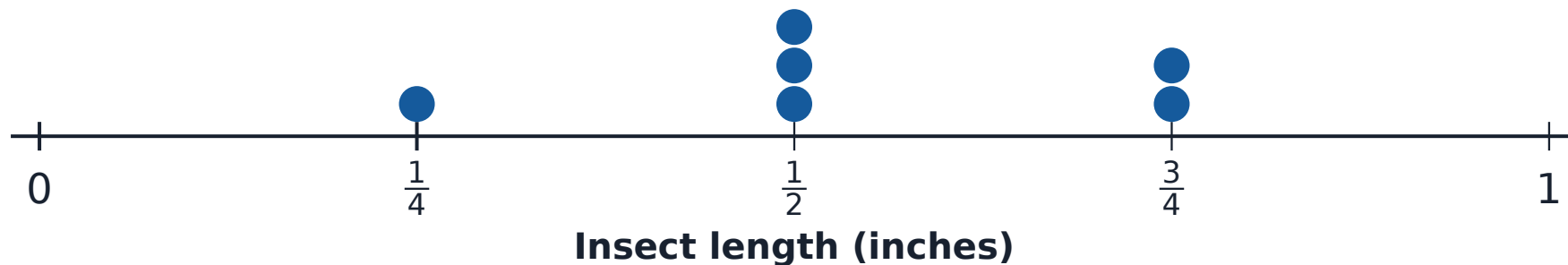
A. 4

B. 3

C. 5

D. 6

What is the total length of the $\frac{1}{2}$ -inch insects?



Total: _____

SET 44 • 5.MD.B.2

BACKWARD 2

Adding and subtracting fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$\frac{1}{8} + \frac{1}{4}$$

Sum: _____

Add.

$$\frac{3}{8} + \frac{1}{2}$$

Sum: _____

True or false?

$$\frac{1}{4} + \frac{1}{4} + \frac{1}{8} = \frac{3}{16}$$

A. True

B. False

$$\frac{1}{2} + \frac{3}{8} + \frac{1}{4} = ?$$

A. $1\frac{3}{8}$

B. $\frac{7}{8}$

C. $1\frac{1}{8}$

D. $\frac{5}{14}$

Subtract.

$$\frac{1}{2} - \frac{1}{8}$$

Difference: _____

SET 44 • 5.MD.B.2

BACKWARD 3

Sharing a total equally • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

3 liters of water are shared equally among 8 beakers.
How much water does each beaker get?

Liters: _____

True or false?

5 cups shared equally among 4 jars gives $\frac{4}{5}$ cup per jar.

- A.** True
- B.** False

2 pounds of clay are shared equally by 6 students.

How much clay does each student get?

A. $\frac{1}{3}$ pound

B. $\frac{1}{6}$ pound

C. 3 pounds

D. $\frac{2}{3}$ pound

7 meters of rope are cut into 4 equal pieces.

How long is each piece?

Length: _____

Write $9 \div 4$ as a mixed number.

Mixed number: _____

SET 44 • 5.MD.B.2

FORWARD 1

Find the mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

3, 5, 6, 8, 8

Mean: _____

Find the mean.

12, 15, 21

Mean: _____

Four friends have 2, 4, 5, and 9 marbles. They share the marbles equally.

How many marbles does each friend get?

A. 4.5

B. 5

C. 20

D. 4

The dot plot shows the goals scored in 6 games.
What is the mean number of goals?



Mean: _____

True or false?

The mean of 4, 4, and 10 is 4.

A. True

B. False

SET 44 • 5.MD.B.2

FORWARD 2

Compare two populations using means • Grade 7 • Statistics & Probability •
7.SP.B.4

Random samples: Sample A has a mean of 15 minutes.
Sample B has a mean of 22 minutes.

How much greater is the mean of Sample B?

Difference: _____

Two random samples of plant heights have means of 18 cm and 24 cm. Both samples have similar variability.

Which conclusion is best?

- A.** The populations are the same.
- B.** Plants in the second population tend to be taller.
- C.** Plants in the first population tend to be taller.
- D.** No conclusion is possible.

True or false?

Comparing the means of random samples is one way to compare two populations.

- A.** True
- B.** False

In random samples, Class X sleeps a mean of 7.5 hours and Class Y sleeps a mean of 8.25 hours.

Which class probably sleeps more?

Class: _____

Sample A has a mean of 40 and a MAD of 5. Sample B has a mean of 50 and a MAD of 5.

The difference in means is how many times the MAD?

Times the MAD: _____

Understand a unit cube and a cubic unit of volume

MAIN	Grade 5 main question	5.MD.C.3.a
BACKWARD 1	Unit squares and square units	3.MD.C.5.a
BACKWARD 2	Faces, edges, and vertices of a cube	2.G.A.1
FORWARD 1	Volume of cubes with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms	7.G.B.6

SET 45 • 5.MD.C.3.a

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.3.a

True or false?

A cube with edges that are 1 unit long is called a unit cube.

- A.** True
- B.** False

What is the volume of a unit cube?

- A.** 1 square unit
- B.** 6 square units
- C.** 1 cubic unit
- D.** 3 cubic units

What is the volume of a cube with edges that are 1 cm long?

Volume: _____

Which object has a volume closest to 1 cubic inch?

- A.** A shoebox
- B.** A sugar cube
- C.** A bathtub
- D.** A school bus

True or false?

Volume measures the amount of space inside a solid figure.

- A.** True
- B.** False

SET 45 • 5.MD.C.3.a

BACKWARD 1

Unit squares and square units • Grade 3 • Measurement & Data • 3.MD.C.5.a

True or false?

A square with sides that are 1 unit long is called a unit square.

A. True

B. False

What is the area of a unit square?

- A.** 1 square unit
- B.** 1 cubic unit
- C.** 2 square units
- D.** 4 square units

How many unit squares cover a 3-by-3 square?

Unit squares: _____

True or false?

Area is measured in cubic units.

A. True

B. False

A square has sides that are 1 inch long.

What is its area?

Area: _____

SET 45 • 5.MD.C.3.a

BACKWARD 2

Faces, edges, and vertices of a cube • Grade 2 • Geometry • 2.G.A.1

How many faces does a cube have?

Faces: _____

True or false?

All the faces of a cube are squares.

A. True

B. False

How many edges does a cube have?

- A.** 6
- B.** 12
- C.** 4
- D.** 8

How many vertices (corners) does a cube have?

Vertices: _____

True or false?

All the edges of a cube are the same length.

A. True

B. False

SET 45 • 5.MD.C.3.a

FORWARD 1

Volume of cubes with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

What is the volume of a cube with edges that are $\frac{1}{2}$ inch long?

Volume: _____

How many cubes with $\frac{1}{2}$ -inch edges fit in a cube with 1-inch edges?

Cubes: _____

A cube has edges that are $\frac{1}{3}$ foot long.

What is its volume?

A. $\frac{1}{3}$ cubic foot

B. $\frac{1}{9}$ cubic foot

C. $\frac{1}{27}$ cubic foot

D. 1 cubic foot

A box is packed with 16 cubes that each have $\frac{1}{2}$ -cm edges.

What is the volume of the box?

Volume: _____

True or false?

Four cubes with $\frac{1}{2}$ -unit edges have a total volume of 1 cubic unit.

- A.** True
- B.** False

SET 45 • 5.MD.C.3.a

FORWARD 2

Volume of right prisms • Grade 7 • Geometry • 7.G.B.6

A prism has a base area of 24 square cm and a height of 5 cm.

What is its volume?

Volume: _____

A cube has edges that are 3 m long.

What is its volume?

Volume: _____

A triangular prism has a triangle base with an area of 6 square inches. The prism is 10 inches long.

What is its volume?

A. 60 cubic inches

B. 30 cubic inches

C. 16 cubic inches

D. 600 cubic inches

A box is 2.5 cm by 4 cm by 6 cm.

What is its volume?

Volume: _____

True or false?

Volume is measured in cubic units.

A. True

B. False

A solid packed with n unit cubes has a volume of n cubic units

MAIN

Grade 5 main question

5.MD.C.3.b

BACKWARD 1

Measuring area by counting unit squares

3.MD.C.6

BACKWARD 2

Arrays and repeated addition

2.OA.C.4

FORWARD 1

Pack prisms with fractional unit cubes

6.G.A.2

FORWARD 2

Solve real-world volume problems

7.G.B.6

SET 46 • 5.MD.C.3.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.3.b

A box is filled with 30 unit cubes with no gaps or overlaps.

What is its volume?

Volume: _____

True or false?

If unit cubes fill a box but leave gaps, the number of cubes is less than the volume of the box.

- A. True
- B. False

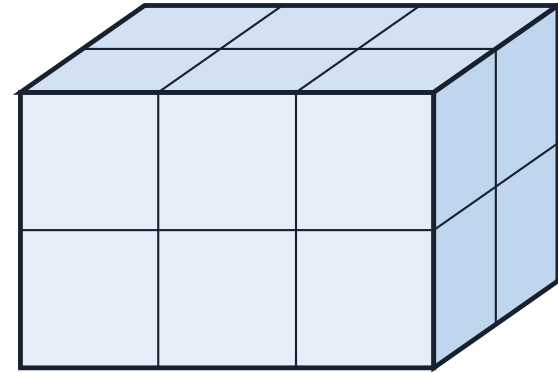
A solid is built from 18 unit cubes with no gaps or overlaps.

What is its volume?

- A.** 18 square units
- B.** 18 cubic units
- C.** 54 cubic units
- D.** 6 cubic units

Each cube is 1 cubic unit.

What is the volume of the solid?



Volume: _____

A tower is made of 9 cubes stacked with no gaps. Each cube is 1 cubic inch.

What is the volume of the tower?

Volume: _____

SET 46 • 5.MD.C.3.b

BACKWARD 1

Measuring area by counting unit squares • Grade 3 • Measurement & Data •
3.MD.C.6

A shape is covered by 12 unit squares with no gaps or overlaps.

What is its area?

Area: _____

True or false?

Overlapping squares give a wrong count for area.

- A. True
- B. False

A rectangle is covered by 20 unit squares.

What is its area?

- A.** 10 square units
- B.** 40 square units
- C.** 20 cubic units
- D.** 20 square units

Each square is 1 square cm. A rectangle is covered by 15 of these squares.

What is its area?

Area: _____

True or false?

Area can be found by counting the unit squares that cover a shape.

- A. True
- B. False

SET 46 • 5.MD.C.3.b

BACKWARD 2

Arrays and repeated addition • Grade 2 • Operations & Algebraic Thinking •
2.OA.C.4

An array has 4 rows with 5 in each row.

How many are there in all?

Total: _____

True or false?

3 rows of 6 make 9.

A. True

B. False

Which equation matches an array with 2 rows of 7?

A. $2 + 7 = 9$

B. $7 + 7 = 14$

C. $7 - 2 = 5$

D. $2 + 2 = 4$

An array has 5 rows with 3 in each row.

How many are there in all?

Total: _____

Write an addition equation for 3 rows of 4.

Equation: _____

SET 46 • 5.MD.C.3.b

FORWARD 1

Pack prisms with fractional unit cubes • Grade 6 • Geometry • 6.G.A.2

A box is packed with cubes that have $\frac{1}{2}$ -inch edges. It is 4 cubes long, 2 cubes wide, and 2 cubes tall.

What is the volume of the box?

Volume: _____

How many cubes with $\frac{1}{4}$ -inch edges fill a cube with 1-inch edges?

Cubes: _____

A prism is packed with 27 cubes that each have $\frac{1}{3}$ -foot edges.

What is its volume?

A. 27 cubic feet

B. 9 cubic feet

C. 1 cubic foot

D. 3 cubic feet

A box is $1\frac{1}{2}$ in. by 1 in. by $\frac{1}{2}$ in.

How many cubes with $\frac{1}{2}$ -inch edges fill it?

Cubes: _____

True or false?

40 cubes with $\frac{1}{2}$ -unit edges have a total volume of 20 cubic units.

- A. True
- B. False

SET 46 • 5.MD.C.3.b

FORWARD 2

Solve real-world volume problems • Grade 7 • Geometry • 7.G.B.6

A fish tank is 30 cm by 20 cm by 25 cm.

What is its volume?

Volume: _____

A prism has a volume of 96 cubic inches and a base area of 12 square inches.

What is its height?

Height: _____

A triangular prism has a triangle base with a base of 4 m and a height of 3 m. The prism is 7 m long.

What is its volume?

A. 84 cubic meters

B. 42 cubic meters

C. 21 cubic meters

D. 14 cubic meters

A storage box is 1.5 m by 1 m by 0.8 m.

What is its volume?

Volume: _____

True or false?

Two prisms with the same base area and the same height have the same volume.

A. True

B. False

Measure volume by counting unit cubes

MAIN	Grade 5 main question	5.MD.C.4
BACKWARD 1	Counting the squares in one layer	3.MD.C.7.a
BACKWARD 2	Equal groups	3.OA.A.1
FORWARD 1	Volume with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms	7.G.B.6

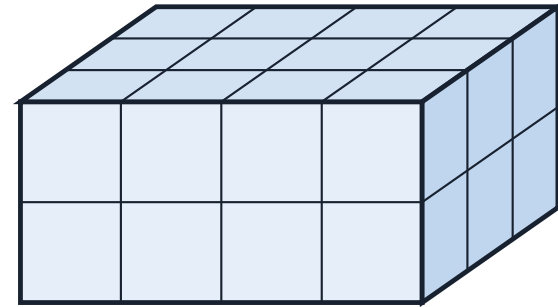
SET 47 • 5.MD.C.4

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.4

Each cube is 1 cubic cm.

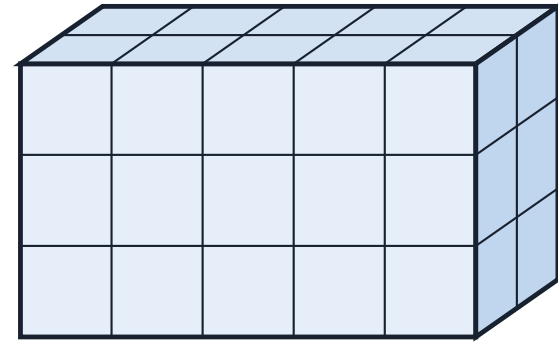
What is the volume of the prism?



Volume: _____

Each cube is 1 cubic inch.

What is the volume of the prism?

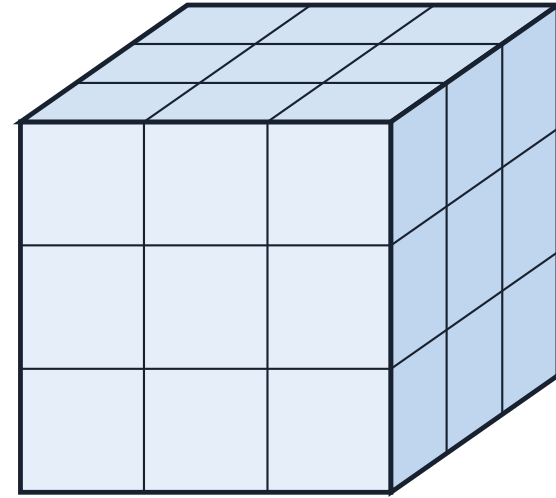


Volume: _____

Each cube is 1 cubic foot.

What is the volume of the prism?

- A.** 18 cubic feet **B.** 54 cubic feet
C. 27 cubic feet **D.** 9 cubic feet



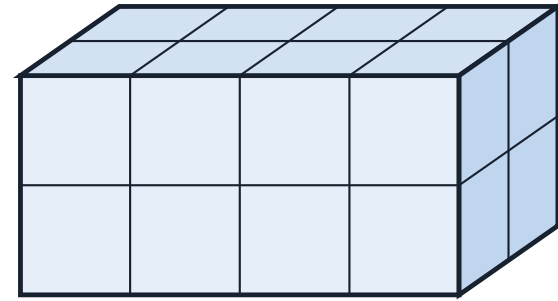
A box is filled with number cubes. The bottom layer has 6 rows of 4 cubes, and there are 3 layers.

What is the volume of the box, measured in number cubes?

Volume: _____

True or false?

Each cube is 1 cubic cm.
The volume of the prism is
16 cubic cm.



- A.** True
- B.** False

SET 47 • 5.MD.C.4

BACKWARD 1

Counting the squares in one layer • Grade 3 • Measurement & Data •
3.MD.C.7.a

The bottom layer of a box has 5 rows of 4 cubes.
How many cubes are in the layer?

Cubes: _____

True or false?

A layer with 3 rows of 6 cubes has 18 cubes.

A. True

B. False

How many unit squares cover a 4-by-6 rectangle?

- A.** 10
- B.** 46
- C.** 24
- D.** 20

A rectangle is tiled with 7 rows of 3 unit squares.
How many squares are there?

Squares: _____

True or false?

The number of unit squares that tile a rectangle is its length times its width.

- A. True
- B. False

SET 47 • 5.MD.C.4

BACKWARD 2

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

There are 3 layers with 12 cubes in each layer.
How many cubes are there in all?

Cubes: _____

True or false?

4 layers of 10 cubes is 14 cubes.

A. True

B. False

Which expression shows 5 layers of 8 cubes?

A. $8 - 5$

B. 5×8

C. $5 + 8$

D. 58

Multiply.

$$6 \times 9$$

Product: _____

There are 2 layers with 15 cubes in each layer.
How many cubes are there in all?

Cubes: _____

SET 47 • 5.MD.C.4

FORWARD 1

Volume with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

A prism is 3 cubes long, 2 cubes wide, and 2 cubes tall.
Each cube has $\frac{1}{2}$ -inch edges.

What is the volume of the prism?

Volume: _____

A box is $2\frac{1}{2}$ cm by 2 cm by 1 cm.

What is its volume?

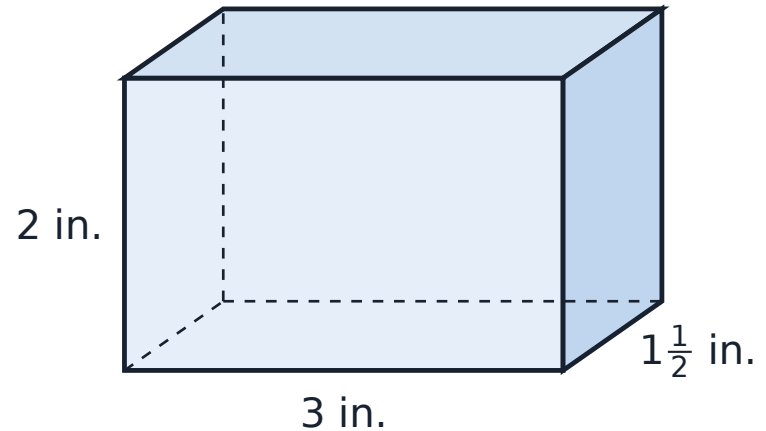
Volume: _____

How many cubes with $\frac{1}{2}$ -foot edges fill a box that is 2 ft by 1 ft by 1 ft?

- A.** 4
- B.** 8
- C.** 16
- D.** 2

Find the volume of the prism.

Volume: _____



True or false?

A cube with edges of $1\frac{1}{2}$ feet has a volume of $2\frac{1}{4}$ cubic feet.

- A. True
- B. False

SET 47 • 5.MD.C.4

FORWARD 2

Volume of right prisms • Grade 7 • Geometry • 7.G.B.6

A prism has a rectangular base that is 5 units by 4 units.
It is 6 units tall.

What is its volume?

Volume: _____

A box has a volume of 120 cubic cm. Its base is 6 cm by 5 cm.

What is its height?

Height: _____

A prism has a base area of 15 square feet and a height of 4 feet.

What is its volume?

- A.** 60 cubic feet
- B.** 30 cubic feet
- C.** 120 cubic feet
- D.** 19 cubic feet

A triangular prism has a base area of 9 square cm and a length of 12 cm.

What is its volume?

Volume: _____

True or false?

A box that is 10 in. by 2 in. by 3 in. has a volume of 15 cubic inches.

A. True

B. False

Relate packing with unit cubes to the formulas $V = l \times w \times h$ and $V = b \times h$

MAIN	Grade 5 main question	5.MD.C.5.a
BACKWARD 1	Area by tiling	3.MD.C.7.a
BACKWARD 2	Multiplying three numbers in any grouping	3.OA.B.5
FORWARD 1	Apply the volume formulas with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms using $V = b \times h$	7.G.B.6

SET 48 • 5.MD.C.5.a

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.a

A prism is packed with unit cubes. It is 5 cubes long, 3 cubes wide, and 4 layers tall.

What is its volume?

Volume: _____

The bottom layer of a prism has 12 unit cubes. There are 5 layers.

Which expression gives the volume?

A. $12 \times 12 \times 5$

B. 12×5

C. 5×5

D. $12 + 5$

True or false?

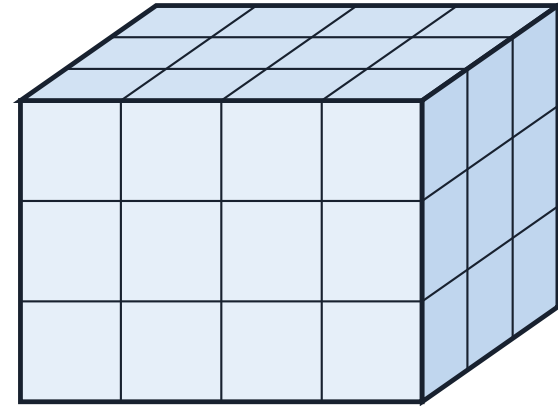
For a prism that is 4 by 3 by 2, $(4 \times 3) \times 2$ and $4 \times (3 \times 2)$ give the same volume.

A. True

B. False

Each cube is 1 cubic unit.

Find the volume by
multiplying the area of the
base by the height.



Volume: _____

A prism has a base area of 18 square units and a height of 4 units.

What is its volume?

Volume: _____

SET 48 • 5.MD.C.5.a

BACKWARD 1

Area by tiling • Grade 3 • Measurement & Data • 3.MD.C.7.a

A rectangle is tiled with unit squares in 3 rows of 4.
What is its area?

Area: _____

True or false?

The area of a 5-by-2 rectangle is 10 square units.

A. True

B. False

Which expression gives the area of a 6-by-3 rectangle?

A. $6 - 3$

B. $6 + 3$

C. 6×3

D. $2 \times (6 + 3)$

How many unit squares tile a 9-by-2 rectangle?

Squares: _____

True or false?

Tiling a rectangle and multiplying its side lengths give the same area.

A. True

B. False

SET 48 • 5.MD.C.5.a

BACKWARD 2

Multiplying three numbers in any grouping • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$(2 \times 3) \times 4 = 2 \times 3 + 4$$

A. True

B. False

Fill in the blank.

$$(5 \times 2) \times 3 = 5 \times (2 \times \underline{\quad})$$

Blank: _____

Which expression is equal to $4 \times 5 \times 2$?

A. 9×2

B. $20 + 2$

C. $4 + 5 + 2$

D. 4×10

Multiply.

$$3 \times 2 \times 5$$

Product: _____

True or false?

$$6 \times (2 \times 3) = 6 + 6$$

A. True

B. False

SET 48 • 5.MD.C.5.a

FORWARD 1

Apply the volume formulas with fractional edge lengths • Grade 6 • Geometry •
6.G.A.2

Use $V = l \times w \times h$ to find V when $l = 3\frac{1}{2}$ in., $w = 2$ in.,
and $h = 1\frac{1}{2}$ in.

Volume: _____

Use $V = b \times h$ to find the volume of a prism with $b = 6\frac{1}{4}$ square cm and $h = 4$ cm.

Volume: _____

A prism is packed with $\frac{1}{2}$ -unit cubes: 6 cubes long, 4 cubes wide, and 2 cubes tall.

Which shows its volume?

A. $6 + 4 + 2 = 12$ cubic units

B. $3 \times 2 \times 1 = 6$ cubic units

C. $6 \times 4 \times 2 = 48$ cubic units

D. $\frac{1}{2} \times 48 = 24$ cubic units

A cube has edges that are $\frac{3}{4}$ foot long.

What is its volume?

Volume: _____

True or false?

For a rectangular prism, $V = l \times w \times h$ and $V = b \times h$ give the same volume.

- A. True
- B. False

SET 48 • 5.MD.C.5.a

FORWARD 2

Volume of right prisms using $V = b \times h$ • Grade 7 • Geometry • 7.G.B.6

A triangular prism has a triangle base with a base of 6 cm and a height of 4 cm. The prism is 10 cm long.

What is its volume?

Volume: _____

A prism has a trapezoid base with an area of 14 square inches. The prism is 5 inches tall.

What is its volume?

Volume: _____

A prism has a volume of 84 cubic meters and a base area of 12 square meters.

What is its height?

A. 1,008 m

B. 7 m

C. 72 m

D. 96 m

A rectangular prism is 8 by 5 by 2.5.

What is its volume?

Volume: _____

True or false?

$V = b \times h$ works for triangular prisms as well as rectangular prisms.

A. True

B. False

Apply $V = l \times w \times h$ and $V = b \times h$ to solve real-world problems

MAIN

Grade 5 main question

5.MD.C.5.b

BACKWARD 1

Multiplying multi-digit numbers

4.NBT.B.5

BACKWARD 2

The area of the base

4.MD.A.3

FORWARD 1

Real-world volume with fractional edge lengths

6.G.A.2

FORWARD 2

Real-world problems with volume and surface area

7.G.B.6

SET 49 • 5.MD.C.5.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.b

A fish tank is 24 inches long, 12 inches wide, and 15 inches tall.

What is its volume?

Volume: _____

A sandbox has a base area of 30 square feet. It is filled with sand to a depth of 2 feet.

How much sand does it hold?

Volume: _____

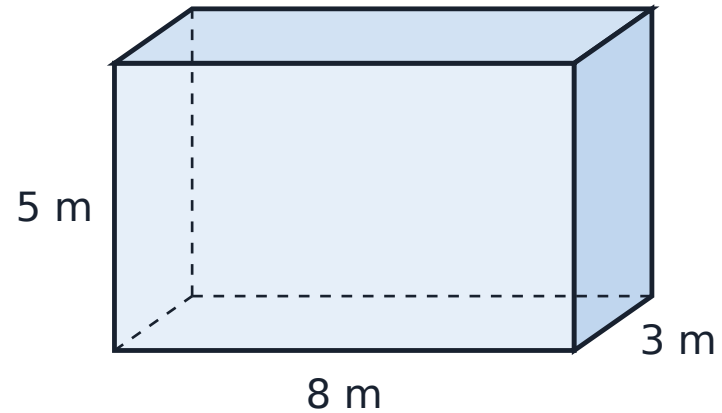
A box is 9 cm long, 4 cm wide, and 5 cm tall.

What is its volume?

- A.** 90 cubic cm
- B.** 36 cubic cm
- C.** 180 cubic cm
- D.** 18 cubic cm

Find the volume of the prism.

Volume: _____



A cube-shaped box has edges that are 7 inches long.
What is its volume?

Volume: _____

SET 49 • 5.MD.C.5.b

BACKWARD 1

Multiplying multi-digit numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Multiply.

$$24 \times 12$$

Product: _____

Multiply.

$$288 \times 5$$

Product: _____

True or false?

$$36 \times 5 = 150$$

A. True

B. False

$$49 \times 7 = ?$$

- A.** 56
- B.** 343
- C.** 283
- D.** 353

Multiply.

$$15 \times 24$$

Product: _____

SET 49 • 5.MD.C.5.b

BACKWARD 2

The area of the base • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 24 in. by 12 in.

What is its area?

Area: _____

True or false?

A 9-by-4 rectangle has an area of 36 square units.

A. True

B. False

The base of a box is 8 m by 3 m.

What is the area of the base?

- A.** 11 square meters
- B.** 24 square meters
- C.** 22 square meters
- D.** 48 square meters

A square has sides that are 7 inches long.
What is its area?

Area: _____

A rectangle has an area of 30 square feet and a length of 6 feet.

What is its width?

Width: _____

SET 49 • 5.MD.C.5.b

FORWARD 1

Real-world volume with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

A box is $1\frac{1}{2}$ ft by 1 ft by $\frac{3}{4}$ ft.

What is its volume?

Volume: _____

A prism has a base area of $12\frac{1}{2}$ square inches and a height of 3 inches.

What is its volume?

Volume: _____

A cube has edges that are $2\frac{1}{2}$ cm long.

What is its volume?

A. $6\frac{1}{4}$ cubic cm

B. 15 cubic cm

C. $7\frac{1}{2}$ cubic cm

D. $15\frac{5}{8}$ cubic cm

A drawer is 2 ft by $1\frac{1}{2}$ ft by $\frac{1}{2}$ ft.

What is its volume?

Volume: _____

True or false?

A box that is 3 by $\frac{1}{2}$ by 4 has a volume of 12 cubic units.

- A.** True
- B.** False

SET 49 • 5.MD.C.5.b

FORWARD 2

Real-world problems with volume and surface area • Grade 7 • Geometry •
7.G.B.6

A tent is a triangular prism. Its triangle has a base of 2 m and a height of 1.5 m. The tent is 3 m long.

What is its volume?

Volume: _____

A pool is 10 m long, 5 m wide, and 2 m deep. It is filled to $\frac{3}{4}$ of its depth.

How much water is in the pool?

Volume: _____

A box has a volume of 360 cubic inches. Its base is 12 in. by 6 in.

What is its height?

A. 3 inches

B. 30 inches

C. 5 inches

D. 60 inches

A rectangular prism is 4.5 cm by 2 cm by 3 cm.
What is its surface area?

Surface area: _____

True or false?

A triangular prism with a base area of 8 square feet and a length of 5 feet holds 40 cubic feet.

A. True

B. False

Find a missing dimension when the volume is known

MAIN	Grade 5 main question	5.MD.C.5.b
BACKWARD 1	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Finding an unknown factor	3.OA.A.4
FORWARD 1	Solve one-step equations of the form $px = q$	6.EE.B.7
FORWARD 2	Solve two-step equations	7.EE.B.4.a

SET 50 • 5.MD.C.5.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.b

A box has a volume of 96 cubic inches. It is 6 inches long and 4 inches wide.

How tall is it?

Height: _____

A prism has a volume of 150 cubic cm and a height of 6 cm.

What is the area of its base?

Base area: _____

A tank holds 240 cubic feet. Its base is 8 feet by 5 feet.

How deep is the tank?

- A.** 6 feet
- B.** 30 feet
- C.** 40 feet
- D.** 227 feet

A rectangular prism has a volume of 84 cubic meters. It is 7 m long and 4 m tall.

How wide is it?

Width: _____

True or false?

A box with a volume of 60 cubic units and a base area of 12 square units is 5 units tall.

A. True

B. False

SET 50 • 5.MD.C.5.b

BACKWARD 1

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$96 \div 4$$

Quotient: _____

Divide.

$$150 \div 6$$

Quotient: _____

True or false?

$$240 \div 8 = 3$$

A. True

B. False

$$84 \div 7 = ?$$

A. 12

B. 77

C. 11

D. 13

Divide.

$$60 \div 5$$

Quotient: _____

SET 50 • 5.MD.C.5.b

BACKWARD 2

Finding an unknown factor • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.4

What number goes in the box?

$$6 \times \square = 48$$

Number: _____

True or false?

If $\square \times 5 = 35$, then $\square = 7$.

A. True

B. False

What number goes in the box?

$$4 \times \square = 36$$

A. 9

B. 8

C. 32

D. 40

What number goes in the box?

$$\square \times 7 = 63$$

Number: _____

True or false?

If $8 \times \square = 56$, then $\square = 6$.

A. True

B. False

SET 50 • 5.MD.C.5.b

FORWARD 1

Solve one-step equations of the form $px = q$ • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for h.

$$24h = 96$$

$$h = \underline{\hspace{10cm}}$$

Solve for b.

$$6b = 150$$

$$b = \underline{\hspace{10cm}}$$

Solve for w.

$$28w = 84$$

- A.** 56
- B.** 2,352
- C.** 112
- D.** 3

A prism has a volume of 52.5 cubic cm and a base area of 15 square cm.

Write and solve an equation to find its height h .

Equation and solution: _____

True or false?

The solution of $\frac{3}{4}x = 12$ is $x = 9$.

A. True

B. False

SET 50 • 5.MD.C.5.b

FORWARD 2

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A box is 4 in. wide, 5 in. tall, and $(x + 2)$ in. long. Its volume is 120 cubic inches.

Find x .

$x =$ _____

Solve for x.

$$6(x + 3) = 54$$

x = _____

Solve for h.

$$12h + 8 = 68$$

A. 5

B. 6.3

C. 60

D. 4

Two boxes have a total volume of 100 cubic feet. One box holds 40 cubic feet. The other has a base area of 12 square feet.

Write and solve an equation to find the height h of the other box.

Equation and solution: _____

True or false?

The solution of $3(x - 2) = 21$ is $x = 9$.

A. True

B. False

Find the volume of a figure made of two rectangular prisms

MAIN	Grade 5 main question	5.MD.C.5.c
BACKWARD 1	Area of figures made of rectangles	3.MD.C.7.d
BACKWARD 2	Volume of one rectangular prism	5.MD.C.5.b
FORWARD 1	Volume of composite figures with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of composite solids	7.G.B.6

SET 51 • 5.MD.C.5.c

MAIN

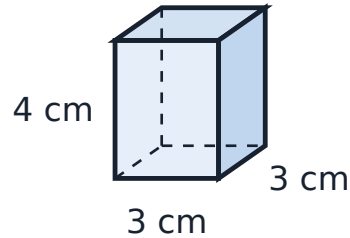
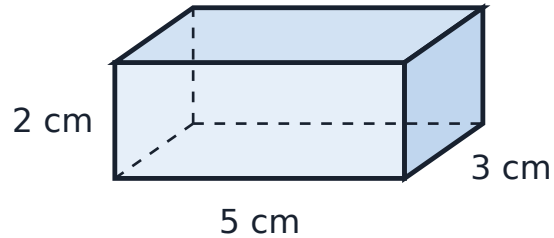
Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.c

A figure is made of two rectangular prisms that do not overlap. One is 6 ft by 4 ft by 3 ft. The other is 2 ft by 4 ft by 5 ft.

What is the total volume?

Volume: _____

A figure is made of the two prisms shown, which do not overlap.
What is the total volume?



Volume: _____

An L-shaped step is made of a 4-by-2-by-1 prism with a 2-by-2-by-1 prism on top.

What is its volume?

- A.** 12 cubic units
- B.** 8 cubic units
- C.** 10 cubic units
- D.** 16 cubic units

A building is a 10 m by 8 m by 6 m prism with a 4 m by 8 m by 3 m prism attached to one side.

What is the total volume?

Volume: _____

True or false?

The volume of a figure made of two prisms that do not overlap is the sum of their volumes.

- A. True
- B. False

SET 51 • 5.MD.C.5.c

BACKWARD 1

Area of figures made of rectangles • Grade 3 • Measurement & Data •
3.MD.C.7.d

A figure is made of a 5-by-3 rectangle and a 2-by-4 rectangle that do not overlap.

What is its area?

Area: _____

True or false?

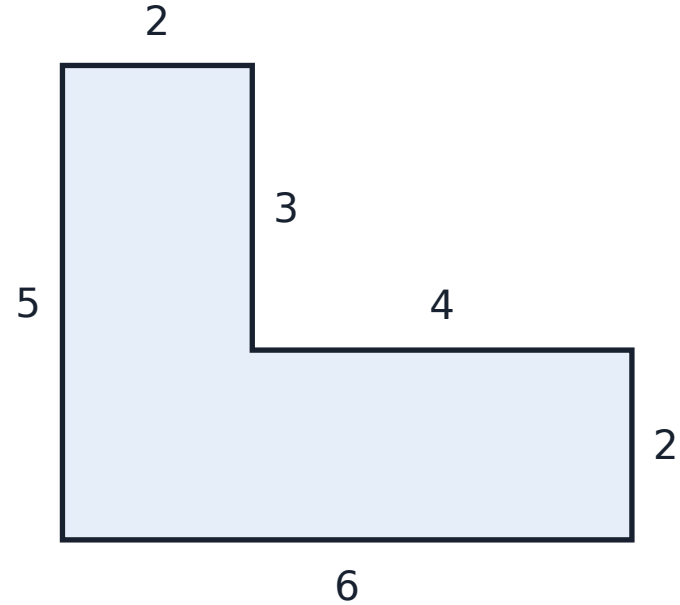
The area of a figure made of two non-overlapping parts is the sum of the areas of the parts.

A. True

B. False

What is the area of the figure?

- A.** 18 square units
- B.** 22 square units
- C.** 12 square units
- D.** 30 square units



A figure is made of two squares that do not overlap.
Their sides are 3 units and 5 units long.

What is the total area?

Area: _____

A 2-by-2 square is cut out of an 8-by-6 rectangle.
What is the area of the part that is left?

Area: _____

SET 51 • 5.MD.C.5.c

BACKWARD 2

Volume of one rectangular prism • Grade 5 • Measurement & Data •
5.MD.C.5.b

What is the volume of a prism that is 6 by 4 by 3?

Volume: _____

What is the volume of a prism that is 2 by 4 by 5?

Volume: _____

True or false?

A prism with a base area of 32 square units and a height of 3 units has a volume of 35 cubic units.

- A. True
- B. False

What is the volume of a prism that is 4 by 8 by 3?

- A.** 96 cubic units
- B.** 15 cubic units
- C.** 120 cubic units
- D.** 32 cubic units

A cube has edges that are 5 units long.

What is its volume?

Volume: _____

SET 51 • 5.MD.C.5.c

FORWARD 1

Volume of composite figures with fractional edge lengths • Grade 6 • Geometry
• 6.G.A.2

A figure is made of two prisms that do not overlap: $2\frac{1}{2}$ ft by 2 ft by 1 ft, and 1 ft by 2 ft by $\frac{1}{2}$ ft.

What is the total volume?

Volume: _____

Two prisms are joined. One has a volume of $4\frac{1}{2}$ cubic inches. The other is 3 in. by 1 in. by $\frac{1}{2}$ in.

What is the total volume?

Volume: _____

A step is made of a 3-by-1-by- $\frac{1}{2}$ prism and a 1-by-1-by- $\frac{1}{2}$ prism that do not overlap.

What is its volume?

A. 2 cubic units

B. $\frac{1}{2}$ cubic unit

C. 4 cubic units

D. $1\frac{1}{2}$ cubic units

A block that is 1 in. by 1 in. by $2\frac{1}{2}$ in. is cut out of a box that is 4 in. by 3 in. by $2\frac{1}{2}$ in.

What volume is left?

Volume: _____

True or false?

A figure made of two cubes with $\frac{1}{2}$ -foot edges has a volume of 1 cubic foot.

- A. True
- B. False

SET 51 • 5.MD.C.5.c

FORWARD 2

Volume of composite solids • Grade 7 • Geometry • 7.G.B.6

A figure is a 6-by-4-by-3 rectangular prism with a triangular prism on top. The triangle has a base of 4 and a height of 2, and the triangular prism is 6 long.

What is the total volume?

Volume: _____

A concrete step is made of two prisms: 1.5 m by 0.5 m by 0.2 m and 1.5 m by 0.25 m by 0.4 m.

What is the total volume?

Volume: _____

A 10-by-10-by-10 cube has a 2-by-2-by-10 hole cut all the way through it.

What volume remains?

- A.** 960 cubic units
- B.** 800 cubic units
- C.** 1,000 cubic units
- D.** 40 cubic units

A shed is a 3 m by 4 m by 2.5 m prism with a roof shaped like a triangular prism. The triangle has a base of 3 m and a height of 1 m, and the roof is 4 m long.

What is the total volume?

Volume: _____

True or false?

The volume of a composite solid can be found by adding the volumes of its parts.

A. True

B. False

Understand the coordinate plane: axes, origin, and coordinates

MAIN	Grade 5 main question	5.G.A.1
BACKWARD 1	Whole numbers on a number line	2.MD.B.6
BACKWARD 2	Perpendicular lines	4.G.A.1
FORWARD 1	Name the coordinates of points in all four quadrants	6.NS.C.6.c
FORWARD 2	Interpret a point on the graph of a proportional relationship	7.RP.A.2.d

SET 52 • 5.G.A.1

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.1

What is the point where the x-axis and the y-axis meet called?

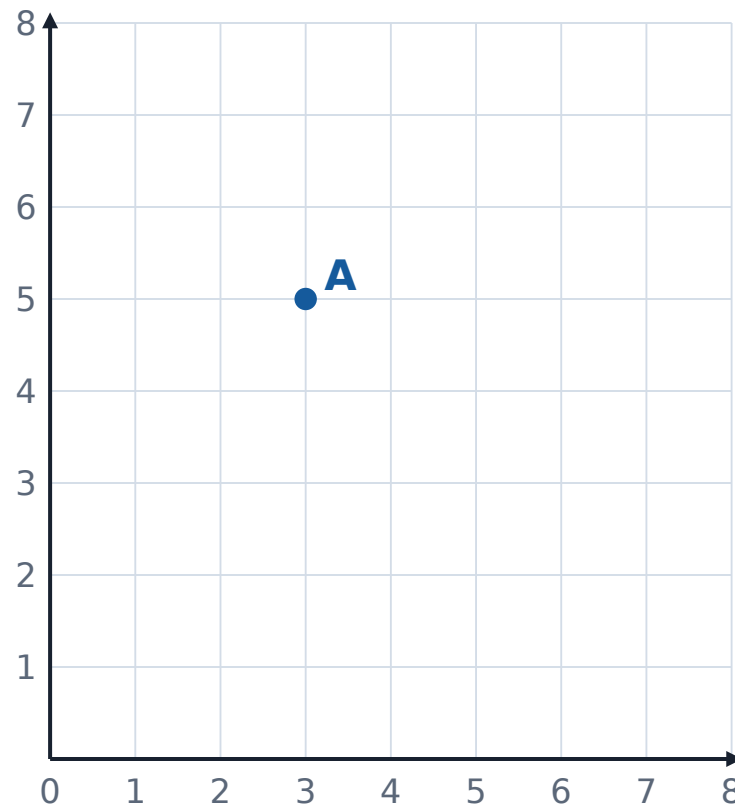
- A.** The y-axis
- B.** The origin
- C.** The x-coordinate
- D.** A quadrant

In the ordered pair $(7, 2)$, what does the 7 tell you?

Answer: _____

What are the coordinates of point A?

A = _____



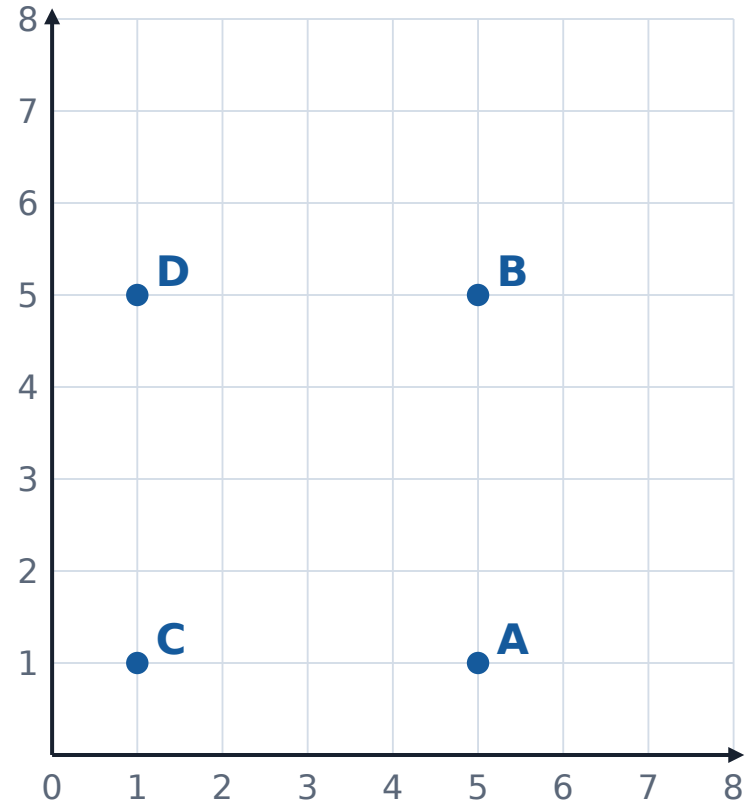
True or false?

To reach $(0, 6)$, you start at the origin and move 6 units to the right along the x-axis.

- A. True
- B. False

Which point is at (5, 1)?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D



SET 52 • 5.G.A.1

BACKWARD 1

Whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point P?



P = _____

True or false?

On a number line, numbers increase from left to right.

- A.** True
- B.** False

Which number is 4 units to the right of 0?

- A.** 4
- B.** 8
- C.** 40
- D.** 0

How many units is it from 0 to 9 on a number line?

Units: _____

True or false?

On a number line, 3 is farther from 0 than 7 is.

A. True

B. False

SET 52 • 5.G.A.1

BACKWARD 2

Perpendicular lines • Grade 4 • Geometry • 4.G.A.1

True or false?

Perpendicular lines meet to form right angles.

A. True

B. False

The x-axis and the y-axis are ____.

- A.** perpendicular
- B.** parallel
- C.** curved
- D.** the same line

How many degrees are in a right angle?

Degrees: _____

True or false?

Parallel lines never meet.

A. True

B. False

Which letter is made of two perpendicular line segments?

A. S

B. L

C. O

D. C

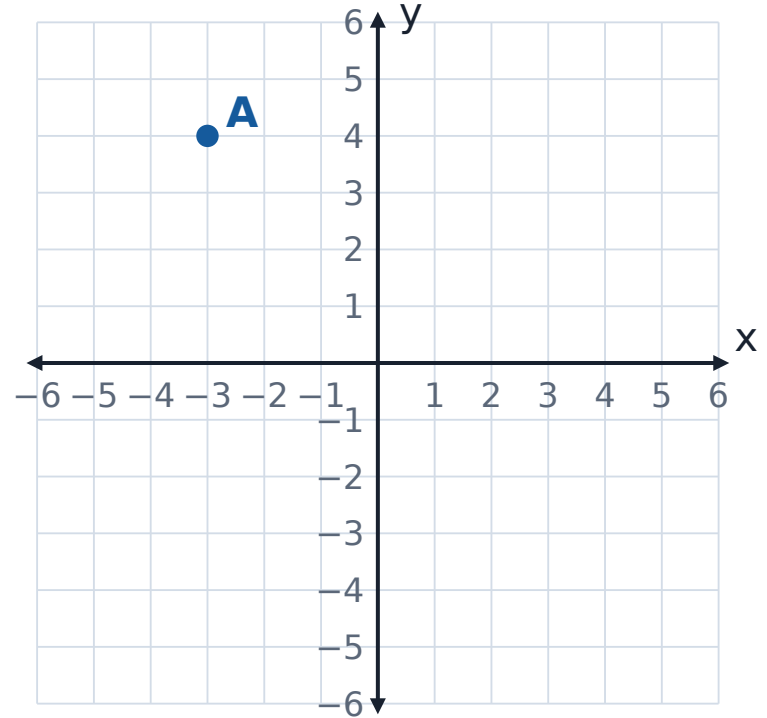
SET 52 • 5.G.A.1

FORWARD 1

Name the coordinates of points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

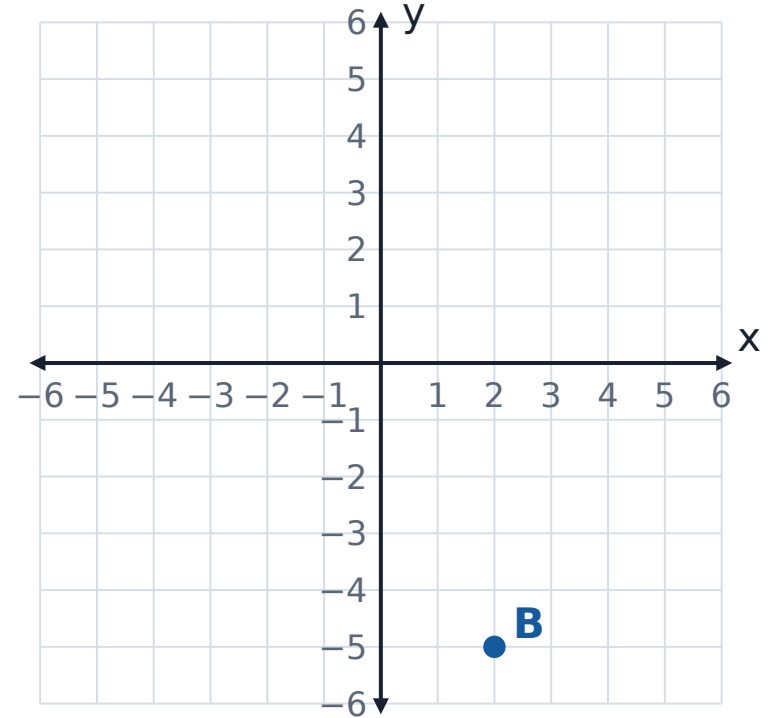
What are the coordinates of point A?

A = _____



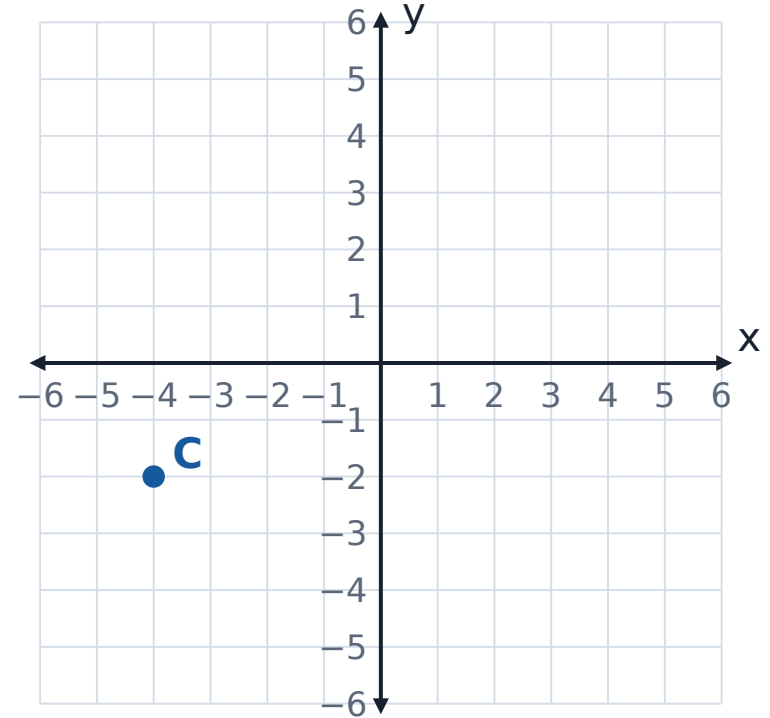
What are the coordinates of point B?

B = _____



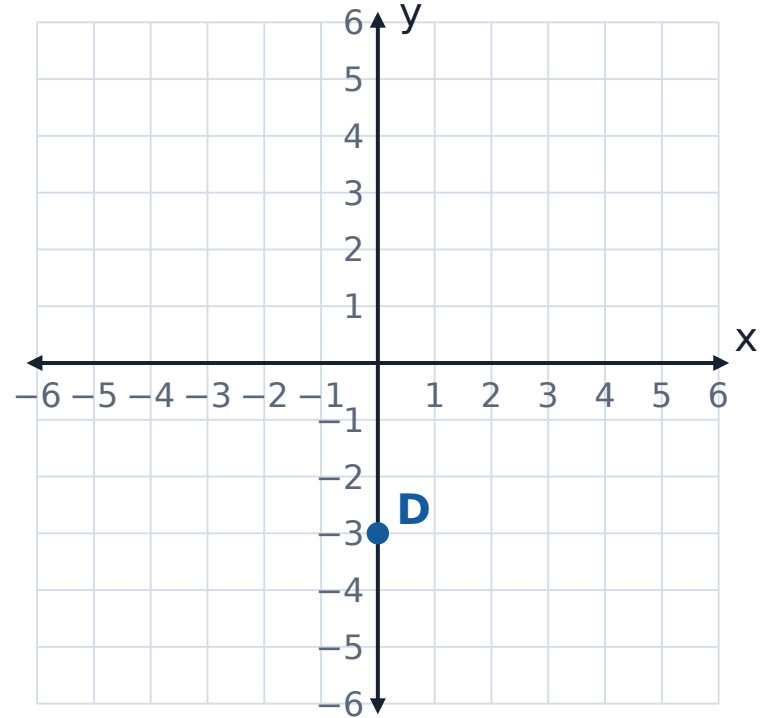
Which ordered pair names point C?

- A.** $(-4, -2)$
- B.** $(-2, -4)$
- C.** $(4, 2)$
- D.** $(-4, 2)$



What are the coordinates of point D?

D = _____

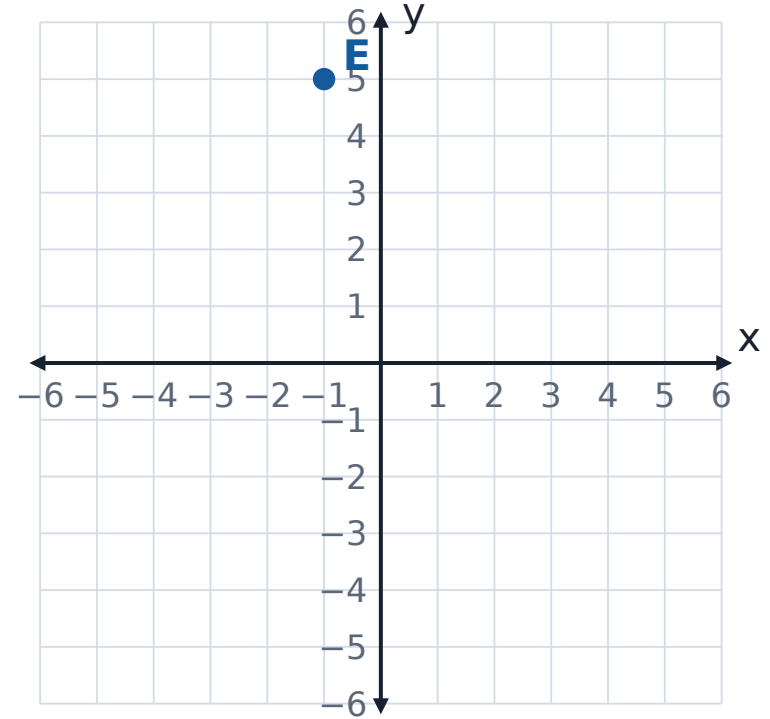


True or false?

Point E is located at $(5, -1)$.

A. True

B. False



SET 52 • 5.G.A.1

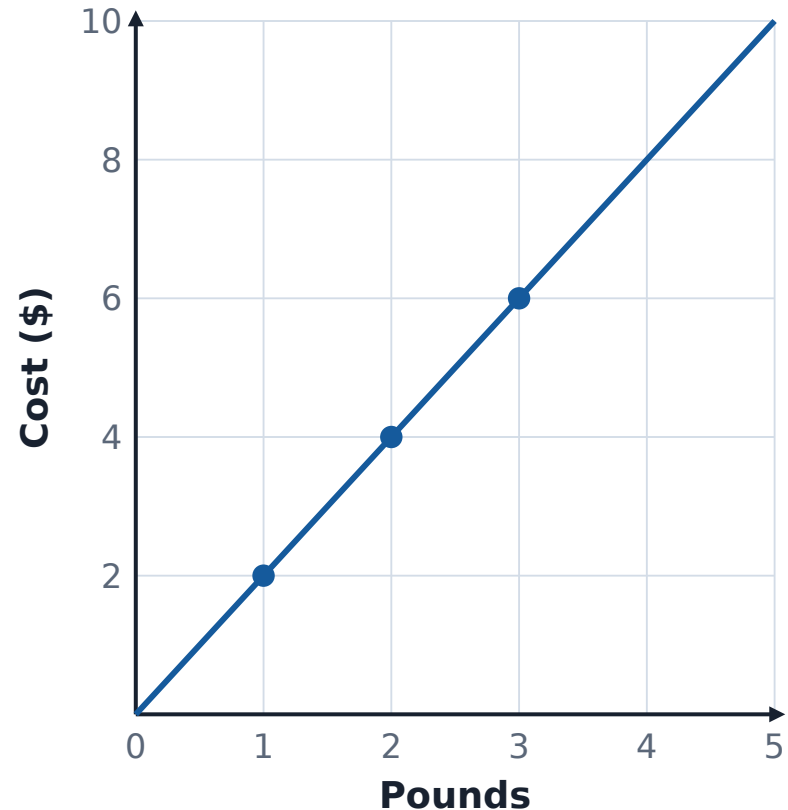
FORWARD 2

Interpret a point on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph shows the cost of pears.

What does the point (3, 6) represent?

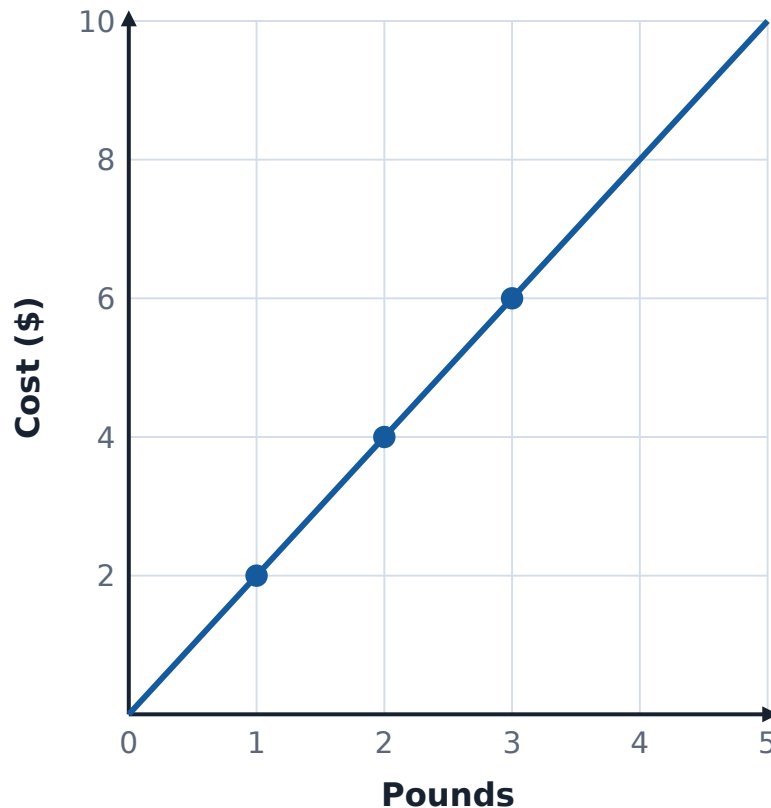
Meaning: _____



The graph shows the cost of pears.

What does the point (1, 2) represent?

- A.** The cost starts at \$2.
- B.** 2 pounds cost \$1.
- C.** \$1 buys 2 pounds.
- D.** 1 pound costs \$2.



True or false?

On the graph of a proportional relationship between pounds and cost, the point $(0, 0)$ means 0 pounds cost \$0.

- A.** True
- B.** False

The graph of a proportional relationship passes through $(1, r)$ and $(4, 20)$.

What is r ?

$r =$ _____

A graph shows the distance y in miles after x hours. It passes through $(2, 90)$.

What does this point mean?

- A.** The speed is 2 miles per hour.
- B.** In 90 hours, the distance is 2 miles.
- C.** The speed is 90 miles per hour.
- D.** In 2 hours, the distance is 90 miles.

Graph points in the first quadrant

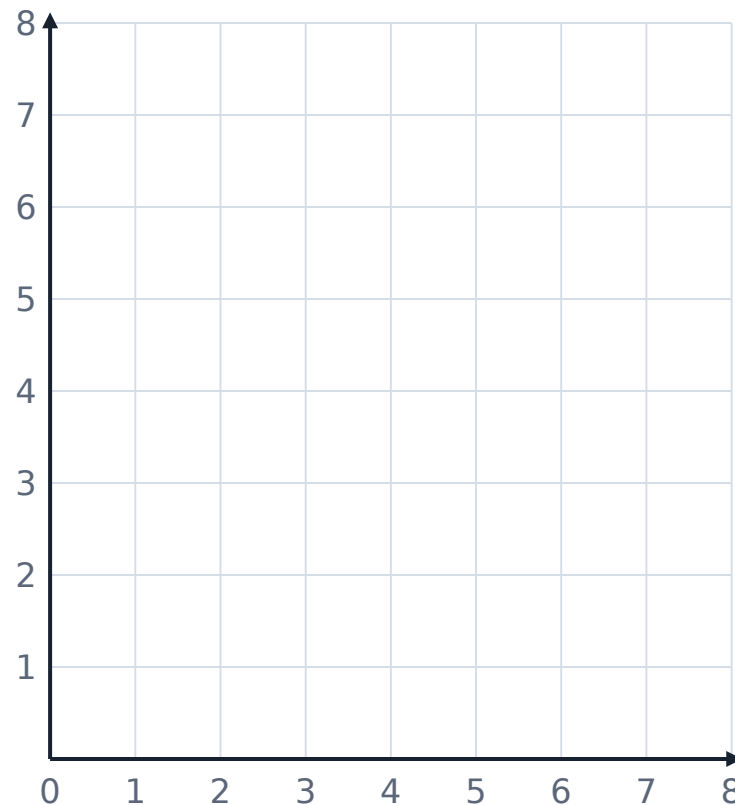
MAIN	Grade 5 main question	5.G.A.2
BACKWARD 1	Reading coordinates	5.G.A.1
BACKWARD 2	Reading the scale on an axis	3.MD.B.3
FORWARD 1	Plot and reflect points in all four quadrants	6.NS.C.6.b
FORWARD 2	Graph pairs to decide whether a relationship is proportional	7.RP.A.2.a

SET 53 • 5.G.A.2

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.2

Plot and label the points $A(2, 7)$, $B(6, 3)$, and $C(0, 5)$.

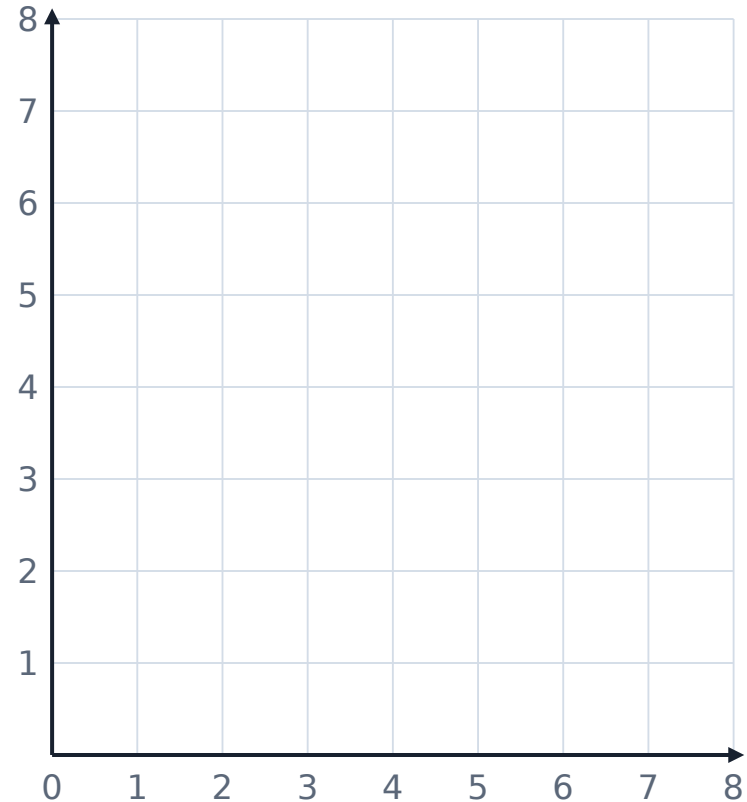


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

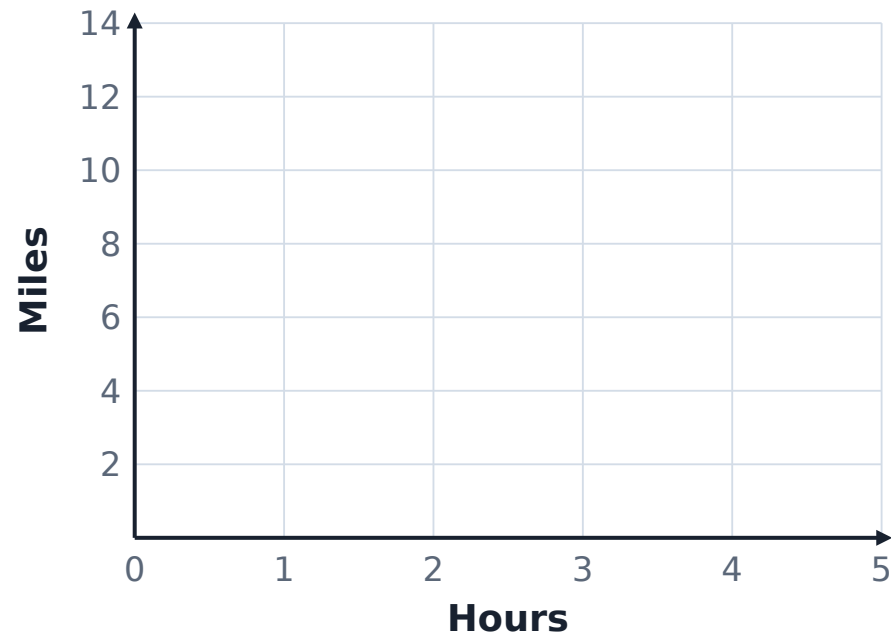
Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q1/5

Plot and label the points D(4, 4), E(7, 0), and F(1, 6).

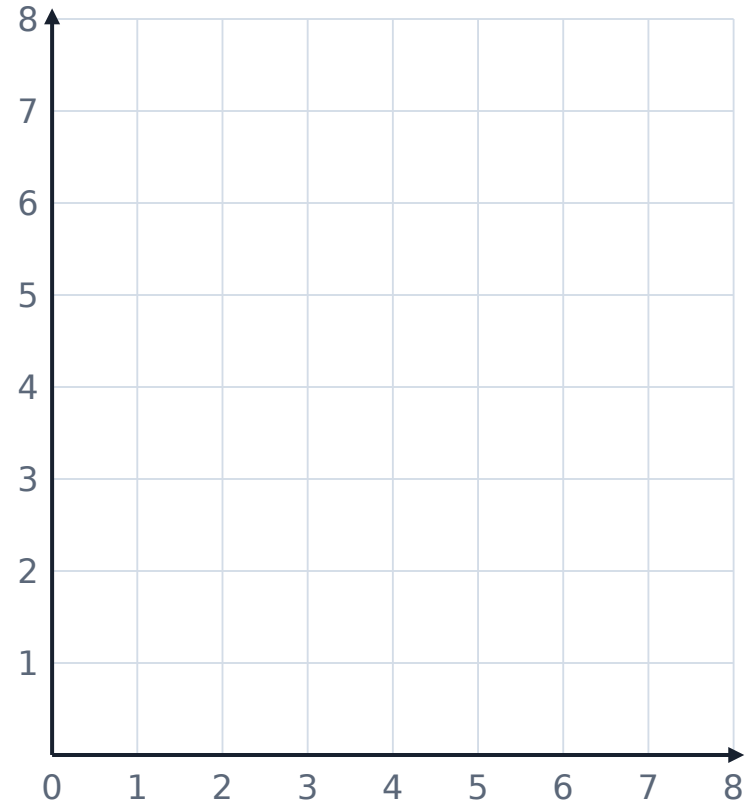


Plot the points from the table.

Hours	1	2	3	4
Miles	3	6	9	12



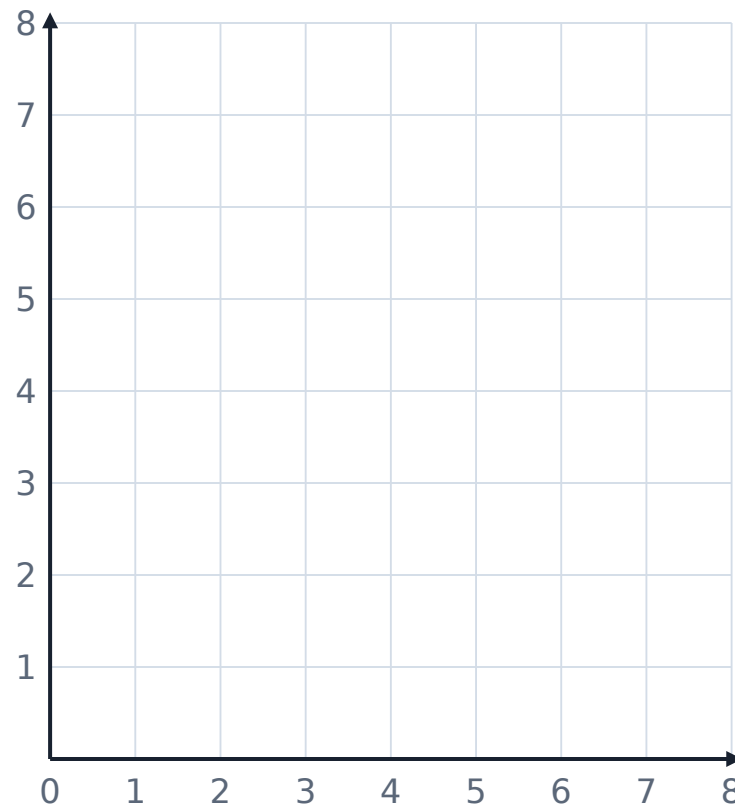
Plot and label the points $G(3, 2\frac{1}{2})$ and $H(5\frac{1}{2}, 4)$.



Plot $P(2, 3)$ and $Q(2, 7)$.

How far apart are the two points?

Distance: _____



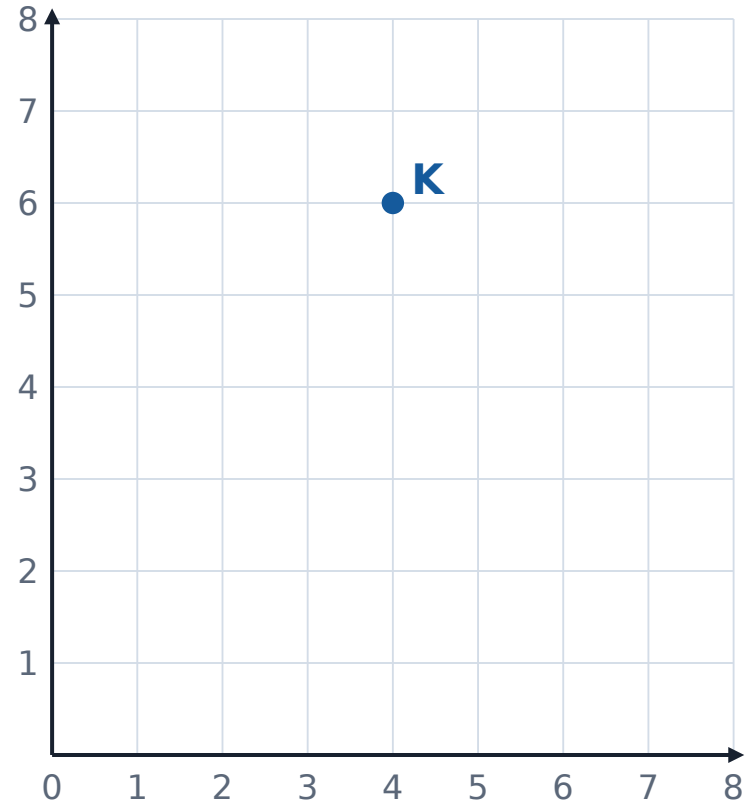
SET 53 • 5.G.A.2

BACKWARD 1

Reading coordinates • Grade 5 • Geometry • 5.G.A.1

What are the coordinates of point K?

K = _____



True or false?

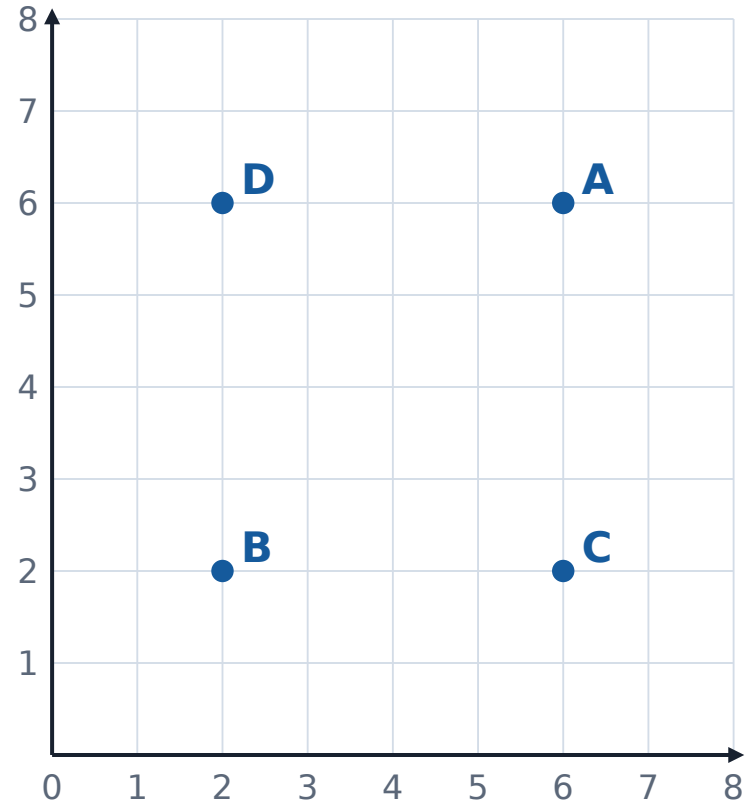
To plot (3, 8), you move up 3 units.

A. True

B. False

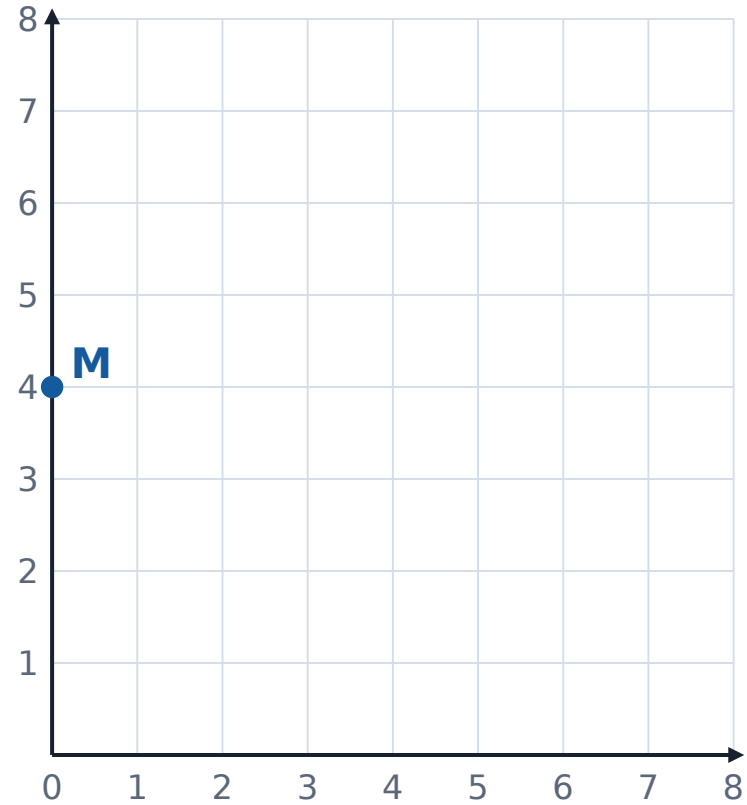
Which point is at (6, 2)?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D



What are the coordinates of point M?

M = _____



True or false?

The point $(5, 0)$ is on the y -axis.

A. True

B. False

SET 53 • 5.G.A.2

BACKWARD 2

Reading the scale on an axis • Grade 3 • Measurement & Data • 3.MD.B.3

An axis is numbered by 5s: 0, 5, 10, 15, ...

What value is 4 grid lines above 0?

- A.** 15
- B.** 25
- C.** 20
- D.** 4

True or false?

On an axis numbered by 2s, the value 7 is halfway between 6 and 8.

- A. True
- B. False

An axis is numbered by 10s.

What value is at the 3rd grid line above 0?

Value: _____

An axis is numbered by 3s.

Where is the value 12?

- A.** At the 4th grid line
- B.** At the 3rd grid line
- C.** Between the 3rd and 4th grid lines
- D.** At the 12th grid line

True or false?

On an axis numbered by 5s, the value 12 is on a grid line.

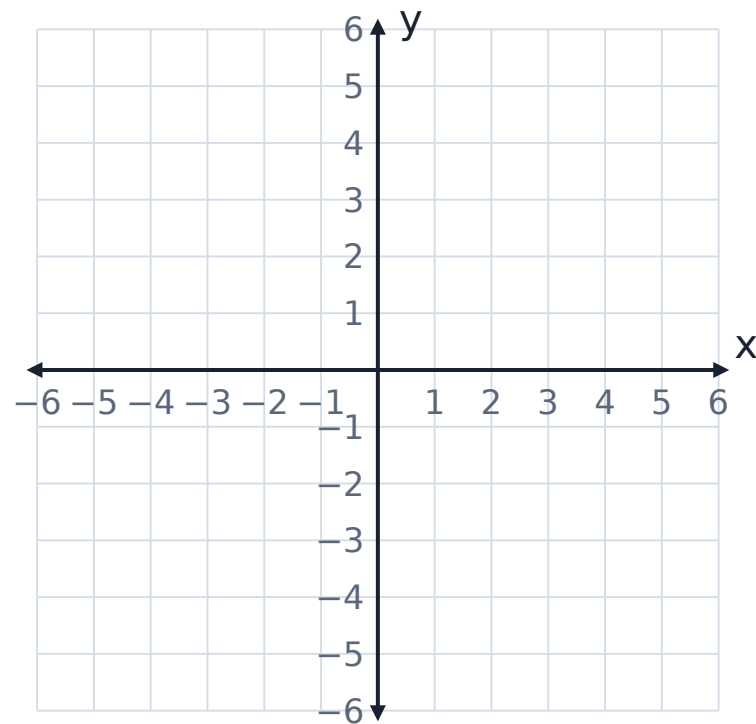
- A.** True
- B.** False

SET 53 • 5.G.A.2

FORWARD 1

Plot and reflect points in all four quadrants • Grade 6 • The Number System •
6.NS.C.6.b

Plot and label the points
 $P(-3, 4)$ and $Q(3, -4)$.



The point $(5, 2)$ is reflected across the x -axis.
What are the coordinates of its image?

Image: _____

In which quadrant is the point $(-2, -6)$?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

The point $(-4, 1)$ is reflected across the y -axis.
What are the coordinates of its image?

Image: _____

True or false?

The points $(3, 5)$ and $(-3, 5)$ are reflections of each other across the x-axis.

A. True

B. False

SET 53 • 5.G.A.2

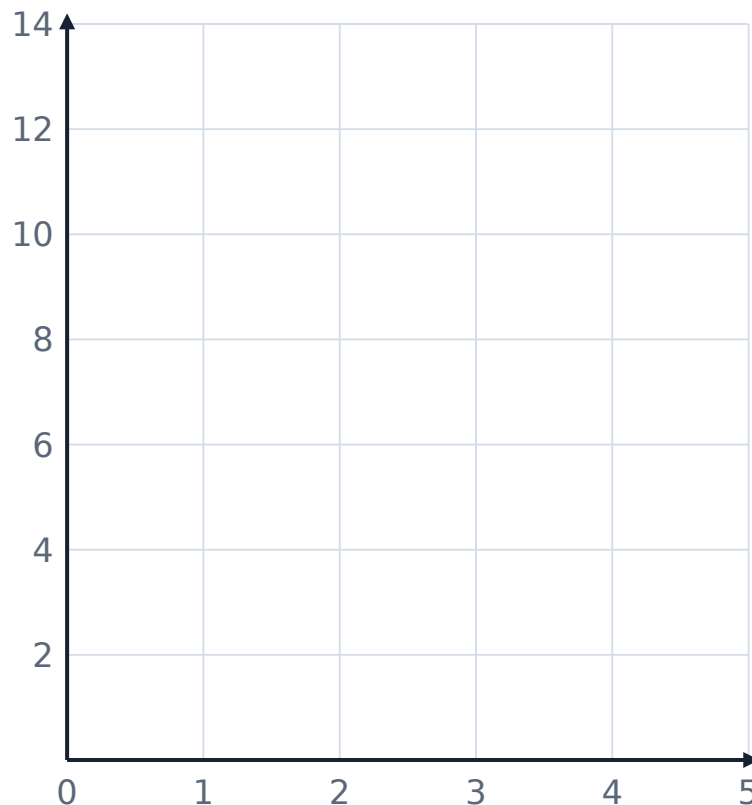
FORWARD 2

Graph pairs to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

Graph the pairs (1, 3), (2, 6),
and (4, 12).

Is the relationship
proportional?

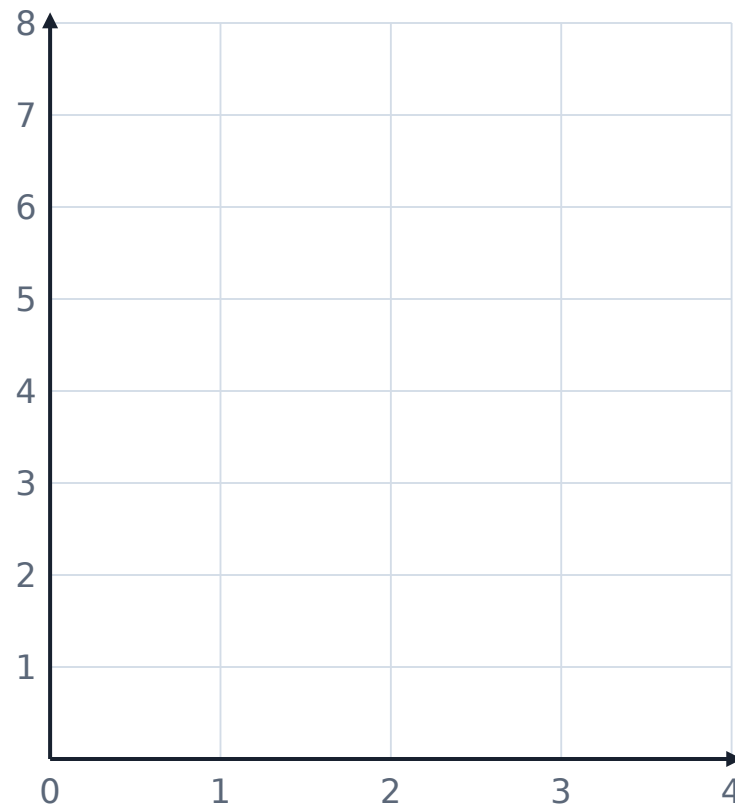
Answer: _____



Graph the pairs $(0, 2)$, $(1, 4)$,
and $(2, 6)$.

Is the relationship
proportional?

Answer: _____



Which graph shows a proportional relationship?

- A.** A curve through $(0, 0)$
- B.** Points that are not on a line
- C.** A straight line through $(0, 3)$
- D.** A straight line through $(0, 0)$

True or false?

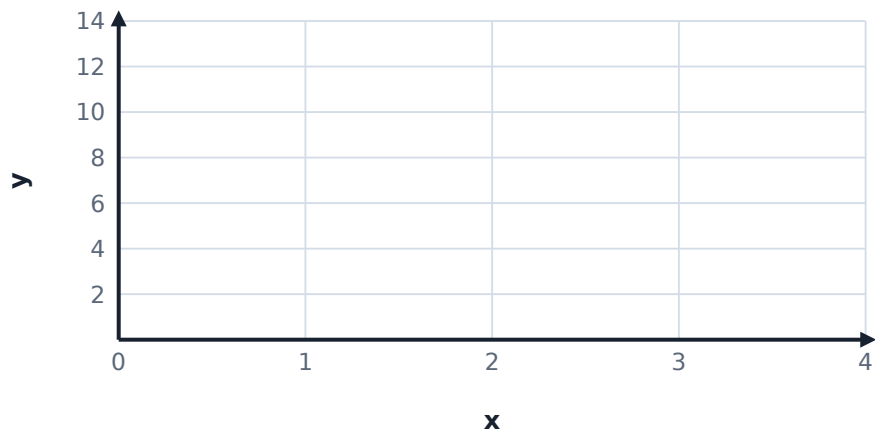
The points $(2, 1)$, $(4, 2)$, and $(6, 3)$ show a proportional relationship.

- A. True
- B. False

Graph the pairs from the table.

Is the relationship proportional?

x	1	2	3
y	4	8	12



Answer: _____

Solve real-world problems by graphing points and interpreting coordinates

MAIN	Grade 5 main question	5.G.A.2
BACKWARD 1	The coordinate system	5.G.A.1
BACKWARD 2	Distance between whole numbers on a number line	2.MD.B.6
FORWARD 1	Find distances in all four quadrants	6.NS.C.8
FORWARD 2	Use a scale on a map grid	7.G.A.1

SET 54 • 5.G.A.2

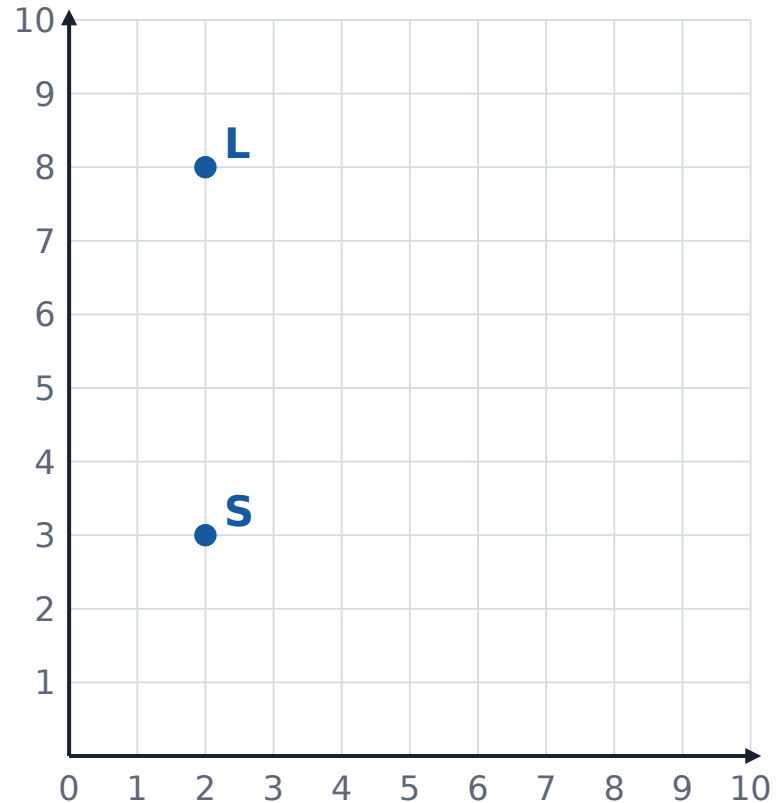
MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.2

The grid is a map. Each unit is 1 block.

How many blocks is it from the school S to the library L?

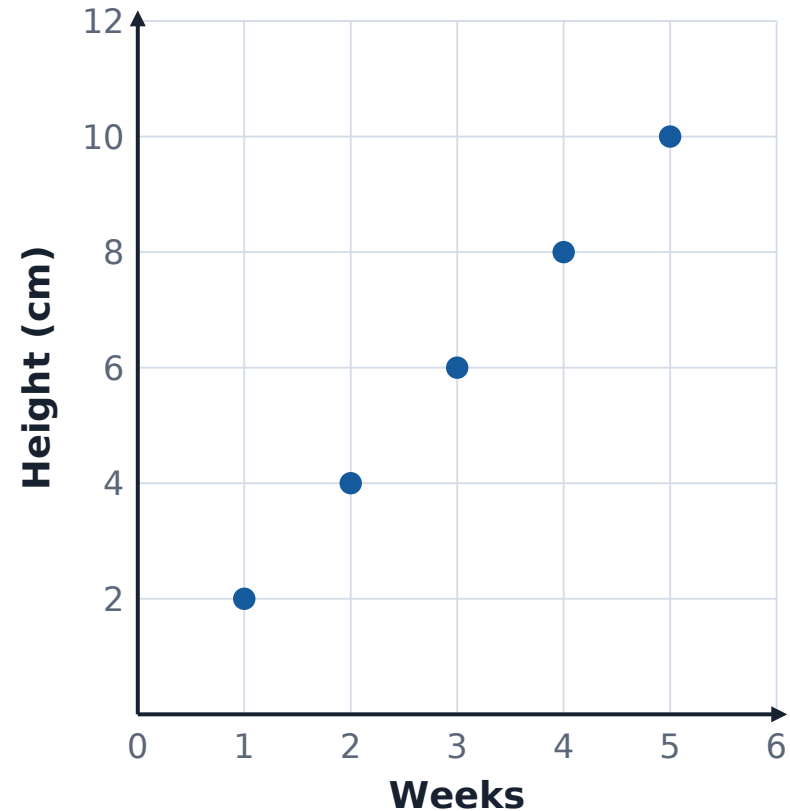
Blocks: _____



The graph shows the height of a plant.

How tall was the plant after 4 weeks?

Height: _____



The point (3, 12) is on a graph of hours worked (x) and dollars earned (y).

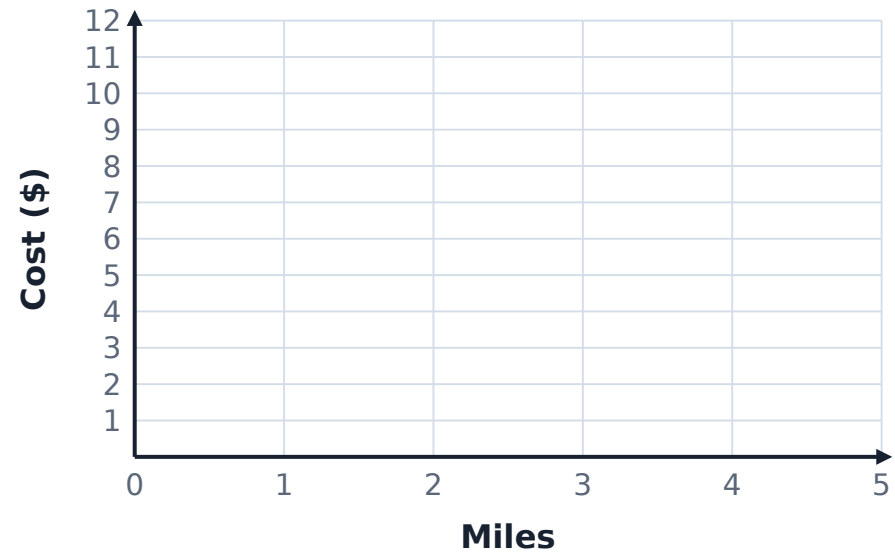
What does the point mean?

- A.** Working 3 hours earns \$12.
- B.** Working 12 hours earns \$3.
- C.** The pay starts at \$12.
- D.** Each hour earns \$3.

The table shows the cost of a taxi ride.

Graph the points.

Miles	1	2	3	4
Cost (\$)	5	7	9	11



On a map grid, the park is at (1, 6) and the pool is at (7, 6). Each unit is 100 meters.

How far apart are the park and the pool?

Distance: _____

SET 54 • 5.G.A.2

BACKWARD 1

The coordinate system • Grade 5 • Geometry • 5.G.A.1

What are the coordinates of the origin?

Origin: _____

True or false?

The first number in an ordered pair tells how far to move along the x-axis.

A. True

B. False

Point R is 4 units to the right of the origin and 1 unit up.

What are its coordinates?

A. (5, 0)

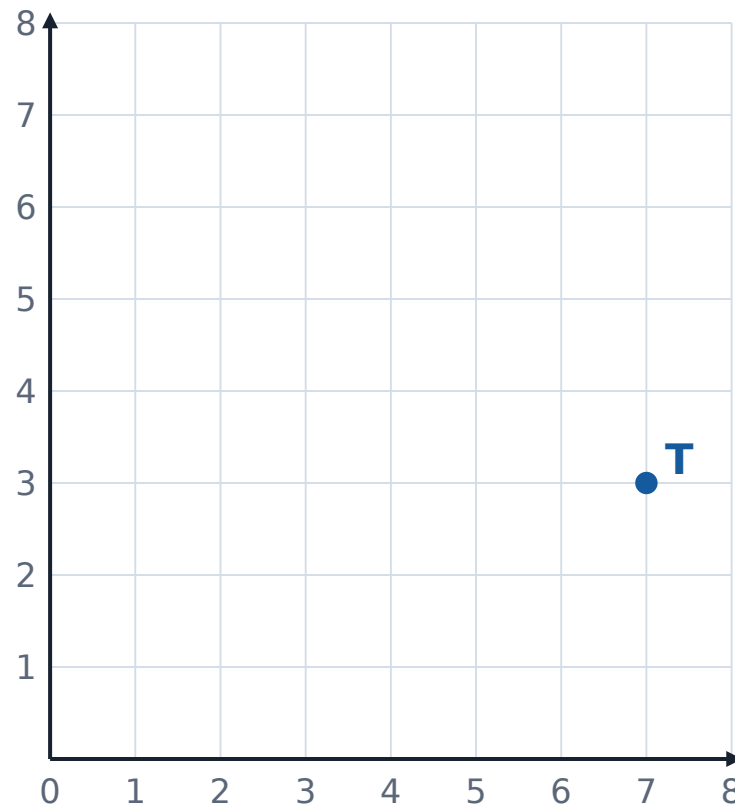
B. (1, 4)

C. (4, 4)

D. (4, 1)

What are the coordinates of point T?

T = _____



True or false?

$(0, 2)$ and $(2, 0)$ name the same point.

A. True

B. False

SET 54 • 5.G.A.2

BACKWARD 2

Distance between whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What is the distance from 2 to 7 on a number line?

Distance: _____

True or false?

The distance from 3 to 10 is 13 units.

A. True

B. False

How many units apart are 1 and 8 on a number line?

A. 9

B. 8

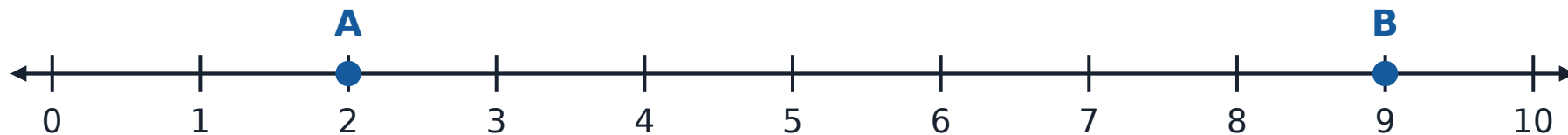
C. 1

D. 7

What is the distance from 6 to 6 on a number line?

Distance: _____

How many units apart are points A and B?



Units: _____

SET 54 • 5.G.A.2

FORWARD 1

Find distances in all four quadrants • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(-4, 2)$ and $(5, 2)$.

Distance: _____

Find the distance between $(3, -6)$ and $(3, 1)$.

Distance: _____

On a map, the store is at $(-2, -3)$ and the bank is at $(-2, 4)$. Each unit is 1 block.

How far apart are they?

A. 7 blocks

B. 1 block

C. 4 blocks

D. 6 blocks

A rectangle has vertices at $(-3, 1)$, $(2, 1)$, $(2, -2)$, and $(-3, -2)$.

What is its perimeter?

Perimeter: _____

True or false?

The distance between $(-5, 0)$ and $(5, 0)$ is 0.

A. True

B. False

SET 54 • 5.G.A.2

FORWARD 2

Use a scale on a map grid • Grade 7 • Geometry • 7.G.A.1

On a map grid, each unit is 2 miles. Two towns are at (1, 3) and (6, 3).

What is the actual distance between them?

Distance: _____

On a floor plan, each unit is $1\frac{1}{2}$ feet. A wall goes from (0, 0) to (0, 8).

How long is the real wall?

Length: _____

A scale drawing uses 1 grid unit = 5 m. A path on the grid is 7 units long.

How long is the real path?

A. 12 m

B. 35 m

C. 2 m

D. 7 m

A garden plan has corners at $(0, 0)$, $(6, 0)$, $(6, 4)$, and $(0, 4)$. Each unit is 3 feet.

What are the real length and width?

Length and width: _____

True or false?

If each grid unit is 10 km, the points (2, 5) and (2, 9) are 4 km apart.

- A. True
- B. False

Attributes of a category belong to all of its subcategories

MAIN	Grade 5 main question	5.G.B.3
BACKWARD 1	Quadrilaterals and their attributes	3.G.A.1
BACKWARD 2	Parallel lines, perpendicular lines, and angles	4.G.A.1
BACKWARD 3	Classifying shapes by sides and angles	4.G.A.2
FORWARD 1	Draw polygons in the coordinate plane	6.G.A.3
FORWARD 2	Decide whether shapes with given conditions are possible	7.G.A.2

SET 55 • 5.G.B.3

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.3

True or false?

All rectangles have four right angles. Squares are rectangles, so all squares have four right angles.

A. True

B. False

All parallelograms have two pairs of parallel sides.

Which shape must also have two pairs of parallel sides?

- A.** Kite
- B.** Pentagon
- C.** Triangle
- D.** Rhombus

A rhombus is a quadrilateral with 4 sides of equal length.

Is every square a rhombus? Explain.

Answer: _____

Explanation: _____

True or false?

All quadrilaterals have 4 right angles.

A. True

B. False

Which statement is true?

- A.** Every parallelogram is a square.
- B.** Every square is a rectangle.
- C.** Every quadrilateral is a rectangle.
- D.** Every rectangle is a square.

SET 55 • 5.G.B.3

BACKWARD 1

Quadrilaterals and their attributes • Grade 3 • Geometry • 3.G.A.1

How many sides does every quadrilateral have?

Sides: _____

True or false?

Triangles, rectangles, and squares are all quadrilaterals.

A. True

B. False

Which shape is NOT a quadrilateral?

- A.** Rectangle
- B.** Square
- C.** Rhombus
- D.** Triangle

True or false?

A square has 4 sides of equal length.

A. True

B. False

Name a quadrilateral that always has 4 right angles.

Shape: _____

SET 55 • 5.G.B.3

BACKWARD 2

Parallel lines, perpendicular lines, and angles • Grade 4 • Geometry • 4.G.A.1

True or false?

Parallel lines are always the same distance apart.

- A. True
- B. False

Which angle measure is a right angle?

- A.** 90°
- B.** 180°
- C.** 45°
- D.** 120°

What do we call two lines that meet to form right angles?

Answer: _____

True or false?

An obtuse angle is smaller than a right angle.

A. True

B. False

How many pairs of parallel sides does a rectangle have?

A. 4

B. 0

C. 2

D. 1

SET 55 • 5.G.B.3

BACKWARD 3

Classifying shapes by sides and angles • Grade 4 • Geometry • 4.G.A.2

True or false?

A shape with no parallel sides can still be a quadrilateral.

A. True

B. False

Which kind of triangle has a right angle?

- A.** An equilateral triangle
- B.** An acute triangle
- C.** A right triangle
- D.** An obtuse triangle

How many right angles does a square have?

Right angles: _____

True or false?

A rhombus always has 4 right angles.

A. True

B. False

Which shape always has two pairs of parallel sides?

A. Rectangle

B. Kite

C. Triangle

D. Pentagon

SET 55 • 5.G.B.3

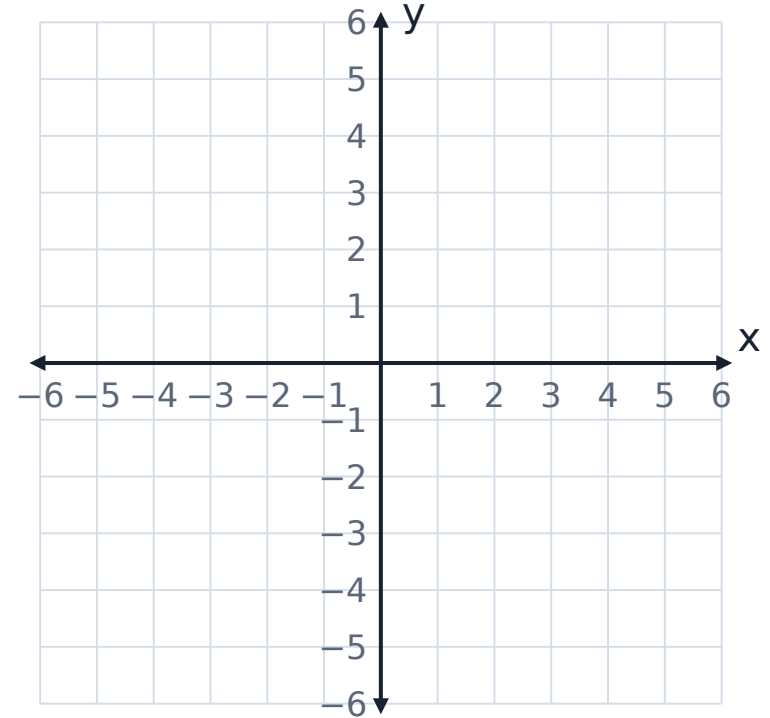
FORWARD 1

Draw polygons in the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Plot $A(-2, 1)$, $B(3, 1)$, $C(3, -3)$, and $D(-2, -3)$. Connect them in order.

What polygon did you draw?

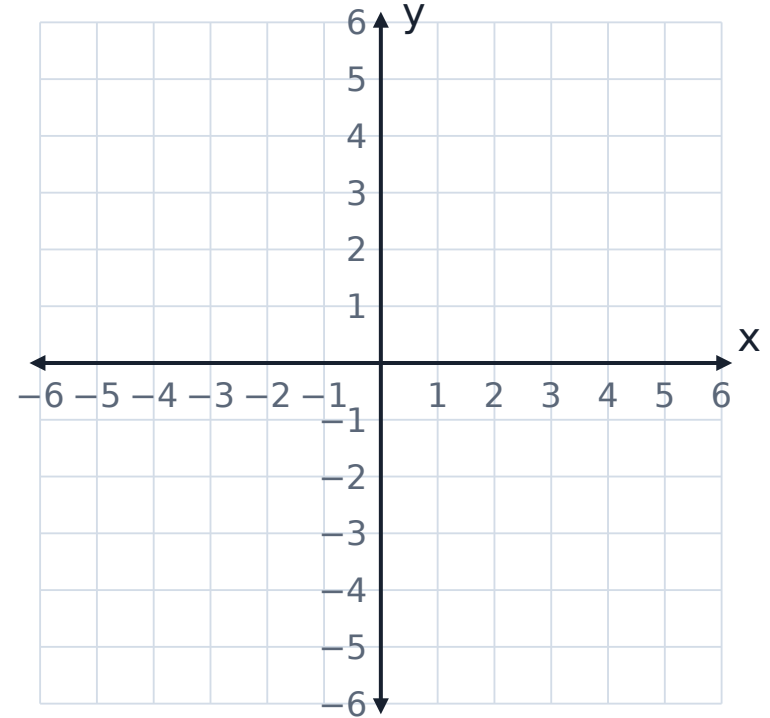
Polygon: _____



Plot $E(0, 3)$, $F(3, 0)$, $G(0, -3)$, and $H(-3, 0)$. Connect them in order.

What polygon did you draw?

Polygon: _____



The points $(-1, 2)$, $(4, 2)$, and $(4, -1)$ are three vertices of a rectangle.

What is the fourth vertex?

A. $(1, -1)$

B. $(-1, -1)$

C. $(-1, 4)$

D. $(4, 2)$

What is the length of the side from $(-2, 5)$ to $(4, 5)$?

Length: _____

True or false?

The points $(0, 0)$, $(4, 0)$, $(4, 2)$, and $(0, 2)$ are the vertices of a square.

- A. True
- B. False

SET 55 • 5.G.B.3

FORWARD 2

Decide whether shapes with given conditions are possible • Grade 7 •
Geometry • 7.G.A.2

True or false?

A triangle can have side lengths of 3 cm, 4 cm, and 5 cm.

A. True

B. False

Which three lengths can form a triangle?

- A.** 4, 5, 7
- B.** 5, 5, 10
- C.** 1, 1, 3
- D.** 2, 3, 6

Can a triangle have side lengths of 2 cm, 2 cm, and 5 cm? Write yes or no.

Answer: _____

True or false?

A triangle can have two right angles.

A. True

B. False

Can a quadrilateral have exactly 3 right angles? Write yes or no.

Answer: _____

Classify quadrilaterals in a hierarchy based on their properties

MAIN	Grade 5 main question	5.G.B.4
BACKWARD 1	Quadrilaterals	3.G.A.1
BACKWARD 2	Classifying by parallel sides and right angles	4.G.A.2
FORWARD 1	Area of parallelograms, rhombuses, and trapezoids	6.G.A.1
FORWARD 2	Shapes with given conditions	7.G.A.2

SET 56 • 5.G.B.4

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.4

Which list goes from the most general category to the most specific?

- A.** Square, rectangle, parallelogram, quadrilateral
- B.** Rectangle, square, quadrilateral, parallelogram
- C.** Quadrilateral, parallelogram, rectangle, square
- D.** Parallelogram, quadrilateral, square, rectangle

True or false?

A square is a rectangle, a rhombus, and a parallelogram.

A. True

B. False

Which of these categories does every rhombus belong to?

quadrilateral, parallelogram, rectangle, square

Categories: _____

A shape has 4 sides of equal length and 4 right angles.
What is the most specific name for it?

- A.** Rectangle
- B.** Rhombus
- C.** Square
- D.** Quadrilateral

True or false?

Every parallelogram is a rhombus.

A. True

B. False

SET 56 • 5.G.B.4

BACKWARD 1

Quadrilaterals • Grade 3 • Geometry • 3.G.A.1

True or false?

A rectangle is a quadrilateral.

A. True

B. False

Which shape always has 4 sides of equal length?

- A.** Triangle
- B.** Rectangle
- C.** Rhombus
- D.** Kite

How many angles does a quadrilateral have?

Angles: _____

True or false?

Every quadrilateral is a square.

A. True

B. False

Name a shape that is a quadrilateral but is not a parallelogram.

Shape: _____

SET 56 • 5.G.B.4

BACKWARD 2

Classifying by parallel sides and right angles • Grade 4 • Geometry • 4.G.A.2

True or false?

A quadrilateral with 2 pairs of parallel sides is a parallelogram.

A. True

B. False

Which shape always has 4 right angles?

- A.** Rectangle
- B.** Parallelogram
- C.** Kite
- D.** Rhombus

How many pairs of parallel sides does a parallelogram have?

Pairs: _____

True or false?

The sides of a square that meet at a corner are perpendicular.

- A. True
- B. False

Which description fits every rhombus?

- A.** No parallel sides
- B.** 3 sides
- C.** 4 sides of equal length
- D.** 4 right angles

SET 56 • 5.G.B.4

FORWARD 1

Area of parallelograms, rhombuses, and trapezoids • Grade 6 • Geometry •
6.G.A.1

A parallelogram has a base of 10 cm and a height of 6 cm.

What is its area?

Area: _____

A rhombus has a base of 7 in. and a height of 5 in.

What is its area?

Area: _____

A square has sides that are 9 m long.

What is its area?

- A.** 36 square meters
- B.** 81 square meters
- C.** 90 square meters
- D.** 18 square meters

A trapezoid has bases of 6 ft and 10 ft and a height of 4 ft.

What is its area?

Area: _____

True or false?

A parallelogram has a greater area than a rectangle with the same base and the same height.

A. True

B. False

SET 56 • 5.G.B.4

FORWARD 2

Shapes with given conditions • Grade 7 • Geometry • 7.G.A.2

Can you draw a rectangle that is not a square? Write yes or no.

Answer: _____

A quadrilateral has 4 sides of equal length but no right angles.

What is it?

- A.** A rhombus that is not a square
- B.** A trapezoid
- C.** A rectangle
- D.** A square

True or false?

Only one triangle can be drawn with angles of 40° , 60° , and 80° .

A. True

B. False

Is there more than one triangle shape with sides of 6 cm, 6 cm, and 6 cm? Write yes or no.

Answer: _____

True or false?

A triangle with three sides of 5 cm also has three equal angles.

- A. True
- B. False

Classify triangles by their sides and angles

MAIN	Grade 5 main question	5.G.B.4
BACKWARD 1	Right, acute, and obtuse triangles	4.G.A.2
BACKWARD 2	Kinds of angles	4.G.A.1
BACKWARD 3	Triangles	2.G.A.1
FORWARD 1	Area of triangles	6.G.A.1
FORWARD 2	Complementary, supplementary, and vertical angles	7.G.B.5

SET 57 • 5.G.B.4

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.4

A triangle has sides of 6 cm, 6 cm, and 6 cm.

Which name fits it?

- A.** Obtuse
- B.** Scalene
- C.** Equilateral
- D.** Right

A triangle has angles of 90° , 50° , and 40° .
Classify it by its angles.

Classification: _____

True or false?

Every equilateral triangle is also an isosceles triangle.

A. True

B. False

A triangle has sides of 3, 5, and 7 units and an angle of 120° .

Which name fits it?

A. Acute equilateral

B. Right scalene

C. Right isosceles

D. Obtuse scalene

A triangle has angles of 70° , 70° , and 40° .
Classify it by its angles and by its sides.

By angles: _____

By sides: _____

SET 57 • 5.G.B.4

BACKWARD 1

Right, acute, and obtuse triangles • Grade 4 • Geometry • 4.G.A.2

True or false?

A right triangle has exactly one right angle.

A. True

B. False

Which angle measure is obtuse?

- A.** 10°
- B.** 45°
- C.** 130°
- D.** 90°

A triangle has one angle greater than 90° .
What kind of triangle is it?

Kind: _____

True or false?

In an acute triangle, all three angles are less than 90° .

A. True

B. False

Which angle measure is acute?

A. 100°

B. 180°

C. 60°

D. 90°

SET 57 • 5.G.B.4

BACKWARD 2

Kinds of angles • Grade 4 • Geometry • 4.G.A.1

True or false?

A right angle measures 90° .

A. True

B. False

What kind of angle measures 35° ?

- A.** Right
- B.** Straight
- C.** Obtuse
- D.** Acute

What kind of angle measures 150° ?

Kind: _____

True or false?

Two right angles together make 360° .

A. True

B. False

What kind of angle measures exactly 90° ?

Kind: _____

SET 57 • 5.G.B.4

BACKWARD 3

Triangles • Grade 2 • Geometry • 2.G.A.1

How many sides does a triangle have?

Sides: _____

True or false?

A triangle has 3 angles.

A. True

B. False

Which shape has 3 sides?

- A.** Square
- B.** Hexagon
- C.** Pentagon
- D.** Triangle

True or false?

All triangles have sides of equal length.

A. True

B. False

How many vertices (corners) does a triangle have?

Vertices: _____

SET 57 • 5.G.B.4

FORWARD 1

Area of triangles • Grade 6 • Geometry • 6.G.A.1

A right triangle has legs that are 6 cm and 8 cm long.
What is its area?

Area: _____

A triangle has a base of 10 in. and a height of 7 in.
What is its area?

Area: _____

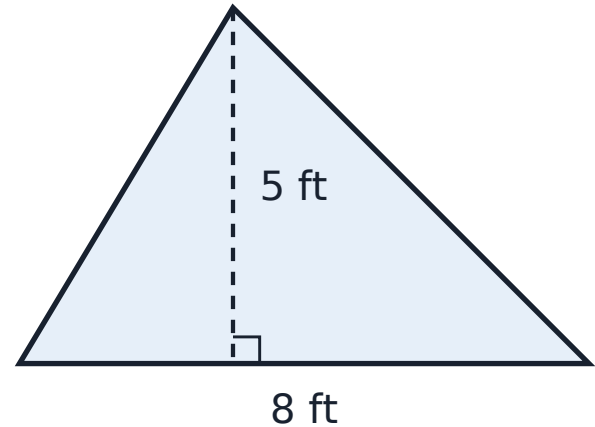
An obtuse triangle has a base of 5 m. Its height, measured outside the triangle, is 4 m.

What is its area?

- A.** 20 square meters
- B.** 40 square meters
- C.** 10 square meters
- D.** 9 square meters

Find the area of the triangle.

Area: _____



True or false?

Two triangles with the same base and the same height have the same area.

A. True

B. False

SET 57 • 5.G.B.4

FORWARD 2

Complementary, supplementary, and vertical angles • Grade 7 • Geometry •
7.G.B.5

Two angles are complementary. One measures 35° .
What is the measure of the other angle?

Angle: _____

Two angles are supplementary. One measures 110° .
What is the measure of the other angle?

Angle: _____

Vertical angles are always ____.

- A.** right angles
- B.** complementary
- C.** equal
- D.** supplementary

An angle and a 72° angle together form a straight line.
What is the measure of the angle?

Angle: _____

True or false?

Two adjacent angles that together form a right angle are complementary.

- A. True
- B. False